

SEMIORTHOGONAL DECOMPOSITIONS OF PERFECT COMPLEXES ON SPLIT SMOOTH PROJECTIVE TORIC SCHEMES OVER $\mathbb{Z}$

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ABSTRACT. Let X be a split smooth projective toric scheme over $S = \operatorname{Spec} \mathbb{Z}$. We prove that $\operatorname{Perf}(X)$ admits a finite S -linear semiorthogonal decomposition whose components are all equivalent to $\operatorname{Perf}(\mathbb{Z})$, represented by relative perfect-complex blocks $M \mapsto \pi^* M \otimes E_i$. As a consequence, for every split base change X_R one has

$$\operatorname{K}(X_R) \simeq \bigoplus_{i=1}^N \operatorname{K}(R)$$

for nonconnective algebraic K -theory. The proof is an integral Cox–VGIT argument using finite monomial descent and a directed path from an outside point, lying outside the effective cone and hence having empty semistable locus, to the target chamber; the auxiliary quotient stacks have finite split diagonalizable stabilizers, although their acting groups are usually tori.

1. INTRODUCTION

Let $S = \operatorname{Spec} \mathbb{Z}$, and let $\pi : X \rightarrow S$ be a split smooth projective toric scheme. The question is whether $\operatorname{Perf}(X)$, with its natural $\operatorname{Perf}(S)$ -module structure, admits a finite semiorthogonal decomposition whose pieces are all equivalent to $\operatorname{Perf}(S)$. We prove this by an integral Cox–VGIT induction. The “outside” starting point used below is simply a rational character outside the effective cone; its semistable locus is empty, so the induction starts from the zero category.

The proof has the following shape. First, we write a split smooth projective toric scheme as a Cox chamber quotient. The chamber quotient attached to the given ample class is a scheme, but the wall-crossing path used to reach it naturally passes through auxiliary quotient stacks. Second, we prove a characteristic-free finite monomial quotient formalism: for split diagonalizable quotients, quasi-coherent complexes are graded complexes, finite monomial covers satisfy Čech descent, and character lines generate the perfect category. This is the point at which non-étale stabilizers such as μ_p in characteristic p are handled.

Third, we prove the local wall theorem in two stages. The residual wall theorem treats the vector-bundle model over a corrected wall center and computes the window decomposition by weighted-projective pushforward and Koszul peeling. The actual wall theorem then compares a genuine Cox wall with this residual model. The comparison is made on monomial opens:

$$U_0 = V \cup C, \quad U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C,$$

so Hom calculations reduce to the residual open while generation is proved globally from character-line generation and Koszul triangles.

Finally, we choose a generic directed segment from an outside point with empty semistable locus to the target chamber. At each wall the direction is chosen so that the chamber already constructed is the source side of the $r \geq 0$ wall theorem. The corrected wall centers are smaller in a lexicographic induction: two-sided centers have smaller dimension, and a one-sided center with only one normal variable has smaller rational Picard rank. Thus the induction starts

Date: June 17, 2026.

2020 Mathematics Subject Classification. 14M25, 14F08, 14A20, 18G80.

Key words and phrases. toric varieties, perfect complexes, semiorthogonal decompositions, Cox construction, VGIT, integral models.

from the zero category outside the effective cone, crosses finitely many walls, and ends at the desired Cox chamber quotient.

The result should be viewed in the context of derived toric geometry and derived VGIT. Kawamata proved foundational field-case results on derived categories of toric varieties and stacks, giving exceptional-type semiorthogonal decompositions under toric hypotheses [Kaw, Theorem 1.1]; BFK and Halpern-Leistner developed general window methods for derived categories of GIT quotients [BFK, §3.5][HL, Theorem 2.10]. The present paper uses a special monomial Cox–VGIT argument to keep the construction relative to $\mathrm{Spec} \mathbb{Z}$. The additional work is to control the auxiliary quotient stacks appearing along the wall-crossing path, including finite split diagonalizable stabilizers that need not be étale on positive-characteristic fibers, and to extract relative $\mathrm{Perf}(\mathbb{Z})$ -linear blocks represented by perfect complexes. The monomial Cox setting lets us write the relevant opens, centers, Hom cones, and Koszul windows explicitly in graded coordinates over $\mathbb{Z}$; this is the mechanism used in place of a general VGIT window theorem. The field-case and VGIT results just cited provide the comparison framework for the theorem, while the proof below constructs the relative objects directly over $\mathbb{Z}$ and then takes fibers. The output is an integral relative block decomposition, rather than a strong exceptional collection of line bundles; this distinction is compatible with Hille–Perling’s counterexample to King’s strong line-bundle exceptional collection conjecture [HP]. The final Cox quotient attached to a smooth toric scheme is a scheme, but the proof passes through auxiliary quotient stacks, and the finite stabilizer hypotheses refer to those intermediate chamber quotients and wall centers.

After base change to a field k , the theorem gives

$$\mathrm{Perf}(X_k) = \langle E_{1,k}, \dots, E_{N,k} \rangle,$$

where each $E_{i,k}$ is an exceptional perfect object and N is forced by the rank of $K_0(X_k)$, equivalently by the number of maximal cones in the smooth complete fan. Thus the field-fiber consequences are fibers of a single relative $\mathrm{Perf}(\mathbb{Z})$ -linear construction; the integral part of the argument is the production of the global objects E_i and the relative Ext identities before any base change.

The block decomposition also gives standard additive-invariant consequences. For every commutative ring R , the split toric base change X_R satisfies

$$K(X_R) \simeq \bigoplus_{i=1}^N K(R)$$

for nonconnective algebraic K -theory. More generally, any Morita-invariant additive invariant takes the decomposition to a direct sum of N copies of its value on R . This includes THH and the topological cyclic homology functor TC for small stable categories modeled by spectral categories, as in the localization theorem of Blumberg–Mandell [BM, Theorems 5.9, 7.1, and 7.2].

For navigation, the paper is organized as follows. Section 2 sets up the quotient-stack and support-cone language and proves the finite monomial descent package. Sections 3 and 4 are geometric: they show that the corrected centers appearing at one-sided and two-sided walls are again proper split toric quotient stacks of smaller inductive complexity. Section 5 proves the basic generation theorem for character lines on finite monomial quotient stacks. Section 6 proves the residual wall theorem, the local vector-bundle calculation from which the block counts and windows come. Section 7 is the comparison section: it identifies the actual Cox wall with the residual model on a two-open monomial cover and then proves the actual wall crossing theorem. Section 8 assembles these local wall crossings along the directed path, and Section 9 applies the resulting Cox quotient theorem to split smooth projective toric schemes and records the base-change and additive invariant consequences. The load-bearing assertions are the finite monomial quotient formalism (Proposition 2.5), the character-line generation theorem (Theorem 5.4), the residual wall theorem (Theorem 6.15), the actual–residual comparison package (Proposition 7.14), the actual wall theorem (Proposition 7.26),

and the directed induction theorem (Theorem 8.10). The remaining lemmas are arranged to make the hypotheses of these assertions checkable in the integral Cox setting.

The dependency graph is acyclic. It has four layers. First, the finite monomial quotient formalism gives character-line generation. Second, center closure supplies the recursive input for corrected wall centers. Third, character-line generation together with the residual weighted-projective and Koszul calculations gives the residual wall theorem. Fourth, the residual wall theorem and the actual–residual open/Hom comparison give the actual wall theorem; the directed global induction then iterates that theorem to prove the main result. The actual wall theorem itself does not assume the global induction. In the global induction, the only recursive calls are to corrected wall centers, and Propositions 3.2 and 4.2 prove that these centers have smaller lexicographic complexity.

Theorem 1.1 (Main theorem). *Let X be a split smooth projective toric scheme over $S = \operatorname{Spec} \mathbb{Z}$. Then there exist perfect complexes*

$$E_1, \dots, E_N \in \operatorname{Perf}(X)$$

such that the functors

$$\Phi_i : \operatorname{Perf}(S) \longrightarrow \operatorname{Perf}(X), \quad M \longmapsto \pi^* M \otimes E_i$$

are fully faithful and their images form a finite S -linear semiorthogonal decomposition

$$\operatorname{Perf}(X) = \langle \Phi_1 \operatorname{Perf}(S), \dots, \Phi_N \operatorname{Perf}(S) \rangle.$$

Moreover

$$\mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_i, E_i) \simeq \mathcal{O}_S, \quad \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_j, E_i) \simeq 0 \quad (j > i),$$

and $E_1, \dots, E_N$ generate $\operatorname{Perf}(X)$.

Remark 1.2. Although admissibility is not included in our standing convention for semiorthogonal decompositions, the blocks obtained in Theorem 1.1 are admissible: after the object-extraction step they are images of relative block functors $M \mapsto \pi^* M \otimes E_i$, and Lemma 9.3 applies. This admissibility is the reason the later additive-invariant corollaries give split exact filtrations and direct-sum decompositions, despite the fact that admissibility is not part of our definition of a semiorthogonal decomposition. The ordering of the blocks and the chosen representatives are determined by the directed VGIT path and window choices used in the construction. The paper only needs the finiteness of N : it is produced by adding the wall block counts along that path, and no canonical closed formula for N is asserted. This is compatible with the standard computation, after base change to a field, that the rank of K_0 of a smooth complete toric variety is the number of maximal cones of the fan; see, for example, the cellular description of Chow groups and K_0 for smooth complete toric varieties in [CLS, Theorem 12.3.12 and Chapter 13]. The construction below determines N through the chosen directed wall-crossing path; after applying the K -theory corollary, N is also forced nonconstructively by the rank of K_0 on any geometric fiber. Thus, for the toric schemes of the main theorem, this integer agrees with the number of maximal cones of the underlying smooth complete fan, although the proof below does not use this closed formula to construct the decomposition.

2. PRELIMINARIES

2.1. Quotient conventions. If A is a finitely generated abelian group, let $\mathbb{Z}[A]$ denote the group algebra with basis elements $\mathbf{e}^\alpha$, $\alpha \in A$, and Hopf algebra structure

$$\Delta(\mathbf{e}^\alpha) = \mathbf{e}^\alpha \otimes \mathbf{e}^\alpha, \quad \epsilon(\mathbf{e}^\alpha) = 1, \quad S(\mathbf{e}^\alpha) = \mathbf{e}^{-\alpha}.$$

We write

$$D(A) = \operatorname{Spec} \mathbb{Z}[A]$$

for the corresponding split diagonalizable group scheme over $\mathbb{Z}$. Its character group is canonically A : the character indexed by $\alpha \in A$ is the morphism $D(A) \rightarrow \mathbb{G}_m$ corresponding

to the Hopf algebra map $\mathbb{Z}[t, t^{-1}] \rightarrow \mathbb{Z}[A]$, $t \mapsto e^\alpha$. The assignment $A \mapsto D(A)$ is contravariant in A . For example, $D(\mathbb{Z}^r) = \mathbb{G}_m^r$ and $D(\mathbb{Z}/n) = \mu_n$; hence $D(A)$ can be non-smooth over primes dividing the torsion of A .

If

$$R = \bigoplus_{\alpha \in A} R_\alpha$$

is an A -graded ring, then quasi-coherent sheaves on $[\mathrm{Spec} R/D(A)]$ are A -graded R -modules. The character line $\mathcal{O}(\alpha)$ corresponds to the graded free module $R(\alpha)$, with

$$R(\alpha)_\gamma = R_{\gamma+\alpha}.$$

Equivalently, the standard generator of $R(\alpha)$ has degree $-\alpha$. With this convention

$$\mathbf{R} \mathrm{Hom}(\mathcal{O}(\alpha), M) \simeq M_{-\alpha}, \quad \mathbf{R} \mathrm{Hom}(\mathcal{O}(\alpha), \mathcal{O}(\beta)) \simeq R_{\beta-\alpha}.$$

This remains true when A has torsion, because comodules over a split diagonalizable group are the same as graded modules: a comodule structure over $\mathbb{Z}[A]$ is exactly the direct-sum decomposition into A -homogeneous pieces. Homogeneous localization, tensor products, internal Homs, and finite Čech totalizations are all taken degreewise in this graded category.

All derived categories below are used in their stable ∞ -categorical, or equivalently dg-enhanced, form. For an affine graded ring R , $D_{\mathrm{qc}}([\mathrm{Spec} R/D(A)])$ is the unbounded derived stable ∞ -category of the Grothendieck abelian category of A -graded R -modules, obtained by inverting quasi-isomorphisms of graded complexes (equivalently, by using any standard dg enhancement such as h-injective graded complexes). Its homotopy category is the usual triangulated unbounded derived category of graded modules. No coderived or completed derived category is used. For a finite monomial open U , $D_{\mathrm{qc}}([U/D(A)])$ is computed from the finite Čech diagram of these affine graded derived stable ∞ -categories. This is the fppf quotient/flat Hopf-algebroid model attached to the split diagonalizable action. We are not using a general assertion that, for an arbitrary algebraic stack, the unbounded derived category of quasi-coherent sheaves automatically agrees with a separately defined D_{qc} . In the present quotient charts the equivalence is the explicit graded-module equivalence below.

The subcategory $\mathrm{Perf}([U/D(A)]) \subset D_{\mathrm{qc}}([U/D(A)])$ is the stack-theoretic perfect subcategory. Equivalently, an object is perfect if it restricts on every affine homogeneous chart to a bounded complex of finitely generated graded projective modules; this equivalence is proved in Proposition 2.8 and is the standing model used afterwards. The only external stack-theoretic input in this identification is the standard fppf locality of perfectness; equivalently, perfectness may be checked after the faithfully flat quotient atlas. For the underlying perfect-complex criterion on affine rings and ringed spaces we use [Sta, Tags 0656 and 08CL], and for the stack-theoretic compact-generation framework we use Hall–Rydh [HR, Theorem A]. After pullback to the affine quotient charts, the comparison with bounded finite graded projective complexes and the equality with compact objects are proved directly in the graded module model below. The monomial opens U are quasi-compact and quasi-separated, and their finite intersections are again monomial affine opens. For such finite covers, restrictions and mapping complexes are computed by the ordinary finite graded Čech totalization. Whenever a quasi-compact open immersion j occurs, j_* is used only on D_{qc} , where it preserves filtered colimits in this qcqs setting; perfect objects are handled by descent and local perfectness.

Lemma 2.1 (graded model for split diagonalizable quotients). *Let A be a finitely generated abelian group, let $G = D(A)$, and let $\mathrm{Spec} R$ be an affine scheme with G -action corresponding to an A -grading on R . Then $\mathrm{QCoh}([\mathrm{Spec} R/G])$ is the category of A -graded R -modules, and $D_{\mathrm{qc}}([\mathrm{Spec} R/G])$ is the derived category of the Grothendieck abelian category of A -graded R -modules. Under this identification, the character line $\mathcal{O}(\alpha)$ is $R(\alpha)$ with the shift convention $R(\alpha)_\gamma = R_{\gamma+\alpha}$; homogeneous localization, tensor products, internal Homs, and finite Čech totalizations are computed degreewise. Perfect objects on affine quotient charts are precisely*

the compact objects in this graded derived category. Equivalently,

$$\mathrm{Perf}([\mathrm{Spec} R/D(A)]) = D_{\mathrm{qc}}([\mathrm{Spec} R/D(A)])^c = \mathrm{thick}\{R(\alpha) : \alpha \in A\}.$$

Proof. By the definition above, $G = D(A) = \mathrm{Spec} \mathbb{Z}[A]$. A G -equivariant quasi-coherent module on $\mathrm{Spec} R$ is therefore an R -module M with a compatible $\mathbb{Z}[A]$ -comodule structure. Such a comodule structure is equivalent to a decomposition $M = \bigoplus_{\alpha \in A} M_\alpha$ for which $R_\beta M_\alpha \subset M_{\alpha+\beta}$. This gives the abelian category of graded R -modules; this is the standard description of representations of split diagonalizable groups, see [SGA3, Exposé I, §4]. This reference treats diagonalizable groups with arbitrary abelian character group; in particular it includes torsion character groups such as $A = \mathbb{Z}/n$, for which $D(A) = \mu_n$ need not be smooth over $\mathbb{Z}$. This is an exact equivalence of Grothendieck abelian categories: kernels, cokernels, extensions, filtered colimits, and tensor products are computed on the underlying graded R -modules. Consequently passing to unbounded derived stable ∞ -categories gives the stated model for $D_{\mathrm{qc}}([\mathrm{Spec} R/G])$; there is no additional derived invariants functor or group-cohomology construction hidden in this identification. All listed operations are defined by the same underlying algebraic operations on graded modules, hence degreewise. The objects $R(\alpha)$, $\alpha \in A$, are compact projective generators for the graded module category: maps out of $R(\alpha)$ pick the $(-\alpha)$ -homogeneous component, and the family detects zero objects. By the standard compact-generation criterion for a derived category with a set of compact projective generators, the compact objects are exactly $\mathrm{thick}\{R(\alpha) : \alpha \in A\}$.

This thick subcategory is also the affine graded perfect subcategory. A finite graded free module is a finite direct sum of shifts $R(\alpha)$, and a finitely generated graded projective module is a retract of such a finite graded free module. Hence bounded complexes of finitely generated graded projectives, together with retracts, are precisely the finite-cell objects generated by the character shifts; conversely every object in the displayed thick subcategory is represented by such a bounded graded perfect complex.

Finally this affine graded perfect subcategory agrees with the stack-theoretic perfect subcategory. Perfectness for the quotient stack is checked after the faithfully flat quotient atlas; this is the fppf-local form of the perfect-complex criterion, cf. [Sta, Tags 0656 and 08CL]. The atlas $\mathrm{Spec} R \rightarrow [\mathrm{Spec} R/D(A)]$ is faithfully flat and locally of finite presentation. Pullback of $R(\alpha)$ to the atlas is a free rank-one module with equivariant grading, and bounded finite graded projective complexes are exactly the descent data for equivariant perfect complexes on the quotient. The atlas need not be smooth when A has torsion, and no smoothness of stabilizers is used. The objects $R(\alpha)$ are the equivariant vector bundles attached to characters. No étale averaging or invariant-exactness argument is involved; torsion in A only changes the grading group, and all constructions are performed directly in the split graded category. For the general stack-theoretic compact-generation framework for local quotient stacks, see Hall–Rydh [HR, Theorem A]; the equality used here is proved in the explicit affine graded model above. Equivalently, this is the standard compact-generation statement for affine quotients by split diagonalizable groups: the shifts of the equivariant structure sheaf form a set of compact projective generators, and perfect complexes are the compact objects in the resulting graded derived category. $\square$

Lemma 2.2 (degree-zero pushforward for affine diagonalizable quotients). *Let $R = \bigoplus_{\alpha \in A} R_\alpha$ be an A -graded ring and put $R_0 = R^{D(A)}$. For the quotient morphism*

$$q : [\mathrm{Spec} R/D(A)] \longrightarrow \mathrm{Spec} R_0$$

and a graded complex $M \in D_{\mathrm{qc}}([\mathrm{Spec} R/D(A)])$, one has

$$\mathbf{R}q_* M \simeq M_0$$

as a complex of R_0 -modules. More generally, on a homogeneous degree-zero principal open $D(f) \subset \mathrm{Spec} R_0$, the sections are

$$\mathbf{R}\Gamma(D(f), \mathbf{R}q_* M) \simeq (M[f^{-1}])_0.$$

Proof. By Lemma 2.1, M is a complex in the abelian category of A -graded R -modules. The morphism q is induced by the inclusion $R_0 \subset R$. On an affine open $D(f) \subset \operatorname{Spec} R_0$, pulling back to the quotient corresponds to the homogeneous localization $R[f^{-1}]$ with f of degree zero. Quasi-coherent sections on the quotient over that open are therefore the degree-zero part $(M[f^{-1}])_0$. Taking the zero-degree summand is an exact projection in the category of graded modules: kernels, cokernels, limits, colimits, and complex cohomology are computed degree by degree. Thus the displayed formula is a direct graded-module computation, not a derived invariant functor calculation. $\square$

In this paper a “split toric quotient stack” means a quotient stack $[U/D(A)]$ over $\mathbb{Z}$, where A is a finitely generated abelian group and U is a finite homogeneous monomial open in a split affine Cox space. The acting group may have both torus factors and finite diagonalizable factors. When we say that such a quotient has finite split diagonalizable stabilizers, we mean the stabilizer group schemes of geometric points; the acting group $D(A)$ itself need not be finite. Its toric character line bundles are the line bundles coming from characters of $D(A)$, equivalently from elements of A ; on homogeneous Cox charts they are graded free rank-one character shifts. The word “split” means that the diagonalizable groups, their character groups, and all Cox coordinates are defined over $\mathbb{Z}$. For an A -graded Cox ring B , we write

$$B_0 = B^{D(A)}$$

for the degree-zero, equivalently invariant, subring. This equality is a graded statement: $B^{D(A)}$ denotes the homogeneous summand of degree zero in the Cox grading. For a homogeneous ideal $I \subset B$, we write

$$D(I) = \bigcup_{f \in I} D(f).$$

Thus $D(0) = \emptyset$, and for a finitely generated monomial ideal $(m_1, \dots, m_s)$ the open $D(I)$ is the finite monomial open $\bigcup_i D(m_i)$. This convention is used both for irrelevant ideals and for the normal-variable ideals in wall crossings. The notation is overloaded but context-separated: $D(A)$ always denotes the split diagonalizable group with character group A , while $D(I)$ or $D(f)$ denotes an open subset cut out by an ideal or element of a Cox ring. For a quotient $[U/D(A)]$ with U open in an affine space with n Cox coordinates and with finite stabilizers on the chamber under discussion, we use

$$\dim([U/D(A)]/S) = n - \operatorname{rk} A_{\mathbb{Q}}$$

for its relative stack dimension over $S = \operatorname{Spec} \mathbb{Z}$. This agrees with the usual relative dimension because the acting torus has dimension $\operatorname{rk} A_{\mathbb{Q}}$ and the stabilizers are finite. When such a quotient is called smooth, this means smooth as an Artin stack over the base. In the chamber quotients below the acting group $D(A)$ has A free, hence is a split torus, and the atlas from the smooth affine open is smooth. The stabilizers of the action may be finite split diagonalizable group schemes that are not étale after reduction modulo primes; smoothness of the stack is not a statement that these stabilizers are smooth group schemes. We use three nested classes of such quotients. A finite monomial quotient stack is any $[U/D(A)]$ with U a finite union of homogeneous monomial principal opens. A Cox chamber quotient is obtained by taking $U = U_{\theta}$ for a character θ in a full-dimensional chamber of the support-cone decomposition. The residual wall quotients are finite monomial quotient stacks over a corrected wall center; they are used only as local wall models. This terminology records hypotheses, not additional structure.

Remark 2.3 (terminology summary). The following terms are internal shorthand used throughout the paper.

- (i) A *finite monomial quotient stack* is a quotient $[U/D(A)]$ where U is a finite union of homogeneous monomial principal opens in a split affine Cox space.
- (ii) A *Cox chamber quotient* is $[U_{\theta}/D(A)]$ for a full-dimensional GIT chamber character θ .

- (iii) A *proper-full chamber* is a nonempty full-dimensional chamber quotient satisfying the hypotheses of Lemma 2.29; the term is not an additional structure.
- (iv) At a wall, the *corrected center* is the zero-weight Cox quotient obtained from the wall character after deleting the nonzero normal variables.
- (v) The *residual wall* is the vector-bundle model over this center, and the *actual wall* is the corresponding wall in the original Cox affine space.

This summary is only a guide to the notation; the precise hypotheses are always those stated in the lemma or proposition where the term is used.

We use the following convention for semiorthogonal decompositions. A finite semiorthogonal decomposition

$$\mathcal{C} = \langle \mathcal{A}_1, \dots, \mathcal{A}_N \rangle$$

of an idempotent-complete stable category means that the $\mathcal{A}_i$ are full stable idempotent-complete subcategories, that $\text{Map}_{\mathcal{C}}(\mathcal{A}_j, \mathcal{A}_i) = 0$ for $j > i$, and that the smallest thick subcategory of $\mathcal{C}$ containing all $\mathcal{A}_i$ is $\mathcal{C}$. Admissibility is not part of this convention unless stated explicitly. In the arguments below the components are images of explicit fully faithful S -linear functors. The zero category is regarded as having the empty semiorthogonal decomposition. If $\mathcal{C}$ is a $\text{Perf}(S)$ -module category, such a decomposition is called S -linear when each $\mathcal{A}_i$ is a $\text{Perf}(S)$ -submodule subcategory, equivalently when the inclusion $\mathcal{A}_i \hookrightarrow \mathcal{C}$ is $\text{Perf}(S)$ -linear after identifying $\mathcal{A}_i$ with its essential image. In the main theorem and in the wall crossings this is verified by constructing the components as images of explicit $\text{Perf}(S)$ -linear fully faithful functors.

The following finite Čech descent statement is used repeatedly. It is a standard form of descent for perfect complexes on quasi-compact quasi-separated schemes and quotient stacks; see [Sta, Tags 09UC, 09M0] for descent of finiteness properties of complexes and the compact/perfect comparison in the qcqs scheme case. In the special monomial quotient situation below we also give the direct graded-module verification by the finite Čech complex. Together with Proposition 2.5, this is the basic technology used later: all descent, compactness, and weight-summand arguments are degreewise statements in the graded derived category, not arguments with an invariants functor.

Proposition 2.4 (finite monomial derived and perfect descent). *Let $B = \mathbb{Z}[x_1, \dots, x_n]$ be graded by a finitely generated abelian group A , put $G = D(A)$, and let*

$$U = \bigcup_{i=1}^s D(m_i) \subset \text{Spec } B$$

be a finite union of homogeneous monomial principal opens. For $\emptyset \neq I \subset \{1, \dots, s\}$, write

$$m_I = \prod_{i \in I} m_i, \quad U_I = D(m_I).$$

Then restriction gives an equivalence

$$D_{\text{qc}}([U/G]) \simeq \text{holim}_{\emptyset \neq I} D_{\text{qc}}([U_I/G]).$$

An object of $D_{\text{qc}}([U/G])$ is perfect if and only if all its restrictions to the U_i 's are perfect. Hence restriction also gives an equivalence

$$\text{Perf}([U/G]) \simeq \text{holim}_{\emptyset \neq I} \text{Perf}([U_I/G]).$$

Moreover the same finite Čech diagram computes mapping complexes, and the first-level restrictions are conservative on D_{qc} and on Perf .

Proof. The scheme U is quasi-compact and quasi-separated, with finite affine open cover by the principal opens $D(m_i)$. For quasi-coherent complexes, ordinary finite Čech descent gives the first displayed equivalence. In the present quotient situation this descent is completely degreewise: localization at a homogeneous monomial is exact in graded modules, the finite

Čech complex is the ordinary Čech complex in each homogeneous degree, and the totalization defining descent is taken in the category of graded complexes. Lemma 2.1 identifies this graded derived category with $D_{\text{qc}}([U/G])$, so this is exactly the quotient-stack D_{qc} -descent statement.

Perfectness is local for this finite open cover after the quasi-coherent descent object has been reconstructed: on the affine quotient charts it is the condition of lying in the thick subcategory generated by graded free modules, and this condition is checked after restriction to the finite principal cover. Thus the perfect descent statement is obtained in two steps. Given a compatible Čech family of perfect objects on the U_I 's, D_{qc} -descent first produces a unique object $P \in D_{\text{qc}}([U/G])$. Since the restrictions $P|_{U_i}$ are perfect and perfectness is local for the finite monomial cover, P is perfect. Conversely, a perfect object restricts to a compatible family of perfect objects. Hence the full subcategory of perfect objects is exactly the homotopy limit of the perfect subcategories inside the already established D_{qc} -Čech limit. The mapping-complex formula follows by applying the same finite totalization to the internal Hom complexes on the cover. Because the Čech diagram is finite and all maps preserve A -degree, taking a fixed homogeneous summand commutes with this totalization; this is the form used in the weight estimates below. Conservativity follows because an object whose restrictions to the first level opens are zero has zero Čech totalization. $\square$

Proposition 2.5 (finite monomial quotient package). *The preceding proposition has the following concrete consequences. Whenever a finite monomial quotient stack $[U/D(A)]$ is covered by homogeneous principal opens U_i , we use the following statements.*

- (a) *Objects of $D_{\text{qc}}([U/D(A)])$ are reconstructed from the finite graded Čech diagram of the U_i 's, and the same is true for perfect objects after reconstruction in D_{qc} .*
- (b) *For $P, Q \in \text{Perf}([U/D(A)])$, the mapping complex is the finite totalization of the chartwise mapping complexes. Since all restriction maps preserve the A -grading, a fixed homogeneous summand may be taken before or after the finite totalization; see Lemma 2.11.*
- (c) *Compactness of perfect objects follows chartwise from Lemma 2.1: finite totalizations commute with filtered colimits of complexes, so a perfect object whose restrictions are compact on the finite affine quotient cover is compact globally. Conversely compact objects are perfect because their restrictions to the affine quotient charts are compact, hence perfect, and perfectness is local for the finite cover.*
- (d) *The first-level restrictions are conservative. This is used only as object-level conservativity; generation statements are proved separately by character-line generation and Koszul triangles.*

All these statements are degree-wise statements about graded modules; no étale averaging or invariant-exactness argument for finite stabilizers is used.

Lemma 2.6 (filtered colimits for monomial open pushforward). *Let $U \subset \text{Spec } B$ be a finite homogeneous monomial open, let $U' \subset U$ be another finite homogeneous monomial open, and write*

$$j : [U'/D(A)] \hookrightarrow [U/D(A)]$$

for the induced quasi-compact open immersion. Then the right adjoint

$$j_* : D_{\text{qc}}([U'/D(A)]) \longrightarrow D_{\text{qc}}([U/D(A)])$$

preserves filtered colimits.

Proof. Choose a finite homogeneous principal monomial cover $U = \bigcup_i D(m_i)$ such that $U' \cap D(m_i)$ is either empty or a finite union of homogeneous principal monomial opens in $D(m_i)$. Refining once more, we may compute j_* on the finite Čech diagrams associated with these covers. More explicitly, write

$$U' \cap D(m_i) = \bigcup_{a \in S_i} D(n_a)$$

with S_i finite and with every n_a homogeneous in $B[m_i^{-1}]$. If $U' \cap D(m_i) = \emptyset$, we take $S_i = \emptyset$ and the restriction of j_*M to $[D(m_i)/D(A)]$ is zero; the formula below is then the empty Čech totalization. For $M \in D_{\text{qc}}([U'/D(A)])$, the defining property of the open-immersion right adjoint gives

$$(j_*M)|_{[D(m_i)/D(A)]} \simeq \mathbf{R}\Gamma^{\text{gr}}(U' \cap D(m_i), M) \simeq \text{Tot}_{\emptyset \neq J \subset S_i} M|_{[D(n_J)/D(A)]}, \quad n_J = \prod_{a \in J} n_a,$$

viewed as a complex of A -graded $B[m_i^{-1}]$ -modules. Here $\mathbf{R}\Gamma^{\text{gr}}$ denotes graded sections before taking a degree-zero invariant summand; equivalently, it is the restriction of the quotient-stack object to the affine quotient chart $[D(m_i)/D(A)]$. The last equivalence is the finite monomial Čech formula on the affine chart. Thus the chartwise restriction of j_* is computed by finite totalizations of homogeneous localizations

$$M \mapsto M[n^{-1}]$$

over the monomial opens $D(n) \subset U' \cap D(m_i)$. Homogeneous localization of graded modules is exact and commutes degreewise with filtered colimits. Filtered colimits of graded modules are exact degreewise, so they commute with the finite kernels, cokernels, fibers, and cofibers appearing in a finite Čech totalization. Therefore the chartwise formula for j_* preserves filtered colimits.

The chartwise constructions are compatible on overlaps because all restriction maps are homogeneous localizations. Proposition 2.4 identifies the global D_{qc} categories with these finite graded Čech limits. Since restriction to every chart and every homogeneous localization in the finite diagram preserves filtered colimits, filtered colimits in the global finite Čech limit are detected chartwise. Hence the global right adjoint j_* , obtained by gluing the chartwise pushforwards, preserves filtered colimits. $\square$

Proof of Proposition 2.5. Parts (a), (b), and (d) are exactly the derived descent, mapping-complex, and conservativity assertions of Proposition 2.4, after identifying the quotient-stack derived categories with graded derived categories by Lemma 2.1. The compatibility with homogeneous summands follows because the Čech diagrams are finite and all restriction maps are homogeneous of degree zero.

For (c), let $P \in \text{Perf}([U/D(A)])$. Its restrictions to the finitely many affine quotient intersections are compact by Lemma 2.1. If $\{M_\alpha\}$ is a filtered diagram in $D_{\text{qc}}([U/D(A)])$, then

$$\mathbf{R}\text{Hom}(P, \text{colim}_\alpha M_\alpha)$$

is the finite totalization of the corresponding chartwise Hom complexes. Restriction commutes with filtered colimits, the chartwise Hom functors commute with filtered colimits by compactness, and finite totalizations of complexes commute with filtered colimits. Hence P is compact globally. Conversely, if Q is compact in $D_{\text{qc}}([U/D(A)])$, then its restriction to each affine quotient chart is compact for the following reason. For a quasi-compact monomial open immersion

$$j_I : [U_I/D(A)] \hookrightarrow [U/D(A)],$$

the functor j_{I*} on D_{qc} preserves filtered colimits by Lemma 2.6. Hence

$$\mathbf{R}\text{Hom}_{[U_I/D(A)]}(j_I^*Q, -) \simeq \mathbf{R}\text{Hom}_{[U/D(A)]}(Q, j_{I*}-)$$

commutes with filtered colimits. Thus j_I^*Q is compact on each affine quotient chart, hence perfect by Lemma 2.1. Proposition 2.4 then implies that Q is perfect on $[U/D(A)]$. Thus $D_{\text{qc}}([U/D(A)])^c = \text{Perf}([U/D(A)])$ for finite monomial quotient stacks. $\square$

Remark 2.7 (how the quotient formalism is used). The preceding results are deliberately stated in a form that matches the later wall-crossing arguments. The paper uses only the following consequences.

- (1) Affine split diagonalizable quotients are treated as ordinary unbounded derived stable ∞ -categories of graded modules (Lemma 2.1). This is where torsion in the character group is allowed; no smoothness or tameness of the stabilizer is used.
- (2) Finite homogeneous monomial covers give finite Čech descent for D_{qc} , for Perf , and for mapping complexes (Proposition 2.4). All later Hom calculations use this finite totalization, together with the fact that homogeneous summands commute with it (Lemma 2.11).
- (3) Perfect objects on finite monomial quotients are exactly compact objects of D_{qc} (Proposition 2.5 and Lemma 2.6). This is the only compactness input in the proof of character-line generation.
- (4) The first-level restrictions in a finite monomial cover are conservative. This is used only to detect that an object is zero after its chartwise restrictions vanish; generation statements are proved separately by character-line generation and Koszul triangles.

Thus Sections 5–7 do not appeal to any additional quotient-stack formalism beyond this package, except for the standard proper perfect base-change input isolated later in Lemma 9.4.

Proposition 2.8 (foundational finite monomial quotient theorem). *Let A be a finitely generated abelian group, let $B = \mathbb{Z}[x_1, \dots, x_n]$ be A -graded, and let*

$$T = [U/D(A)]$$

where $U \subset \text{Spec } B$ is a finite homogeneous monomial open. Then:

- (i) $D_{\text{qc}}(T)$ is computed by the finite graded Čech diagram of any finite homogeneous monomial principal cover of U .
- (ii) $\text{Perf}(T)$ is the full subcategory of objects whose restrictions to the affine quotient charts are bounded finite-cell graded perfect complexes; this condition satisfies finite monomial descent.
- (iii) $D_{\text{qc}}(T)^c = \text{Perf}(T)$.
- (iv) Mapping complexes of perfect objects are finite graded Čech totalizations, and taking any prescribed homogeneous summand commutes with these totalizations.
- (v) For every affine quotient chart $[\text{Spec } R/D(A)]$ in the cover,

$$\text{Perf}([\text{Spec } R/D(A)]) = \text{thick}\{R(\alpha) : \alpha \in A\}.$$

All assertions are valid when A has torsion; in particular no étaleness of the stabilizers is assumed.

Proof. For affine charts this is Lemma 2.1: the quotient category is the derived category of A -graded modules, and perfect objects are the finite-cell graded perfect complexes generated by the shifts $R(\alpha)$. The stack-theoretic use of the word perfect is the local finite-cell/dualizable one, checked after the faithfully flat quotient atlas; compactness is then identified explicitly in the affine graded category by the compact projective generators $R(\alpha)$.

For a finite monomial open U , Proposition 2.4 gives D_{qc} -descent and perfect descent from the finite monomial principal cover. Proposition 2.5 proves compactness and mapping-complex assertions from the same finite Čech diagram. The only use of open pushforward in that proof is on D_{qc} , where the relevant monomial j_* is computed degree-wise by homogeneous localization and finite Čech totalization; it is not used to push perfect objects to perfect objects. Since the diagrams are finite and all transition maps have degree zero, homogeneous summands commute with totalization by Lemma 2.11. Split diagonalizable torsion is recorded solely in the grading group, so the same proof applies to non-reduced stabilizers such as μ_p on characteristic p fibers. This proves the stack-level descent, compactness, and mapping-complex formalism used in the rest of the paper; later sections do not require a separate theorem asserting tameness, linear reductivity, or exactness of invariants for the finite stabilizers. $\square$

Remark 2.9 (standalone foundational quotient formalism). The foundational quotient input for the rest of the paper is the following package:

Proposition 2.8 is assembled from Lemma 2.1 + Lemma 2.2
+ Proposition 2.4 + Proposition 2.5.

Later appeals to finite monomial descent, compactness of perfects, homogeneous mapping-complex totalizations, homogeneous pushforward summands, or conservativity of monomial chart restrictions refer to this package. Thus these later arguments use the split grading rather than an étale averaging argument, and do not require étaleness of finite stabilizers. For example, over a characteristic p fiber a stabilizer such as μ_p is handled by the same split grading: all limits, colimits, Čech totalizations, and weight summands are computed degree by degree in the graded derived category.

Remark 2.10 (the categorical level of the finite quotient formalism). All equivalences in Proposition 2.8 are equivalences of presentable stable ∞ -categories for D_{qc} , and of small idempotent-complete stable ∞ -categories for Perf . The homotopy categories recover the usual triangulated derived categories. The proof is not an appeal to exactness of invariants for a linearly reductive group. Instead, after choosing the split diagonalizable group $D(A)$, every affine quotient chart is the ordinary unbounded derived stable ∞ -category of A -graded modules. The finite monomial cover then gives a finite Čech diagram in that graded model. When A has torsion, the quotient atlas is used as an fppf atlas rather than a smooth atlas; perfectness is checked fppf locally and then expressed in the graded finite-cell model of Lemma 2.1.

Thus the three uses that are most sensitive to non-étale stabilizers are all internal to the graded model:

- (i) D_{qc} -descent is the finite Čech descent of homogeneous localizations of graded complexes;
- (ii) $\text{Perf} = D_{\text{qc}}^c$ is checked by compact projective graded generators on affine charts, plus the explicit monomial-open j_* calculation of Lemma 2.6;
- (iii) mapping complexes and their λ -weight summands are finite graded totalizations, so homogeneous summands commute with them by Lemma 2.11.

This remark is included to fix the categorical model used in later sections: all Hom, totalization, compactness, and descent statements are made before passing to homotopy categories.

Lemma 2.11 (homogeneous summands in finite Čech totalizations). *Let $\mathcal{K}_\bullet$ be a finite diagram of A -graded complexes whose transition maps have degree zero. Then, for every $\delta \in A$,*

$$(\text{Tot } \mathcal{K}_\bullet)_\delta \simeq \text{Tot}((\mathcal{K}_\bullet)_\delta).$$

Consequently, if every term of a finite Čech diagram has no cohomology in a set of homogeneous degrees $\Omega \subset A$, then its totalization has no cohomology in degrees Ω . The same assertion holds for the cofiber of a morphism of finite graded Čech diagrams.

Proof. The category of A -graded complexes is the category of complexes in $\prod_{\delta \in A} \text{Ab}$, with morphisms preserving the degree. Finite limits, finite colimits, fibers, cofibers, and finite products are therefore computed degree by degree. A finite totalization is obtained from finite products and fibers, so taking the δ -homogeneous summand commutes with it. Cohology and cofibers are computed in the same degree-wise category, giving the vanishing statement. $\square$

Lemma 2.12 (refining finite semiorthogonal decompositions). *Let*

$$\mathcal{C} = \langle \mathcal{A}_1, \dots, \mathcal{A}_m \rangle$$

be a finite semiorthogonal decomposition in the above sense, and suppose each $\mathcal{A}_i$ has a finite semiorthogonal decomposition

$$\mathcal{A}_i = \langle \mathcal{A}_{i,1}, \dots, \mathcal{A}_{i,r_i} \rangle.$$

Then

$$\mathcal{C} = \langle \mathcal{A}_{1,1}, \dots, \mathcal{A}_{1,r_1}, \dots, \mathcal{A}_{m,1}, \dots, \mathcal{A}_{m,r_m} \rangle.$$

Proof. Semiorthogonality inside each $\mathcal{A}_i$ is part of its decomposition. If $j > i$, then every object of $\mathcal{A}_j$, hence of every $\mathcal{A}_{j,b}$, is right orthogonal to every object of $\mathcal{A}_i$, hence to every $\mathcal{A}_{i,a}$. The thick subcategory generated by all $\mathcal{A}_{i,a}$'s contains each $\mathcal{A}_i$, and therefore contains $\mathcal{C}$. $\square$

Lemma 2.13 (fully faithful functors transport SODs). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a fully faithful exact functor between idempotent-complete stable categories. If*

$$\mathcal{C} = \langle \mathcal{A}_1, \dots, \mathcal{A}_m \rangle$$

is a finite semiorthogonal decomposition, then the essential image $F(\mathcal{C}) \subset \mathcal{D}$ has the finite semiorthogonal decomposition

$$F(\mathcal{C}) = \langle F(\mathcal{A}_1), \dots, F(\mathcal{A}_m) \rangle.$$

Proof. Full faithfulness identifies mapping complexes in the image with mapping complexes in $\mathcal{C}$, so semiorthogonality is transported from $\mathcal{C}$. The thick subcategory of $F(\mathcal{C})$ generated by the $F(\mathcal{A}_i)$'s contains $F(A)$ for every object $A \in \mathcal{C}$, because the $\mathcal{A}_i$'s thickly generate $\mathcal{C}$ and F is exact and preserves retracts onto its essential image. $\square$

Lemma 2.14 (full faithfulness on thick closures). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an exact functor between stable idempotent-complete categories, and let $\mathcal{S} \subset \mathcal{C}$ be a set of objects. If*

$$\mathrm{Map}_{\mathcal{C}}(S, S') \longrightarrow \mathrm{Map}_{\mathcal{D}}(F(S), F(S'))$$

is an equivalence for all $S, S' \in \mathcal{S}$, then F is fully faithful on $\mathrm{thick}(\mathcal{S})$.

Proof. For fixed $S \in \mathcal{S}$, the objects T for which $\mathrm{Map}_{\mathcal{C}}(S, T) \rightarrow \mathrm{Map}_{\mathcal{D}}(F(S), F(T))$ is an equivalence form a thick subcategory, hence contain $\mathrm{thick}(\mathcal{S})$. Then, for fixed $T \in \mathrm{thick}(\mathcal{S})$, the objects R for which $\mathrm{Map}_{\mathcal{C}}(R, T) \rightarrow \mathrm{Map}_{\mathcal{D}}(F(R), F(T))$ is an equivalence again form a thick subcategory containing $\mathcal{S}$, hence contain $\mathrm{thick}(\mathcal{S})$. $\square$

Lemma 2.15 (checking containment on thick generators). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an exact functor between idempotent-complete stable categories, let $\mathcal{S} \subset \mathcal{C}$, and let $\mathcal{T} \subset \mathcal{D}$ be a thick subcategory. If $F(S) \in \mathcal{T}$ for every $S \in \mathcal{S}$, then $F(\mathrm{thick}(\mathcal{S})) \subset \mathcal{T}$.*

Proof. The full subcategory of objects $C \in \mathcal{C}$ with $F(C) \in \mathcal{T}$ is thick: exactness of F preserves shifts and finite cones, and idempotent completeness of $\mathcal{T}$ handles retracts. It contains $\mathcal{S}$, hence contains $\mathrm{thick}(\mathcal{S})$. $\square$

Lemma 2.16 (compact-generation criterion). *Let $\mathcal{D}$ be a compactly generated presentable stable ∞ -category, and let $\mathcal{S} \subset \mathcal{D}^c$ be a set of compact objects whose localizing subcategory is all of $\mathcal{D}$. Then*

$$\mathcal{D}^c = \mathrm{thick}(\mathcal{S}).$$

Proof. Let $\mathcal{C} = \mathrm{thick}(\mathcal{S})$, regarded as a small idempotent-complete stable category. The inclusion $\mathcal{C} \hookrightarrow \mathcal{D}$ extends uniquely to a colimit-preserving exact functor

$$F : \mathrm{Ind}(\mathcal{C}) \longrightarrow \mathcal{D}.$$

For $C \in \mathcal{C}$, both mapping functors $\mathrm{Map}_{\mathrm{Ind}(\mathcal{C})}(C, -)$ and $\mathrm{Map}_{\mathcal{D}}(F(C), -)$ commute with filtered colimits, and they agree on $\mathcal{C}$. Hence they agree when the second variable is any Ind-object. For a fixed Ind-object X , the functors $\mathrm{Map}_{\mathrm{Ind}(\mathcal{C})}(-, X)$ and $\mathrm{Map}_{\mathcal{D}}(F(-), F(X))$ send filtered colimits in the first variable to limits, so agreement on $\mathcal{C}$ extends to every first variable as well. Thus F is fully faithful. Its essential image is a localizing subcategory of $\mathcal{D}$ containing $\mathcal{S}$, hence is $\mathcal{D}$. Thus F is an equivalence. The compact objects of $\mathrm{Ind}(\mathcal{C})$ are precisely the idempotent completion of $\mathcal{C}$, which is $\mathcal{C}$ by construction. Transporting compact objects across F gives the displayed equality. $\square$

2.2. Cox chambers. Let $B = \mathbb{Z}[x_1, \dots, x_n]$ be graded by a free abelian group A , with $a_i = \deg x_i$. For $I \subset \{1, \dots, n\}$, put

$$\omega_I = \sum_{i \in I} \mathbb{Q}_{\geq 0} a_i \subset A_{\mathbb{Q}}.$$

We use the empty-support convention

$$\omega_{\emptyset} = \text{cone}(\emptyset) = \{0\}, \quad x^{\emptyset} = 1.$$

Thus a monomial ideal generated by a set containing $x^{\emptyset}$ is the unit ideal. A Koszul complex on an empty sequence is the one-term complex $\mathcal{O}$. For $\chi \in \text{Eff}$, set

$$\sigma(\chi) = \bigcap_{\chi \in \omega_I} \omega_I.$$

The cones $\sigma(\chi)$ form the torus GIT fan, or support-cone decomposition, on $\text{Eff} = \sum_i \mathbb{Q}_{\geq 0} a_i$. This is the standard GIT fan for a linear action of a split torus on affine space [CLS, §14.3]; see also [DH]. The relative interiors of these cones are exactly the GIT-equivalence classes: two points lie in the same relative interior precisely when they belong to the same subcollection of the closed support cones ω_I . Thus every ω_I is a union of closed cells, and U_{χ} is constant as χ varies in the relative interior of a cell. At the relative interior of a codimension-one wall face F , the quotient of the local fan by $\text{span}_{\mathbb{Q}} F$ is one-dimensional; hence there are at most two adjacent full-dimensional chambers, one on each side. Actual wall crossings below are taken either between such adjacent chambers or across a boundary facet of Eff , where the outside side has empty semistable locus. Points outside Eff are used only as empty starting points for directed segments; they are not regarded as chambers of the GIT fan. If $\chi \in A$, set

$$U_{\chi} = \bigcup_{d>0, f \in B_{d\chi}} D(f).$$

The corresponding quotient is

$$\mathfrak{X}_{\chi} = [U_{\chi}/D(A)].$$

Lemma 2.17 (outside points have empty semistable locus). *If $\zeta \in A_{\mathbb{Q}} \setminus \text{Eff}$, then the semistable locus for ζ is empty. Equivalently, for any $N > 0$ with $N\zeta \in A$, $U_{N\zeta} = \emptyset$.*

Proof. The semistable locus is the union of coordinate strata with $\zeta \in \omega_I$. Every support cone ω_I is contained in Eff , so no such stratum exists. Positive scaling does not change membership in the cones ω_I , giving the equivalent integral formulation. $\square$

Lemma 2.18 (finite monomiality of semistable opens). *For every integral character $\chi \in A$, the semistable open U_{χ} is a finite union of homogeneous monomial principal opens. Finite unions and finite intersections of such opens are again finite homogeneous monomial opens. In particular, the opens obtained from finitely many chamber or wall semistable loci by union and intersection are finite monomial opens.*

Moreover, if $B_Z = \mathbb{Z}[z_1, \dots, z_m]$ is a coordinate subring and

$$I_Z = (z^K : \theta_0 \in \text{cone}(\deg z_k : k \in K)) \subset B_Z$$

is a wall-center monomial ideal, then $D(I_Z)$, $D(I_Z B)$, and their intersections with finitely generated normal-coordinate opens such as $D(u_1, \dots, u_p)$ and $D(v_1, \dots, v_q)$, are finite monomial opens.

Proof. Let $I_{\chi} \subset B$ be the ideal generated by all homogeneous monomials of degrees $d\chi$, $d > 0$. Then $U_{\chi} = D(I_{\chi})$: a semi-invariant section is a finite sum of monomial semi-invariants, and whenever such a section is nonzero at a point, at least one of its monomial terms is nonzero there. The polynomial ring B is noetherian, so the monomial ideal I_{χ} has finitely many

monomial generators. Hence U_χ is a finite monomial open. A finite union of finite monomial opens is again indexed by the finite union of the two sets of monomials, and

$$\left(\bigcup_i D(m_i)\right) \cap \left(\bigcup_j D(n_j)\right) = \bigcup_{i,j} D(m_i n_j)$$

proves the assertion for intersections. The ideal I_Z is a monomial ideal in the noetherian ring B_Z , hence finitely generated by monomials. Extension to B and intersection with the finitely generated ideals $(u_1, \dots, u_p)$, $(v_1, \dots, v_q)$ preserve finite monomiality by the same union and intersection formula. $\square$

Lemma 2.19 (support-cone cells). *Let C be a relatively open cell of the support-cone decomposition and let ω_J be a support cone. If $C \cap \omega_J \neq \emptyset$, then $C \subset \omega_J$. If F is a wall face contained in $\overline{C}$ and ω_J is closed with $C \subset \omega_J$, then $F \subset \omega_J$. In particular, if θ_0 lies in the relative interior of F and $\theta_0 \in \omega_J$, then $F \subset \omega_J$.*

Proof. Membership in a fixed closed cone ω_J is constant on each relatively open cell by definition of the common refinement: the cell decomposition is obtained by intersecting all support cones and their complements inside their linear spans. Thus if one point of C lies in ω_J , every point of C does. The second follows from closedness:

$$F \subset \overline{C} \subset \omega_J.$$

For the last assertion, the cell containing θ_0 is F ; since ω_J is a union of closed cells and contains a point of the relative interior of F , it contains all of F . $\square$

Lemma 2.20 (local wall fan). *Let F be a codimension-one cell of the support-cone decomposition, and let θ_0 lie in its relative interior. Let $\lambda : A \rightarrow \mathbb{Z}$ be a primitive normal with $\text{span}_{\mathbb{R}} F = \ker(\lambda)_{\mathbb{R}}$. Then, after shrinking to a small neighborhood of θ_0 , the quotient of the support-cone decomposition by $\text{span}_{\mathbb{R}} F$ is the one-dimensional fan with cones $\{0\}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{R}_{\leq 0}$, with one of the two rays possibly missing when $F \subset \partial \text{Eff}$. Consequently, if $\xi \in A_{\mathbb{R}}$ has $\lambda(\xi) > 0$, then $\theta_0 + \varepsilon \xi$ lies in the positive adjacent chamber for all sufficiently small $\varepsilon > 0$, provided the positive side is inside Eff . The same statement with signs reversed holds on the negative side.*

Proof. Let $\text{Star}(F)$ be the subfan of the support-cone decomposition consisting of cells whose closures contain F . Since θ_0 lies in the relative interior of F , after shrinking to a sufficiently small neighborhood of θ_0 , only cells in $\text{Star}(F)$ occur. Quotienting this star by $\text{span}_{\mathbb{R}} F$ gives a one-dimensional rational polyhedral fan. Its open rays are precisely the full-dimensional chambers adjacent to F ; in the interior case there are two such rays, while at a boundary wall face contained in ∂Eff the ray outside Eff is absent. After shrinking around θ_0 , the original decomposition is the inverse image of this quotient fan. The final assertion is exactly this local description applied to the transverse direction ξ . $\square$

Lemma 2.21 (zero-weight chamber criterion). *Let $A_0 \subset A$ be a saturated subgroup whose rational span $(A_0)_{\mathbb{Q}} \subset A_{\mathbb{Q}}$ is a rational hyperplane. Let B_Z be the polynomial subring generated by variables whose degrees lie in A_0 , and let $F \subset (A_0)_{\mathbb{Q}}$ be a support-cone cell with $\text{span}_{\mathbb{Q}} F = (A_0)_{\mathbb{Q}}$. Let θ_0 lie in the relative interior of F . Assume that for every zero-variable support cone*

$$\omega_K^Z = \text{cone}(\deg z_k : k \in K) \subset (A_0)_{\mathbb{Q}},$$

the condition $\theta_0 \in \omega_K^Z$ implies $F \subset \omega_K^Z$. Then θ_0 lies in the relative interior of a full-dimensional cell of the support-cone decomposition for the zero-weight datum (B_Z, A_0) . Its zero-weight semistable locus is precisely the open determined by the zero-variable supports K with $\theta_0 \in \omega_K^Z$.

Proof. For a zero-variable cone ω_K^Z , membership is locally constant near θ_0 inside the relative interior of F . If $\theta_0 \in \omega_K^Z$, this is the hypothesis $F \subset \omega_K^Z$. If $\theta_0 \notin \omega_K^Z$, closedness of ω_K^Z gives a

small relative neighborhood of θ_0 in F disjoint from $\omega_K^{\mathbb{Z}}$. Hence the zero-weight support-cone decomposition has a cell containing such a relative neighborhood of θ_0 in F . Since F spans $(A_0)_{\mathbb{Q}}$, that cell is full-dimensional. The final assertion is the defining support-cone criterion for semistability in the zero-weight Cox datum. $\square$

Corollary 2.22 (zero-weight chamber induced by an ambient wall). *Let $A_0 = \ker \lambda$, and let B_Z be the subring generated by the zero-weight variables. Let $F \subset (A_0)_{\mathbb{Q}}$ be a support-cone cell of the ambient Cox datum with $\text{span}_{\mathbb{Q}} F = (A_0)_{\mathbb{Q}}$, and let θ_0 lie in the relative interior of F . If the zero-weight semistable locus $U_{\theta_0}^{\mathbb{Z}}$ is nonempty, then θ_0 is a full-dimensional chamber character, relative to $(A_0)_{\mathbb{Q}}$, for the zero-weight datum (B_Z, A_0) , and the corresponding semistable locus is exactly $U_{\theta_0}^{\mathbb{Z}}$.*

Proof. For every zero-variable support K , the zero-weight cone $\omega_K^{\mathbb{Z}}$ is the same closed cone as the ambient support cone ω_K generated by that subset of variables, now viewed inside $(A_0)_{\mathbb{Q}}$. If $\theta_0 \in \omega_K^{\mathbb{Z}}$, then $\theta_0 \in \omega_K$ as an ambient support cone. Since θ_0 is in the relative interior of the ambient cell F , Lemma 2.19 gives $F \subset \omega_K$, hence $F \subset \omega_K^{\mathbb{Z}}$. Lemma 2.21 applies. The nonemptiness assumption only rules out the vacuous case in which there is no zero-variable support containing θ_0 . $\square$

Lemma 2.23 (monomial quotient chart normalization). *Let A be free, let $B = \mathbb{Z}[x_1, \dots, x_n]$ be A -graded, and let $f = x^u$ be a homogeneous monomial with support J . Assume that the degrees $\{a_j : j \in J\}$ span $A_{\mathbb{Q}}$. Choose $K \subset J$ such that $\{a_k : k \in K\}$ is a $\mathbb{Q}$ -basis of $A_{\mathbb{Q}}$, let $L_K \subset A$ be the subgroup generated by the a_k 's, and put $H_K = D(A/L_K)$. Then A/L_K is finite and*

$$[D(f)/D(A)] \simeq [S_K/H_K],$$

where

$$S_K = \text{Spec } \mathbb{Z}[y_i \ (i \notin J), \ y_j^{\pm 1} \ (j \in J \setminus K)]$$

with H_K -action through the degree classes modulo L_K . Moreover

$$\Gamma(S_K, \mathcal{O})^{H_K} \simeq B[f^{-1}]^{D(A)}.$$

Proof. Since $\{a_k : k \in K\}$ is a $\mathbb{Q}$ -basis, A/L_K is finite. The map

$$D(A) \longrightarrow \mathbb{G}_m^K$$

defined by the characters a_k , $k \in K$, is a finite flat isogeny with kernel H_K . On $D(f)$, all coordinates indexed by J are invertible. The isogeny therefore lets one set the coordinates indexed by K equal to 1 fppf-locally. Two such normalizations differ by a unique element of H_K , acting on the remaining coordinates through their classes in A/L_K . Equivalently, if $i : S_K \rightarrow D(f)$ is the slice with the K -coordinates equal to 1, the action map descends to an isomorphism of $D(A)$ -schemes

$$D(A) \times^{H_K} S_K \xrightarrow{\sim} D(f),$$

where $D(A) \times^{H_K} S_K$ is the contracted product for the residual H_K -action. This isomorphism gives the displayed quotient-stack isomorphism $[S_K/H_K] \simeq [D(f)/D(A)]$. Taking invariant functions after the normalization gives

$$\Gamma(S_K, \mathcal{O})^{H_K} \simeq B[f^{-1}]^{D(A)},$$

because an H_K -invariant monomial on S_K is exactly the normalization of a $D(A)$ -weight-zero Laurent monomial on $D(f)$, and the normalization process is reversible fppf-locally. $\square$

Lemma 2.24 (finite diagonalizable quotient properness). *Let F be a finite abelian group, let $H = D(F)$, and let C be a finitely generated F -graded $\mathbb{Z}$ -algebra. Then*

$$[\text{Spec } C/H] \longrightarrow \text{Spec } C^H$$

is proper.

Proof. Choose a finite set of algebra generators for C over $\mathbb{Z}$, and replace each generator by its finitely many homogeneous components. Thus C is generated by homogeneous elements $c_1, \dots, c_t$. If N is the exponent of F , then $c_i^N \in C^H$ for every i . Every monomial in the c_i 's can therefore be reduced, as a C^H -multiple, to one with all exponents $< N$. Hence C is a finite C^H -module. The composite $\text{Spec } C \rightarrow \text{Spec } C^H$ is finite, hence proper. The atlas $\text{Spec } C \rightarrow [\text{Spec } C/H]$ is finite flat and surjective because H is finite locally free. The quotient morphism is of finite type by descent from the finite composite $\text{Spec } C \rightarrow \text{Spec } C^H$; its diagonal is finite because it is controlled by the finite group scheme H , hence it is separated. Universal closedness descends from the proper surjective atlas by [Sta, Tag 0CL4, Lemma 101.37.6], applied to

$$\text{Spec } C \longrightarrow [\text{Spec } C/H] \longrightarrow \text{Spec } C^H.$$

Thus the quotient morphism is proper in the sense of [Sta, Tag 0CL4, Definition 101.37.1]. If C is a Laurent polynomial algebra, as in the monomial chart normalization below, one includes both a Laurent variable and its inverse among the homogeneous generators. The same reduction of exponents modulo N then proves finiteness over C^H ; no étale slice or averaging argument is involved. $\square$

Lemma 2.25 (Proj quotient map and monomial charts). *Let A be free, $B = \mathbb{Z}[x_1, \dots, x_n]$ be A -graded, and $\chi \in A$. Put*

$$R_\chi = \bigoplus_{d \geq 0} B_{d\chi}.$$

There is a morphism

$$q_\chi : [U_\chi/D(A)] \longrightarrow \text{Proj } R_\chi$$

induced by the semi-invariant section ring R_χ . If $f \in B_{d\chi}$ is a homogeneous monomial of positive Proj degree d , then

$$q_\chi^{-1}(D_+(f)) = [D(f)/D(A)]$$

and

$$\Gamma(D_+(f), \mathcal{O}_{\text{Proj } R_\chi}) = (R_\chi[f^{-1}])_0 = B[f^{-1}]^{D(A)}.$$

Proof. The open U_χ is the union of the $D(f)$'s for homogeneous semi-invariants $f \in B_{d\chi}$, $d > 0$. On such a principal open, the degree-zero part of the homogeneous localization $R_\chi[f^{-1}]$ defines the standard affine chart $D_+(f) \subset \text{Proj } R_\chi$. Here f has degree d for the ordinary Proj grading on R_χ , while it has Cox A -degree $d\chi$ as an element of B . A fraction h/f^m has Proj degree zero exactly when $h \in B_{md\chi}$, and this is exactly the condition that h/f^m have $D(A)$ -weight zero as an element of $B[f^{-1}]$. Hence

$$(R_\chi[f^{-1}])_0 = B[f^{-1}]^{D(A)}.$$

The affine morphisms $[D(f)/D(A)] \rightarrow D_+(f)$ defined by these invariant rings agree on overlaps because homogeneous localization is functorial. They therefore glue to q_χ , and the displayed preimage identity follows from the same principal-open construction. Thus q_χ is an algebraic morphism locally of finite presentation: on the affine chart $D_+(f)$ it is the quotient morphism $[D(f)/D(A)] \rightarrow \text{Spec } B[f^{-1}]^{D(A)}$ associated with a finitely generated graded algebra. $\square$

Lemma 2.26 (chartwise properness for the Proj quotient map). *Let $q_\chi : [U_\chi/D(A)] \rightarrow \text{Proj } R_\chi$ be the morphism of Lemma 2.25. Let $f_1, \dots, f_N$ be homogeneous monomials in positive Proj degrees such that the opens $D_+(f_i)$ cover $\text{Proj } R_\chi$. Suppose that, for each i , the restricted morphism*

$$q_\chi^{-1}(D_+(f_i)) = [D(f_i)/D(A)] \longrightarrow D_+(f_i)$$

is proper. Then q_χ is proper.

Moreover, if f and g are two such monomials, the two affine descriptions restrict on $D_+(fg) = D_+(f) \cap D_+(g)$ to the same quotient morphism

$$[D(fg)/D(A)] \longrightarrow \text{Spec } B[(fg)^{-1}]^{D(A)}.$$

Proof. For a homogeneous monomial f , Lemma 2.25 identifies $q_\chi^{-1}(D_+(f))$ with $[D(f)/D(A)]$, and the target affine chart with $\text{Spec } B[f^{-1}]^{D(A)}$. If f and g are two monomials, then the overlap $D_+(f) \cap D_+(g)$ is $D_+(fg)$, and pulling the two affine descriptions back to this overlap is exactly homogeneous localization at fg . The invariant rings on both sides are therefore the same ring $B[(fg)^{-1}]^{D(A)}$, and the quotient stacks are the same stack $[D(fg)/D(A)]$. This proves the compatibility and the cocycle condition for the chartwise morphisms.

Properness of morphisms of algebraic stacks is local on the target for the Zariski topology: in the definition [Sta, Tag 0CL4, Definition 101.37.1], finite type, separatedness, and universal closedness may be checked after a Zariski cover of the target. Since the $D_+(f_i)$'s cover $\text{Proj } R_\chi$ and the preimage over each one is proper by hypothesis, q_χ is proper. $\square$

Lemma 2.27 (monomial Proj charts cover). *Let $R = \bigoplus_{d \geq 0} R_d$ be a graded ring generated over R_0 by homogeneous monomials $m_1, \dots, m_N$ of positive degree. Then the opens $D_+(m_i)$ cover $\text{Proj } R$. More generally, if a homogeneous section*

$$g = \sum_t c_t m'_t$$

is written as a finite sum of monomial terms of the same Proj degree, then

$$D_+(g) \subset \bigcup_t D_+(m'_t).$$

Proof. The irrelevant ideal R_+ has radical $\sqrt{(m_1, \dots, m_N)}$. Indeed, every positive-degree homogeneous element is an R_0 -linear combination of monomials in the m_i 's, so any homogeneous prime containing all m_i contains R_+ . Hence no homogeneous prime of $\text{Proj } R$ contains all m_i , and the $D_+(m_i)$'s cover.

For the second assertion, let $\mathfrak{p}$ be a homogeneous prime with $g \notin \mathfrak{p}$. Since g is a finite sum of monomial terms of the same degree and $\mathfrak{p}$ is prime, at least one term m'_t is not in $\mathfrak{p}$. Thus every point of $D_+(g)$ lies in some $D_+(m'_t)$. This proves the inclusion. In particular, nonmonomial semi-invariant sections introduce no additional affine charts beyond the monomial terms. $\square$

Lemma 2.28 (properness of a finitely generated graded Proj). *Let $R = \bigoplus_{d \geq 0} R_d$ be a positively graded R_0 -algebra. Assume that R is generated as an R_0 -algebra by finitely many homogeneous elements of positive degrees. Then $\text{Proj } R$ is proper over $\text{Spec } R_0$.*

Proof. Let $\xi_1, \dots, \xi_s$ be homogeneous R_0 -algebra generators of positive degrees $e_1, \dots, e_s$. Choose $N > 0$ divisible by all e_i . The Veronese subring

$$R^{(N)} = \bigoplus_{m \geq 0} R_{mN}$$

has the same Proj as R . The degree-one piece R_N , viewed as the degree 1 part of $R^{(N)}$, is a finite R_0 -module: it is generated by the finitely many monomials $\xi_1^{a_1} \cdots \xi_s^{a_s}$ satisfying $\sum_i a_i e_i = N$. Let $S \subset R^{(N)}$ be the graded R_0 -subalgebra generated by R_N in degree 1. Then S is a quotient of $\text{Sym}_{R_0}(R_N)$, so $\text{Proj } S$ is projective over $\text{Spec } R_0$.

It remains to see that $R^{(N)}$ is finite as a graded S -module. For a monomial $\xi^a = \xi_1^{a_1} \cdots \xi_s^{a_s}$ of total degree mN , write

$$a_i = q_i(N/e_i) + r_i, \quad 0 \leq r_i < N/e_i.$$

Then $\xi_i^{q_i(N/e_i)} = (\xi_i^{N/e_i})^{q_i}$ is a product of degree-one elements of S , and the residue monomial ξ^r has bounded exponents. Because the original total degree and the removed degrees are multiples of N , the residue total degree is also a multiple of N . There are only finitely many such residue monomials, and they generate $R^{(N)}$ over S . Thus $R^{(N)}$ is finite over S . The induced morphism $\text{Proj } R^{(N)} \rightarrow \text{Proj } S$ is finite, hence proper, and $\text{Proj } S$ is projective over $\text{Spec } R_0$. Therefore $\text{Proj } R = \text{Proj } R^{(N)}$ is proper over $\text{Spec } R_0$. $\square$

Lemma 2.29 (proper full chambers). *Assume that A is free and that $B_0 = B^{D(A)}$ is finite over $\mathbb{Z}$. If χ lies in a full-dimensional chamber and $U_\chi \neq \emptyset$, then $\mathfrak{X}_\chi$ is split, smooth, and proper over $\mathbb{Z}$, and its stabilizers are finite split diagonalizable group schemes.*

Proof. Fixed Cox data. Throughout this lemma the Cox ring B , the grading group A , and the degree-zero invariant ring $B_0 = B^{D(A)}$ are fixed. Varying the chamber character χ changes only the semistable monomial open U_χ , not the invariant ring over which the Proj quotient is constructed.

Finite stabilizers and smoothness. For a geometric point with nonzero coordinate support I , the stabilizer in $D(A)$ is $D(A/A_I)$, where $A_I \subset A$ is generated by $\{a_i : i \in I\}$. If the point is χ -semistable and χ is in a full-dimensional chamber, then the full-dimensional cell containing χ is contained in the closed cone ω_I . Hence ω_I has full linear span in $A_\mathbb{Q}$, so $A_I \otimes \mathbb{Q} = A \otimes \mathbb{Q}$. Thus A/A_I is finite and the stabilizer is finite split diagonalizable. Smoothness follows because A is free, so $D(A)$ is a split torus, U_χ is open in affine space, and $U_\chi \rightarrow [U_\chi/D(A)]$ is a smooth atlas.

It remains to prove properness. Put

$$R_\chi = \bigoplus_{d \geq 0} B_{d\chi}.$$

Finite generation and the projective quotient. The proof has three separate steps. First we construct the Proj quotient map q_χ . Then we identify its monomial affine preimages, after a finite flat diagonalizable normalization, with quotient charts $[S_K/H_K]$ by finite split diagonalizable groups. Finally we use a finite-module argument to prove that each such quotient chart is proper over its invariant affine chart. The monomials in R_χ are indexed by the affine semigroup

$$\{(u, d) \in \mathbb{Z}_{\geq 0}^n \times \mathbb{Z}_{\geq 0} : \sum_i u_i a_i = d\chi\}.$$

This is the set of lattice points in the rational polyhedral cone cut out by $\sum_i t_i a_i = s\chi$ inside $\mathbb{R}_{\geq 0}^{n+1}$. The ambient cone $\mathbb{R}_{\geq 0}^{n+1}$ is pointed: if a vector and its negative both lie in it, then the vector is zero. Hence the subcone is pointed as well, and Gordan's lemma applies. Separating a finite set of semigroup generators according to whether $d = 0$ or $d > 0$, the $d = 0$ generators generate the monomial subring $B_0 = R_{\chi,0}$, while the finitely many positive-degree generators generate R_χ as a graded B_0 -algebra. Indeed, every semigroup element (u, d) is a nonnegative integral sum of the chosen generators; collecting all summands with $d = 0$ gives a coefficient in B_0 , and the remaining summands are products of the positive Proj-degree generators. Thus $\text{Proj } R_\chi$ is proper over $\text{Spec } B_0$ by Lemma 2.28. Hence $\text{Proj } R_\chi$ is proper over $\mathbb{Z}$ because B_0 is finite over $\mathbb{Z}$.

Let $q_\chi : [U_\chi/D(A)] \rightarrow \text{Proj } R_\chi$ be the morphism of Lemma 2.25. *Monomial affine charts and properness.* Choose homogeneous monomial B_0 -algebra generators $m_1, \dots, m_N$ for the positive-degree part of R_χ . Lemma 2.27 shows that the affine opens $D_+(m_i)$ cover $\text{Proj } R_\chi$, and that nonmonomial semi-invariants are covered by their monomial terms. It is therefore enough to check properness over these monomial affine Proj charts, with $f = x^u$. By Lemma 2.25, the preimage of $D_+(f)$ under q_χ is $[D(f)/D(A)]$, and

$$\Gamma(D_+(f), \mathcal{O}_{\text{Proj } R_\chi}) = (R_\chi[f^{-1}])_0 = B[f^{-1}]^{D(A)}.$$

If J is the support of f , then $\chi \in \omega_J$. Let C_χ be the full-dimensional cell containing χ . Since ω_J is a closed cone appearing in the support-cone decomposition and contains χ , Lemma 2.19 gives $C_\chi \subset \omega_J$. Hence ω_J has full linear span $A_\mathbb{Q}$. Choose $K \subset J$ as in Lemma 2.23. That lemma gives an isomorphism

$$[D(f)/D(A)] \simeq [S_K/H_K],$$

where S_K is affine of finite type over $\mathbb{Z}$ and H_K is finite split diagonalizable, and gives

$$\Gamma(S_K, \mathcal{O})^{H_K} \simeq B[f^{-1}]^{D(A)} = (R_\chi[f^{-1}])_0 = \Gamma(D_+(f), \mathcal{O}_{\text{Proj } R_\chi}).$$

Thus, over the affine chart $D_+(f)$, the comparison is the commutative diagram

$$\begin{array}{ccc} [D(f)/D(A)] & \simeq & [S_K/H_K] \\ \downarrow & & \downarrow \\ D_+(f) = \operatorname{Spec} B[f^{-1}]^{D(A)} & = & \operatorname{Spec} \Gamma(S_K, \mathcal{O})^{H_K}. \end{array}$$

The upper horizontal arrow is induced by the finite flat diagonalizable normalization of the variables in K , and the lower equality is the degree-zero invariant-ring identification above. Thus the target affine chart is $\operatorname{Spec} \Gamma(S_K, \mathcal{O})^{H_K}$. By Lemma 2.24, the morphism $[S_K/H_K] \rightarrow \operatorname{Spec} \Gamma(S_K, \mathcal{O})^{H_K} = D_+(f)$ is proper. Hence q_χ is proper over every monomial affine chart of $\operatorname{Proj} R_\chi$. The chart morphisms are compatible on overlaps by Lemma 2.26, and the same lemma shows that q_χ is proper. Since $\operatorname{Proj} R_\chi$ is proper over $\mathbb{Z}$, so is $\mathfrak{X}_\chi$. $\square$

Remark 2.30 (properness inputs in Lemma 2.29). The proof of Lemma 2.29 uses no unmentioned properness theorem for quotient stacks. The argument is the following explicit chain. Gordan's lemma gives finite generation of the section ring R_χ , and Lemma 2.28 makes $\operatorname{Proj} R_\chi$ proper over $\operatorname{Spec} B_0$. Lemma 2.25 constructs

$$q_\chi : [U_\chi/D(A)] \rightarrow \operatorname{Proj} R_\chi$$

and identifies its monomial affine preimages and invariant rings. Lemma 2.23 reduces each such preimage to a quotient by a finite split diagonalizable group. The properness of that finite quotient over its invariant affine chart is exactly Lemma 2.24, whose proof is a finite-module argument plus properness descent for algebraic stacks. Finally Lemma 2.26 glues these chartwise proper morphisms over the Zariski cover of $\operatorname{Proj} R_\chi$.

3. ONE-SIDED WALL CENTERS

Let A be free, $B = \mathbb{Z}[z_1, \dots, z_m, w_1, \dots, w_s]$, and let $\lambda : A \rightarrow \mathbb{Z}$ be primitive. Assume

$$\lambda(\deg z_i) = 0, \quad \lambda(\deg w_j) > 0$$

for all i, j . Put $A_0 = \ker \lambda$ and

$$B_Z = \mathbb{Z}[z_1, \dots, z_m].$$

Let $F \subset \lambda^{-1}(0)$ be a nonzero codimension-one wall face with $\operatorname{span}_{\mathbb{Q}} F = (A_0)_{\mathbb{Q}}$. Let θ_0 be an integral character in its relative interior after a positive scaling.

Lemma 3.1 (one-sided support reduction). *Let C be the nonempty full-dimensional chamber adjacent to F . If ω_J is a support cone with $C \cap \omega_J \neq \emptyset$, then*

$$F \subset \omega_J.$$

Consequently, whenever $\theta_0 \in \omega_J$, the character θ_0 lies in the cone generated by the zero-weight variables in J .

Proof. By Lemma 2.19, $C \cap \omega_J \neq \emptyset$ implies $C \subset \omega_J$. We then use the boundary relation between the wall face and the chamber closure:

$$C \subset \omega_J, \quad \omega_J \text{ closed}, \quad F \subset \overline{C} \implies F \subset \omega_J.$$

For the final assertion, no intersection hypothesis with C is needed. Assume only $\theta_0 \in \omega_J$. Choose a conical expression

$$\theta_0 = \sum_{z_i \in J} a_i \deg z_i + \sum_{w_j \in J} b_j \deg w_j, \quad a_i, b_j \in \mathbb{Q}_{\geq 0}.$$

Applying λ gives

$$0 = \lambda(\theta_0) = \sum_{w_j \in J} b_j \lambda(\deg w_j).$$

All $\lambda(\deg w_j)$ are strictly positive, so all $b_j = 0$. Thus θ_0 is generated by the zero-weight variables in J . $\square$

Proposition 3.2 (one-sided center closure). *Let the nonempty side $[U_\theta/D(A)]$ be split smooth proper over $\mathbb{Z}$, with finite split diagonalizable stabilizers and $B_0 = \mathbb{Z}$. Then the corrected one-sided wall center*

$$\mathfrak{Z} = [U_{\theta_0}^Z/D(A_0)], \quad U_{\theta_0}^Z \subset \text{Spec } B_Z,$$

is split smooth proper over $\mathbb{Z}$, with finite split diagonalizable stabilizers. It is again a Cox quotient with free character lattice, and its A_0 -degree-zero Cox part satisfies $(B_Z)_0 = \mathbb{Z}$. If s is the number of positive normal-weight variables, then

$$\dim[U_\theta/D(A)] - \dim \mathfrak{Z} = s - 1.$$

Thus dimension drops for $s \geq 2$; for $s = 1$ the dimension is unchanged but $\text{rk } A_{0,\mathbb{Q}} = \text{rk } A_{\mathbb{Q}} - 1$.

Proof. Since λ is primitive and A is free, $A_0 = \ker \lambda$ is a free direct summand of rank $\text{rk } A - 1$. Lemma 3.1 shows that the wall character is semistable using only zero-weight variables. Indeed, $\theta_0 \in F \subset \text{Eff}$, so some support cone ω_J contains θ_0 ; the one-sided support lemma then writes θ_0 as a nonnegative combination of the zero-weight variables in that same support. Hence $U_{\theta_0}^Z$ is nonempty. If θ_0 lies in a cone generated by zero-weight variables, then that ambient zero-variable cone contains the relative interior of the wall cell F , hence contains all of F by Lemma 2.19. Since F spans $A_{0,\mathbb{Q}}$, Corollary 2.22 applies to the zero-weight datum (B_Z, A_0) . Thus θ_0 is a full-dimensional chamber character, relative to $(A_0)_{\mathbb{Q}}$, for that datum, and its semistable locus is precisely $U_{\theta_0}^Z$.

If a monomial in B_Z has A_0 -degree zero, then it has A -degree zero. Since $B_0 = \mathbb{Z}$, this monomial is 1. Hence $(B_Z)_0 = \mathbb{Z}$. Lemma 2.29 applies to the zero-weight Cox datum (B_Z, A_0, θ_0) , giving splitness, smoothness, properness, and finite split diagonalizable stabilizers.

The dimensions are computed from the quotient presentations by subtracting the rank of the acting torus from the number of Cox coordinates:

$$\dim[U_\theta/D(A)] = (m + s) - \text{rk } A, \quad \dim \mathfrak{Z} = m - (\text{rk } A - 1),$$

so the difference is $s - 1$. The rank statement follows from $\text{rk } A_0 = \text{rk } A - 1$. $\square$

4. TWO-SIDED WALL CENTERS

Let A be free, let

$$B = \mathbb{Z}[z_1, \dots, z_m, u_1, \dots, u_p, v_1, \dots, v_q],$$

and let $\lambda : A \rightarrow \mathbb{Z}$ be primitive. Assume

$$\lambda(\deg z_i) = 0, \quad \lambda(\deg u_i) = a_i > 0, \quad \lambda(\deg v_j) = -b_j < 0$$

and $p, q \geq 1$. Put $A_0 = \ker \lambda$, $B_Z = \mathbb{Z}[z_1, \dots, z_m]$. Let $F \subset \lambda^{-1}(0)$ be a nonzero codimension-one wall face with $\text{span}_{\mathbb{Q}} F = (A_0)_{\mathbb{Q}}$, and let $\theta_0 \neq 0$ lie in its relative interior. Define

$$I_Z = (z^K : \theta_0 \in \text{cone}(\deg z_k : k \in K)) \subset B_Z, \quad U_{\theta_0}^Z = D(I_Z).$$

Lemma 4.1 (proper toric quotients have no nonconstant invariant monomials). *Let $B = \mathbb{Z}[x_1, \dots, x_n]$ be graded by a free abelian group A , and let θ be a character such that $U_\theta \neq \emptyset$ and $[U_\theta/D(A)]$ is proper over $\mathbb{Z}$. If the dense torus stratum is contained in U_θ , then the only A -degree-zero monomial in B is 1. Hence $B_0 = \mathbb{Z}$.*

Proof. Let x^ν be an A -degree-zero monomial. It descends to a regular function

$$f_\nu : [U_\theta/D(A)] \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

Assume $x^\nu \neq 1$. After base change to an algebraically closed field k , restrict f_ν to the dense quotient torus

$$[(\mathbb{G}_m)^n/D(A)]_k \subset [U_\theta/D(A)]_k.$$

The character lattice of this quotient torus is the kernel of the degree map $\mathbb{Z}^n \rightarrow A$. Since x^ν is a nonconstant degree-zero monomial, its exponent vector gives a nonzero character of the quotient torus. A nonzero character of a torus has image $\mathbb{G}_m \subset \mathbb{A}_k^1$.

On the other hand, the morphism $f_{\nu,k}$ is proper onto its image. The source is proper over k , and $\mathbb{A}_k^1$ is separated, so the graph of $f_{\nu,k}$ is a closed immersion into $[U_\theta/D(A)]_k \times_k \mathbb{A}_k^1$; composing with the projection to $\mathbb{A}_k^1$ is proper. Hence the scheme-theoretic image is a closed subscheme of $\mathbb{A}_k^1$ proper over k : the scheme-theoretic image of a proper morphism is proper over the base, equivalently properness descends along the induced proper surjection from the source to its image. Its underlying closed set contains $\mathbb{G}_m$, hence is all of $\mathbb{A}_k^1$, impossible because $\mathbb{A}_k^1$ is not proper over k . Thus no nonconstant degree-zero monomial exists.

Since monomials form a $\mathbb{Z}$ -basis of B , every degree-zero element is a $\mathbb{Z}$ -linear combination of degree-zero monomials. The only such monomial is 1, so $B_0 = \mathbb{Z}$. $\square$

Proposition 4.2 (two-sided center closure). *Assume one adjacent full-dimensional chamber quotient $[U_\theta/D(A)]$ is split smooth proper over $\mathbb{Z}$, with finite split diagonalizable stabilizers. Then $B_0 = \mathbb{Z}$. If the corrected two-sided wall center*

$$\mathfrak{Z} = [U_{\theta_0}^Z/D(A_0)]$$

is nonempty, then it is a split smooth proper Cox quotient over $\mathbb{Z}$, with free character lattice A_0 , finite split diagonalizable stabilizers, and $(B_Z)_0 = \mathbb{Z}$. In this nonempty case,

$$\dim[U_\theta/D(A)] - \dim \mathfrak{Z} = p + q - 1 \geq 1.$$

Proof. Because λ is primitive and A is free, A_0 is a free direct summand of A . First $B_0 = \mathbb{Z}$. Since $U_\theta \neq \emptyset$, the chamber character lies in $\text{Eff} = \text{cone}(\deg z_i, \deg u_i, \deg v_j)$; hence the dense torus stratum, whose support is all variables, is contained in U_θ . Lemma 4.1 gives $B_0 = \mathbb{Z}$. Every A_0 -degree-zero monomial in B_Z has original A -degree zero, so it is 1. Hence $(B_Z)_0 = \mathbb{Z}$.

If $U_{\theta_0}^Z = \emptyset$, there is no nonempty center to prove anything further about. Assume from now on that $U_{\theta_0}^Z \neq \emptyset$. We next check that θ_0 is a full-dimensional chamber character for the zero-weight datum. Since F is a codimension-one wall face with primitive normal λ , its linear span is $(A_0)_\mathbb{Q}$. Corollary 2.22 therefore applies and shows that θ_0 is a full-dimensional chamber character, relative to $(A_0)_\mathbb{Q}$, for the zero-weight datum, with semistable locus exactly $U_{\theta_0}^Z$.

Lemma 2.29, applied to the zero-weight datum (B_Z, A_0) , gives splitness, smoothness, properness, and finite split diagonalizable stabilizers for $\mathfrak{Z}$. Finally,

$$\dim[U_\theta/D(A)] = (m + p + q) - \text{rk } A, \quad \dim \mathfrak{Z} = m - (\text{rk } A - 1),$$

so the difference is $p + q - 1$, which is positive because $p, q \geq 1$. $\square$

5. CHARACTER-LINE GENERATION

Let Λ be a finitely generated abelian group, $H = D(\Lambda)$, and let $B = \mathbb{Z}[x_1, \dots, x_N]$ be a Λ -graded polynomial algebra. Let

$$U = \bigcup_{i=1}^r D(m_i)$$

be a quasi-compact H -stable open covered by homogeneous monomial principal opens, and put $T = [U/H]$.

Lemma 5.1 (affine graded generation). *For a Λ -graded ring R ,*

$$D_{\text{qc}}([\text{Spec } R/D(\Lambda)]) = \text{Loc}\{R(\alpha) : \alpha \in \Lambda\}, \quad \text{Perf}([\text{Spec } R/D(\Lambda)]) = \text{thick}\{R(\alpha) : \alpha \in \Lambda\}.$$

Proof. By Lemma 2.1, the category of quasi-coherent sheaves on the quotient is the category of Λ -graded R -modules. The modules $R(\alpha)$ are compact projective generators: $\mathbf{R} \text{Hom}(R(\alpha), M)$ is the $(-\alpha)$ -homogeneous component of M . If all these mapping complexes vanish, then every homogeneous component of M vanishes, hence $M = 0$. Therefore the $R(\alpha)$'s generate the derived category as a localizing subcategory. Since they are compact, the compact objects are the thick closure of these compact generators. By Lemma 2.1, these compact objects are precisely the perfect objects on the affine quotient. $\square$

Lemma 5.2 (homogeneous localization). *Let $m \in B_\delta$ be homogeneous with $D(m) \subset U$, and put $T_m = [D(m)/H]$. For $P \in D_{\text{qc}}(T)$ and $\beta \in \Lambda$,*

$$\mathbf{R} \operatorname{Hom}_{T_m}(B[m^{-1}](\beta), P|_{T_m}) \simeq \operatorname{hocolim}_{n \geq 0} \mathbf{R} \operatorname{Hom}_T(\mathcal{O}_T(\beta - n\delta), P).$$

Proof. Choose a finite monomial principal cover $U = \bigcup_i D(m_i)$, and write the Čech complex computing $R\Gamma(U, P)$ as

$$C^\bullet(U, P) = \operatorname{Tot}_{\emptyset \neq I} R\Gamma(D(m_I), P).$$

The restricted cover of $D(m)$ is

$$D(m) = \bigcup_i D(mm_i),$$

with finite intersections $D(mm_I)$. Since homogeneous localization is exact in the graded module category and commutes with the finite totalization, the Čech complex for the restricted cover is canonically

$$C^\bullet(D(m), P|_{D(m)}) = \operatorname{Tot}_{\emptyset \neq I} R\Gamma(D(mm_I), P) \simeq C^\bullet(U, P)[m^{-1}].$$

Thus

$$R\Gamma(D(m), P|_{D(m)}) \simeq R\Gamma(U, P)[m^{-1}]$$

as graded complexes. Taking the $(-\beta)$ -homogeneous component after inverting $m \in B_\delta$ gives

$$(R\Gamma(U, P)[m^{-1}])_{-\beta} \simeq \operatorname{hocolim}_{n \geq 0} R\Gamma(U, P)_{-\beta+n\delta},$$

where the transition maps are multiplication by m . In Hom-complex notation these transition maps are induced by precomposition with the coordinate multiplication maps

$$\mathcal{O}_T(\beta - (n+1)\delta) \xrightarrow{m} \mathcal{O}_T(\beta - n\delta).$$

Finally $\mathbf{R} \operatorname{Hom}_T(\mathcal{O}_T(\gamma), P)$ is the $(-\gamma)$ -homogeneous component of $R\Gamma(U, P)$, by the graded shift convention. This is the displayed formula. Equivalently, multiplication by m^n gives an isomorphism on T_m

$$\mathcal{O}_{T_m}(\beta - n\delta) \xrightarrow{\sim} \mathcal{O}_{T_m}(\beta),$$

so the localized Hom complex is the directed localization of the global Hom complexes $\mathbf{R} \operatorname{Hom}_T(\mathcal{O}_T(\beta - n\delta), P)$. The formula above is the chain-level computation of localized sections in the unbounded graded derived category; no compactness of P is used. Equivalently, this proof computes $j_* j^* P$ for $j : T_m \hookrightarrow T$ by derived homogeneous localization of the finite Čech complex of P . No projective-scheme generation statement is being used here; the input is only that $D(m)$ is a principal homogeneous monomial open in the affine Cox space and that Proposition 2.4 computes sections by a finite graded Čech complex. $\square$

Lemma 5.3 (compactness of perfects on finite monomial quotients). *Let $T = [U/D(\Lambda)]$ be as above. Then every perfect object of T is compact in $D_{\text{qc}}(T)$, and*

$$D_{\text{qc}}(T)^c = \operatorname{Perf}(T).$$

Proof. Let $T_i = [D(m_i)/D(\Lambda)]$, and use the finite Čech diagram of Proposition 2.4. On every finite intersection

$$T_I = [D(m_I)/D(\Lambda)]$$

the category $D_{\text{qc}}(T_I)$ is the derived category of graded $B[m_I^{-1}]$ -modules. Lemma 5.1 identifies its compact objects with the perfect objects. No étale averaging or invariant-exactness argument is used here. All constructions are performed in the graded module category attached to the split diagonalizable group $D(\Lambda)$; torsion in Λ , and therefore non-étale fibers such as μ_p , do not affect the compactness argument.

If $P \in \text{Perf}(T)$, then $P|_{T_I}$ is compact for every I . For a filtered diagram $M_\alpha \in D_{\text{qc}}(T)$, mapping complexes are computed by finite Čech totalization:

$$\mathbf{R} \text{Hom}_T(P, M_\alpha) \simeq \text{Tot}_{\emptyset \neq I} \mathbf{R} \text{Hom}_{T_I}(P|_{T_I}, M_\alpha|_{T_I}).$$

Restriction to the finitely many charts commutes with filtered colimits, the chartwise Hom functors commute with filtered colimits by compactness, and finite totalizations commute with filtered colimits of complexes. Hence $\mathbf{R} \text{Hom}_T(P, -)$ commutes with filtered colimits, so P is compact.

Conversely let $K \in D_{\text{qc}}(T)^c$. For each quasi-compact open immersion $j_I : T_I \hookrightarrow T$, the pushforward j_{I*} on D_{qc} preserves filtered colimits of quasi-coherent complexes for qcqs open immersions. In the present monomial quotient model this can be seen directly: on each affine graded chart $D(m) \subset U$, j_{I*} is computed by the corresponding homogeneous localization and by the finite Čech totalization for the cover of the complement; localization and finite totalization commute with filtered colimits degreewise. This is the qcqs quasi-coherent extension-by-zero statement, used here only on D_{qc} . Perfectness is checked after restriction to the finite monomial cover; see [Sta, Tags 09UC, 09M0] for descent/locality of finiteness properties and the compact/perfect comparison in the qcqs scheme case. Hence $j_I^* K$ is compact because

$$\mathbf{R} \text{Hom}_{T_I}(j_I^* K, -) \simeq \mathbf{R} \text{Hom}_T(K, j_{I*} -).$$

Thus $j_I^* K$ is perfect on every affine quotient chart. Perfectness is local for the finite monomial cover by Proposition 2.4, so $K \in \text{Perf}(T)$. $\square$

Theorem 5.4 (character-line generation). *With $T = [U/D(\Lambda)]$ as above,*

$$\text{Perf}(T) = \text{thick}\{\mathcal{O}_T(\alpha) : \alpha \in \Lambda\}.$$

Proof. Let $P \in \text{Perf}(T)$ be right orthogonal to every $\mathcal{O}_T(\alpha)$. On a finite intersection $T_I = [D(m_I)/D(\Lambda)]$, Lemma 5.2 gives

$$\mathbf{R} \text{Hom}_{T_I}(B[m_I^{-1}](\beta), P|_{T_I}) = 0$$

for all β . By Lemma 5.1, the objects $B[m_I^{-1}](\beta)$ generate $\text{Perf}(T_I)$, hence $\text{id}_{P|_{T_I}} = 0$. Thus all restrictions of P to the first-level cover vanish, and Proposition 2.4 gives $P = 0$.

We next prove the corresponding localizing statement in $D_{\text{qc}}(T)$. Let $P \in D_{\text{qc}}(T)$ be right orthogonal to every $\mathcal{O}_T(\alpha)$. The same homogeneous-localization formula gives

$$\mathbf{R} \text{Hom}_{T_I}(B[m_I^{-1}](\beta), P|_{T_I}) = 0$$

for every finite intersection T_I and every β . Now we use the first, localizing, assertion of Lemma 5.1: the objects $B[m_I^{-1}](\beta)$ generate $D_{\text{qc}}(T_I)$ as a localizing subcategory, so $P|_{T_I} = 0$. The first-level restrictions of P therefore vanish, and finite Čech conservativity gives $P = 0$. Thus the right orthogonal to the localizing subcategory generated by the character lines is zero. Since the character lines are compact, the localizing subcategory they generate is coreflective in $D_{\text{qc}}(T)$; equivalently, a proper compactly generated localizing subcategory has a nonzero right orthogonal. Hence the character lines generate all of $D_{\text{qc}}(T)$ as a localizing subcategory.

The character lines are compact and perfect. Lemma 2.16 therefore gives

$$D_{\text{qc}}(T)^c = \text{thick}\{\mathcal{O}_T(\alpha) : \alpha \in \Lambda\}.$$

By Lemma 5.3, this compact subcategory is $\text{Perf}(T)$. The argument is entirely in the graded derived category attached to the split diagonalizable group. In particular, it uses the split grading rather than an étale averaging argument for the stabilizers. $\square$

6. THE RESIDUAL WALL THEOREM

Let $\mathfrak{Z}$ be a split smooth proper toric quotient stack over $\mathbb{Z}$ with finite split diagonalizable stabilizers. We assume throughout this section that $\mathfrak{Z}$ is given by a finite homogeneous monomial Cox presentation in the sense of Section 2; equivalently, its toric character lines and finite homogeneous Cox chart cover are fixed. This is exactly the form supplied by the one-sided and two-sided center-closure propositions in the actual wall applications. If the corrected center is empty, we use the same notation with $\text{Perf}(\mathfrak{Z}) = 0$; then all residual pieces are empty and the statements below reduce to the zero category. Thus the substantive arguments in this section concern the nonempty-center case. Let $L_i^\pm$ be toric character line bundles on $\mathfrak{Z}$, and set

$$E_+ = \bigoplus_{i=1}^p L_i^+, \quad E_- = \bigoplus_{j=1}^q L_j^-.$$

Throughout the residual and actual wall calculations, the phrase “fiber coordinate of L ” is Cox-coordinate notation: it means the coordinate with the indicated character in the graded polynomial algebra. If one uses the geometric convention $\text{Tot}(L) = \text{Spec Sym}(L^\vee)$, this amounts to replacing the displayed line by its dual. All Koszul shifts below are computed with this Cox character convention. Let $\mathbb{G}_m^\lambda$ act on the fiber coordinate x_i of L_i^+ with weight $a_i > 0$, and on the fiber coordinate y_j of L_j^- with weight $-b_j < 0$. Put

$$\eta_+ = \sum_i a_i, \quad \eta_- = \sum_j b_j, \quad r = \eta_+ - \eta_-.$$

Set

$$\mathfrak{V} = [\text{Tot}_{\mathfrak{Z}}(E_+ \oplus E_-)/\mathbb{G}_m^\lambda].$$

The positive residual side $\mathfrak{X}_+$ is the quotient of the locus where the x_i 's are not all zero; if $p = 0$, it is empty. The negative residual side $\mathfrak{X}_-$ is defined analogously using the y_j 's; if $q = 0$, it is empty. The supported residual loci are

$$Y_+ = \{y_1 = \cdots = y_q = 0\} \subset \mathfrak{X}_+, \quad Y_- = \{x_1 = \cdots = x_p = 0\} \subset \mathfrak{X}_-,$$

with the convention that $Y_+ = \mathfrak{X}_+$ if $q = 0$ and $Y_- = \mathfrak{X}_-$ if $p = 0$. Their natural projections to $\mathfrak{Z}$ are weighted projective bundle morphisms whenever the corresponding side is nonempty. Throughout this section, $\mathcal{O}(n)$ denotes the character line of residual $\mathbb{G}_m^\lambda$ -weight n ; with this convention $\deg_\lambda(x_i) = a_i$ and $\deg_\lambda(y_j) = -b_j$. Multiplication by a coordinate of residual weight ℓ is a map

$$\mathcal{O}(d - \ell) \longrightarrow \mathcal{O}(d).$$

The window notation used in the residual and actual wall arguments is:

$$\begin{aligned} \eta_+ &= \sum_i a_i, & \eta_- &= \sum_j b_j, & r &= \eta_+ - \eta_-, \\ &\text{positive window } [w, w + \eta_+), \\ W &= w + \eta_+, & \text{negative window } [W - \eta_-, W) &= [w + r, w + \eta_+). \end{aligned}$$

If $r \geq 0$, the positive-side supported blocks are indexed by $w, w + 1, \dots, w + r - 1$. When $r < 0$, the same table is used after replacing λ by $-\lambda$ and interchanging the two sides. For $r \geq 0$, the wall crossing uses the following bookkeeping:

piece	weights	role
plus window	$[w, w + \eta_+)$	$\text{Perf}(\mathfrak{X}_+)$
blocks	$w, \dots, w + r - 1$	supported center pieces
remainder	$[w + r, w + \eta_+)$	minus window $[W - \eta_-, W)$

When $r = 0$, the block row is empty and the remainder is the common window. The local-cohomology sign convention used for window full faithfulness is:

side	inverted variables	possible cone weights	window exclusion
$\mathfrak{X}_+$	$x_1, \dots, x_p$	$\leq -\eta_+$	$d > -\eta_+$
$\mathfrak{X}_-$	$y_1, \dots, y_q$	$\geq \eta_-$	$d < \eta_-$

Here d denotes the residual weight of the Hom summand being tested. The entries are only mnemonic; Lemma 6.2 gives the precise Čech calculation.

The proof below has three logical steps. Lemma 6.8 identifies each residual side with its ambient window. The proof of that lemma uses the local-cohomology weight bound for full faithfulness and window generation for essential surjectivity. Lemma 6.9 then computes the weighted-projective pushforwards from the supported loci $Y_\pm$. Finally, Lemma 6.14 uses the Koszul resolution of Y_+ or Y_- to remove one residual weight at a time; iterating this peeling proves Theorem 6.15.

Lemma 6.1 (residual local Cox chart dictionary). *After replacing $\mathfrak{Z}$ by any homogeneous monomial Cox chart $[D(t)/D(A_3)]$ on which the toric character lines $L_i^\pm$ are represented by graded free rank-one shifts, every residual calculation below is the following double-graded algebra calculation. The residual ambient stack is represented by*

$$R = R_0[x_1, \dots, x_p, y_1, \dots, y_q],$$

graded by $A_3 \oplus \mathbb{Z}$, where R_0 is the chart algebra of $\mathfrak{Z}$, the second factor is the residual λ -weight, and

$$\deg(x_i) = (\alpha_i, a_i), \quad \deg(y_j) = (\beta_j, -b_j).$$

The positive side is the monomial open $D(x_1, \dots, x_p)$, the negative side is $D(y_1, \dots, y_q)$, and the supported positive locus is the coordinate quotient obtained by setting all $y_j = 0$ and then imposing $D(x_1, \dots, x_p)$. A generator $\rho^*F \otimes \mathcal{O}(n)$ restricts on this chart to a finite complex of graded free R_0 -modules tensored with the residual character shift $(0, n)$. Consequently:

- (i) Hom complexes between such generators decompose by the residual integer weight $m - n$; all A_3 -character twists have residual weight zero.
- (ii) The Hom cone for restricting to the positive side is the $(m - n)$ -weight summand of the Čech cofiber obtained by inverting the x_i 's; the negative side is the analogous y_j -cofiber.
- (iii) The Koszul complex resolving Y_+ is the Koszul complex on the variables y_j . Since y_j has residual weight $-b_j$, its term after tensoring by $\mathcal{O}(a)$ has residual weight $a + b_j$, and a subset I contributes weight $a + b_I$.

All later residual formulas are obtained from this chart calculation by finite Čech totalization over the chosen center chart cover; the transition maps preserve both gradings, so Lemma 2.11 allows the residual weight summands and the finite totalization to be interchanged.

Lemma 6.2 (monomial local-cohomology weight bound). *Let R_0 be a graded algebra on which the residual λ -weight is zero, and put*

$$R = R_0[x_1, \dots, x_p, y_1, \dots, y_q], \quad \lambda(x_i) = a_i > 0, \quad \lambda(y_j) = -b_j < 0.$$

For $p > 0$, the cofiber of

$$R\Gamma(\mathrm{Spec} R, \mathcal{O}) \longrightarrow R\Gamma(D(x_1, \dots, x_p), \mathcal{O})$$

has no homogeneous cohomology of residual weight $> -\eta_+$. For $q > 0$, the analogous cofiber for $D(y_1, \dots, y_q)$ has no homogeneous cohomology of residual weight $< \eta_-$. The same bounds remain true after a finite monomial localization in the residual weight-zero variables and after finite Čech totalization over such localizations.

Proof. Use the nonaugmented Čech complex for the cover of $D(x_1, \dots, x_p)$ by the $D(x_i)$'s and decompose it by exponent vectors in the x_i 's. The cofiber is represented by exponent vectors with every x_i -exponent negative:

$$x_1^{-r_1} \cdots x_p^{-r_p} y_1^{s_1} \cdots y_q^{s_q}, \quad r_i \geq 1, \ s_j \geq 0,$$

up to residual weight-zero factors from R_0 . These monomials have residual weight

$$-\sum_i r_i a_i - \sum_j s_j b_j \leq -\eta_+.$$

Mixed exponent vectors in the Čech complex form simplex-star subcomplexes and are contractible by the add/delete homotopy which inserts a coordinate whose exponent is nonnegative. For the y -statement one applies this calculation to the cover by the $D(y_j)$'s and to the homomorphism $-\lambda$: the local-cohomology monomials have all y -exponents negative, their $-\lambda$ -weights are at most $-\eta_-$, and equivalently their λ -weights are at least η_- . Weight-zero monomial localization preserves homogeneous weights, and finite Čech totalization preserves the bounds by Lemma 2.11. $\square$

Lemma 6.3 (Koszul window generation). *Let $T = [U/D(\Lambda)]$ be a finite monomial quotient stack equipped with a homomorphism $\lambda : \Lambda \rightarrow \mathbb{Z}$. Assume that character lines generate $\text{Perf}(T)$. Let*

$$s_i : \mathcal{O}_T(-\delta_i) \longrightarrow \mathcal{O}_T, \quad i = 1, \dots, p,$$

be homogeneous sections whose common ideal is the unit ideal on T , and put $a_i = \lambda(\delta_i) > 0$ and $\eta = \sum_i a_i$. Then for every $w \in \mathbb{Z}$,

$$\text{Perf}(T) = \text{thick}\{\mathcal{O}_T(\chi) : w \leq \lambda(\chi) < w + \eta\}.$$

The sign-reversed statement holds for sections of negative λ -weights after replacing λ by $-\lambda$.

Proof. Since the s_i 's generate the unit ideal, their Koszul complex is exact. It is therefore enough to show that every character line can be moved into the interval. Let $L = \mathcal{O}_T(\chi)$ and $d = \lambda(\chi)$. If $d \geq w + \eta$, tensor the Koszul complex by L . The non-augmentation terms are character lines $L(-\delta_I)$, where $\delta_I = \sum_{i \in I} \delta_i$ and $I \neq \emptyset$, and their λ -weights lie in $[d - \eta, d - 1]$. If $d < w$, tensor the Koszul complex by $L(\sum_i \delta_i)$. Its top term is L , and all other terms have weights in $[d + 1, d + \eta]$. Induction on the distance from $[w, w + \eta)$ moves L into the displayed window. Since character lines generate $\text{Perf}(T)$, the displayed window generates $\text{Perf}(T)$. For sections of negative λ -weight, apply the preceding paragraph to $-\lambda$. The resulting interval for $-\lambda$ is exactly an interval of the same length in the opposite λ -orientation, so the same Koszul induction moves every character line into the corresponding negative-weight window. $\square$

For a residual piece T , write $\rho_T : T \rightarrow \mathfrak{Z}$ for the projection.

Proposition 6.4 (residual generators). *For every nonempty residual piece T among $\mathfrak{Y}, \mathfrak{X}_+, \mathfrak{X}_-$ and the supported loci Y_+, Y_- ,*

$$\text{Perf}(T) = \text{thick}\{\rho_T^* F \otimes \mathcal{O}_T(n) : F \in \text{Perf}(\mathfrak{Z}), \ n \in \mathbb{Z}\}.$$

Proof. Choose a Cox presentation $\mathfrak{Z} = [U_{\mathfrak{Z}}/D(A_{\mathfrak{Z}})]$. Since the $L_i^{\pm}$ are toric character line bundles, adding the fiber coordinates x_i, y_j gives a single polynomial Cox algebra over the Cox algebra of $\mathfrak{Z}$, graded by $A_{\mathfrak{Z}} \oplus \mathbb{Z}$, where the second factor records the residual $\mathbb{G}_m^{\Lambda}$ -weight. Each nonempty residual piece T is defined in this affine Cox space by homogeneous monomial nonvanishing conditions: none for $\mathfrak{Y}$, the condition that some $x_i \neq 0$ for $\mathfrak{X}_+$, and the condition that some $y_j \neq 0$ for $\mathfrak{X}_-$. For the supported loci Y_+ and Y_- , first pass to the coordinate quotient polynomial Cox algebra obtained by setting respectively all $y_j = 0$ or all $x_i = 0$, and then impose the same monomial open condition that the remaining fiber coordinates are not all zero. Hence Theorem 5.4 applies directly to T , giving generation by all character lines of bidegree $(\chi, n) \in A_{\mathfrak{Z}} \oplus \mathbb{Z}$.

The character $\chi \in A_3$ defines the global toric character line $\mathcal{L}_\chi$ on $\mathfrak{Z}$, and the bidegree (χ, n) character line on T is

$$\rho_T^* \mathcal{L}_\chi \otimes \mathcal{O}_T(n),$$

which is one of the displayed objects with $F = \mathcal{L}_\chi$. Since the displayed family also allows arbitrary $F \in \text{Perf}(\mathfrak{Z})$, it contains the character-line generators and therefore generates $\text{Perf}(T)$. This is the only generation input used below. $\square$

For an interval $I \subset \mathbb{Z}$, define

$$\mathcal{C}_I(T) = \text{thick}\{\rho_T^* F \otimes \mathcal{O}_T(n) : F \in \text{Perf}(\mathfrak{Z}), n \in I\}.$$

Example 6.5 (rank-one weighted projective wall). If $\mathfrak{Z} = S$, $q = 0$, and the positive variables have residual weights $a_1, \dots, a_p > 0$, then

$$\mathfrak{X}_+ = [(\mathbb{A}_S^p \setminus 0)/\mathbb{G}_m]$$

is the weighted projective stack with weights a_i . The number $\eta_+ = a_1 + \dots + a_p$ is exactly the length of the window $[w, w + \eta_+)$. The residual wall theorem below specializes to

$$\text{Perf}(\mathfrak{X}_+) = \langle \mathcal{O}_{\mathfrak{X}_+}(w), \mathcal{O}_{\mathfrak{X}_+}(w+1), \dots, \mathcal{O}_{\mathfrak{X}_+}(w + \eta_+ - 1) \rangle,$$

with the absent negative side interpreted as the zero category. This example is the rank-one base case used later in the directed induction.

Lemma 6.6 (window generation). *If $p > 0$, then for every $w \in \mathbb{Z}$,*

$$\text{Perf}(\mathfrak{X}_+) = \mathcal{C}_{[w, w + \eta_+)}(\mathfrak{X}_+).$$

If $q > 0$, then for every $W \in \mathbb{Z}$,

$$\text{Perf}(\mathfrak{X}_-) = \mathcal{C}_{[W - \eta_-, W)}(\mathfrak{X}_-).$$

If $p = 0$ or $q = 0$, the corresponding side is empty and its perfect category is zero.

Proof. Assume $p > 0$. On $\mathfrak{X}_+$ the functions $x_1, \dots, x_p$ generate the unit ideal, and their residual weights are $a_i > 0$. By Theorem 5.4, character lines generate $\text{Perf}(\mathfrak{X}_+)$. Lemma 6.3 therefore moves all character-line generators into the interval $[w, w + \eta_+)$. A character line of bidegree (χ, n) in that interval is $\rho^* \mathcal{L}_\chi \otimes \mathcal{O}(n)$, hence belongs to $\mathcal{C}_{[w, w + \eta_+)}(\mathfrak{X}_+)$. This proves the positive statement.

For the negative side, use $y_1, \dots, y_q$ and the homomorphism $-\lambda$. These sections then have positive $(-\lambda)$ -weights b_j , and Lemma 6.3 moves the character lines into every interval of length $\eta_- = \sum_j b_j$ for the $(-\lambda)$ -orientation, equivalently into every interval $[W - \eta_-, W)$ for the original λ -orientation. If $p = 0$ or $q = 0$, the condition that the corresponding coordinates are not all zero is impossible. $\square$

Lemma 6.7 (residual Hom cones with base perfect factors). *Let $F, G \in \text{Perf}(\mathfrak{Z})$ and $n, m \in \mathbb{Z}$. If $p > 0$, then the cone of the restriction map*

$$\mathbf{R} \text{Hom}_{\mathfrak{Z}}(\rho^* F \otimes \mathcal{O}(n), \rho^* G \otimes \mathcal{O}(m)) \longrightarrow \mathbf{R} \text{Hom}_{\mathfrak{X}_+}(j_+^*(\rho^* F \otimes \mathcal{O}(n)), j_+^*(\rho^* G \otimes \mathcal{O}(m)))$$

is computed by the finite Čech totalization of the following chartwise complexes. On every homogeneous center Cox chart, and compatibly on finite intersections, the chartwise cone is the $(m - n)$ -residual-weight summand of

$$\mathbf{R} \text{Hom}_{\mathfrak{Z}}(F, G) \otimes R\Gamma_{(x_1, \dots, x_p)}(\mathcal{O}_{\text{Tot}(E_+ \oplus E_-)}).$$

Here the displayed tensor expression is shorthand for the chartwise description in which the summands $E_\pm$ are represented by their corresponding center character-line twists; the twists have residual λ -weight zero and are part of the finite Čech transition data. All nonzero cohomology weights of the local-cohomology factor are $\leq -\eta_+$. Therefore the displayed cone vanishes when

$$m - n > -\eta_+.$$

If $q > 0$, the analogous cone for the restriction to $\mathfrak{X}_-$ has nonzero cohomology only in λ -weights $\geq \eta_-$, and hence vanishes when $m - n < \eta_-$.

Proof. Choose the finite homogeneous Cox chart cover of $\mathfrak{Z}$ used in Lemma 6.1; on it the line bundles $L_i^\pm$ are represented by graded free rank-one character shifts. On a chart, $\mathfrak{V}$ is represented by the double-graded polynomial algebra over the center chart, and $\mathfrak{X}_+$ is obtained by inverting the ideal $(x_1, \dots, x_p)$. The cone of restriction is therefore the local cohomology cone for this ideal. Since F and G are perfect, after refining the fixed finite cover they are represented on every chart and finite intersection by bounded complexes of finite graded free center modules. The chartwise Hom complex between the corresponding residual generators is then

$$\mathbf{R} \operatorname{Hom}_{\mathfrak{Z}}(F, G)_{\text{chart}} \otimes \mathbf{R} \Gamma_{\text{chart}}(R[x_i, y_j] \rightarrow R[x_i, y_j][x_1, \dots, x_p]^{-1}),$$

with the residual character twist contributing the numerical weight $m - n$. On overlaps the transition functions for the center line bundles and for the finite free resolutions have residual λ -weight zero, so the residual grading is preserved by every Čech differential. Since the center cover is finite, Lemma 2.11 allows us to take the $(m - n)$ -residual-weight summand before or after totalization. Thus the global cone is the finite Čech totalization of the stated chartwise cones, equivalently the stated $(m - n)$ -weight summand of the base Hom complex tensored with the local-cohomology complex. This argument uses only local finite free resolutions of the perfect objects; it does not assume that F or G are globally direct sums of character lines.

The local-cohomology cohomology classes for inverting the x_i 's are represented by monomials

$$x_1^{-r_1} \cdots x_p^{-r_p} y_1^{s_1} \cdots y_q^{s_q}, \quad r_i \geq 1, \quad s_j \geq 0,$$

tensored with residual-weight-zero center factors. Their residual weights are

$$-\sum_i r_i a_i - \sum_j s_j b_j \leq -\eta_+,$$

which proves the plus bound. The finite totalization over the center chart cover preserves the residual grading and the displayed bound by Lemma 2.11. The minus statement is the same calculation after replacing λ by $-\lambda$, so the nonzero cone weights are $\geq \eta_-$. $\square$

Lemma 6.8 (residual window restriction). *Let $j_+ : \mathfrak{X}_+ \hookrightarrow \mathfrak{V}$ and $j_- : \mathfrak{X}_- \hookrightarrow \mathfrak{V}$ be the residual open immersions. If $p > 0$, then for every $w \in \mathbb{Z}$*

$$j_+^* : \mathcal{C}_{[w, w+\eta_+]}(\mathfrak{V}) \longrightarrow \operatorname{Perf}(\mathfrak{X}_+)$$

is an equivalence. If $q > 0$, then for every $W \in \mathbb{Z}$

$$j_-^* : \mathcal{C}_{[W-\eta_-, W]}(\mathfrak{V}) \longrightarrow \operatorname{Perf}(\mathfrak{X}_-)$$

is an equivalence.

Proof. We prove the positive statement; the negative one is obtained by replacing λ by $-\lambda$ and using the y_j 's. The assertion is vacuous when $p = 0$, so assume $p > 0$. There are two independent points. Full faithfulness is a Hom-cone vanishing statement, proved from the local-cohomology weight bound. Essential surjectivity is a generation statement on the open side, proved from character-line generation and the Koszul window-generation argument for the unit ideal $(x_1, \dots, x_p)$; it is not deduced from right-orthogonal descent.

It suffices first to prove full faithfulness on generators $\rho^* F \otimes \mathcal{O}(n)$. By Lemma 6.7, the Hom cone for restriction to $\mathfrak{X}_+$ vanishes whenever $m - n > -\eta_+$. If $n, m \in [w, w + \eta_+)$, then $m - n > -\eta_+$, so j_+^* is fully faithful on the plus window by Lemma 2.14.

For essential surjectivity, Lemma 6.6 says that $\operatorname{Perf}(\mathfrak{X}_+)$ is generated by

$$j_+^*(\rho^* F \otimes \mathcal{O}(n)), \quad F \in \operatorname{Perf}(\mathfrak{Z}), \quad w \leq n < w + \eta_+.$$

These are precisely the restrictions of the positive window generators on $\mathfrak{V}$. Thus j_+^* is essentially surjective. This proves the positive window equivalence. For the negative equivalence one repeats the same two checks with $-\lambda$, using the negative half of Lemma 6.7 and Lemma 6.6. $\square$

Lemma 6.9 (weighted-projective pushforward). *Let*

$$q_+ : Y_+ = [(\mathrm{Tot}_3(E_+) \setminus 0)/\mathbb{G}_m^\lambda] \rightarrow \mathfrak{Z}$$

with $p > 0$. Then

$$\mathbf{R}q_{+*}\mathcal{O}_{Y_+} \simeq \mathcal{O}_{\mathfrak{Z}}, \quad \mathbf{R}q_{+*}\mathcal{O}_{Y_+}(d) = 0 \quad (-\eta_+ < d < 0).$$

Similarly, for $q > 0$,

$$\mathbf{R}q_{-*}\mathcal{O}_{Y_-} \simeq \mathcal{O}_{\mathfrak{Z}}, \quad \mathbf{R}q_{-*}\mathcal{O}_{Y_-}(d) = 0 \quad (0 < d < \eta_-).$$

Consequently, for every $F \in \mathrm{Perf}(\mathfrak{Z})$,

$$\mathbf{R}q_{+*}(q_+^*F \otimes \mathcal{O}_{Y_+}(d)) \simeq F \otimes \mathbf{R}q_{+*}\mathcal{O}_{Y_+}(d),$$

and similarly on the negative side. Thus the same vanishing intervals hold after tensoring by arbitrary perfect objects from the base $\mathfrak{Z}$.

Proof. We separate the proof into three points: first the local Čech computation of $Rq_{+*}\mathcal{O}(d)$, then its gluing over the stacky base, and finally the projection formula for arbitrary $F \in \mathrm{Perf}(\mathfrak{Z})$. This keeps the argument inside the split graded module model and avoids any étaleness assumption on the stabilizers.

Step 1: local weighted Čech calculation. The assertion is local on a finite homogeneous toric Cox chart cover of $\mathfrak{Z}$ on which the $L_i^\pm$ are represented by graded free rank-one character shifts. On such a chart $W = [\mathrm{Spec} R/D(A_0)]$, the preimage $q_+^{-1}W$ is the quotient of the punctured affine space

$$\mathrm{Spec} R[x_1, \dots, x_p] \setminus V(x_1, \dots, x_p)$$

by $D(A_0) \times \mathbb{G}_m^\lambda$, with $D(A_0)$ acting through the characters of the L_i^+ 's and $\mathbb{G}_m^\lambda$ acting with weights a_i . The quasi-coherent pushforward to W is therefore computed in the graded module model of Lemmas 2.1 and 2.2: take the finite Čech complex of the punctured affine fiber and then take residual $\mathbb{G}_m^\lambda$ -weight zero after the twist. Thus $\mathbf{R}q_{+*}\mathcal{O}_{Y_+}(d)$ is computed by the unshifted residual λ -weight d summand of that Čech complex. This is a computation of the stack pushforward, not an atlas-level invariant calculation, because the quotient-stack derived category is the graded derived category on each chart and the finite chart intersections are handled by the same descent formalism.

No coprimality condition on the weights is imposed. If the positive weights have common divisor greater than 1, the quotient has a finite diagonalizable generic stabilizer; in characteristics dividing that divisor this stabilizer is not étale, but the calculation remains the same degreewise computation in the split graded module category. The base line-bundle twists only contribute characters of residual weight zero, so they do not affect the numerical intervals in the statement. On each chart and finite intersection, quasi-coherent complexes are graded modules and the claim is the Čech calculation for the cover of $\mathbb{A}^p \setminus 0$ by the $D(x_i)$:

$$\bigoplus_i R[x_1, \dots, x_p, x_i^{-1}] \rightarrow \bigoplus_{i < j} R[x_1, \dots, x_p, x_i^{-1}, x_j^{-1}] \rightarrow \dots \rightarrow R[x_1^{\pm 1}, \dots, x_p^{\pm 1}].$$

Decompose this Čech complex by exponent vectors $\nu = (\nu_1, \dots, \nu_p) \in \mathbb{Z}^p$. If all $\nu_i \geq 0$, the summand is the ordinary Čech cochain of the full simplex with constant value x^ν , hence contributes the polynomial monomial x^ν in H^0 , of nonnegative λ -weight. If some but not all ν_i are negative, then the Čech terms are indexed by subsets containing the nonempty proper set $\{i : \nu_i < 0\}$; this is the cochain complex of a simplex star and is contractible. Explicitly, after choosing k with $\nu_k \geq 0$, the standard Čech homotopy which adds or removes k contracts this star subcomplex, with the usual alternating sign. If all $\nu_i < 0$, the summand occurs only in the top Čech degree and gives the usual top local-cohomology monomial

$$x_1^{-r_1} \dots x_p^{-r_p}, \quad r_i \geq 1,$$

whose weight is at most $-\eta_+$. Hence no cohomology occurs in weights $-\eta_+ < d < 0$, and the weight-zero part is the constant monomial. This also covers $p = 1$: then the punctured fiber is

$\mathbb{G}_m$, the only negative Laurent monomials have weights $\leq -a_1 = -\eta_+$, and the open interval $(-\eta_+, 0)$ is empty of possible weights. For Y_- , the Čech cover is the cover by the $D(y_j)$'s; the nonzero higher local-cohomology monomials have all y_j -exponents negative and therefore have λ -weight at least η_- , while the weight-zero part is again the constant monomial. This gives the stated vanishing for $0 < d < \eta_-$. Since the cover is finite and all transition maps preserve residual weight, this completes the local calculation of $Rq_{\pm*}\mathcal{O}(d)$.

Step 2: gluing over the base. On overlaps of the chosen Cox charts, both sides are obtained by the same homogeneous localization of the A_0 -graded base ring and by the same finite Čech complex in the weighted fiber variables. Therefore the chartwise complexes for $Rq_{\pm*}\mathcal{O}(d)$ agree on all finite intersections. Finite monomial descent, Proposition 2.8, glues the chartwise weight-vanishing and the weight-zero identification to the stacky base $\mathfrak{Z}$. The use of graded modules is legitimate over $\mathbb{Z}$ even for non-étale stabilizers because the acting groups are split diagonalizable.

Step 3: tensoring by perfect objects from the base. Let $F \in \text{Perf}(\mathfrak{Z})$. On each chart and finite intersection, represent F by a bounded finite complex of graded finite projective modules. The pullback $q_{\pm}^*F$ is obtained by tensoring the weighted fiber Čech complex with this finite complex over the base ring. Since the latter has residual λ -weight zero and the Čech complex is finite, tensoring commutes with taking the residual-weight summands above. Thus the chartwise projection formula holds degree by degree, and the same finite descent glues it to

$$\mathbf{R}q_{\pm*}(q_{\pm}^*F \otimes \mathcal{O}(d)) \simeq F \otimes \mathbf{R}q_{\pm*}\mathcal{O}(d).$$

□

Remark 6.10 (two residual sanity checks). When all positive weights are $a_i = 1$, the stack $[(\mathbb{A}^p \setminus 0)/\mathbb{G}_m]$ is the ordinary projective bundle $\mathbb{P}^{p-1}$, and Lemma 6.9 becomes the standard vanishing $Rq_*\mathcal{O}(d) = 0$ for $-p < d < 0$. When $p = 1$ and the single positive weight is a , the positive residual side is $[\mathbb{G}_m/\mathbb{G}_m]$ with stabilizer μ_a ; the interval $[w, w + a)$ contains exactly a residual weights. Thus the rank-one weighted-projective wall produces a grading blocks, including in characteristic dividing a , because the calculation is in the split graded module category rather than in an étale representation category.

Lemma 6.11 (residual Koszul sign convention). *Let s be a residual fiber coordinate of λ -weight ℓ . With the character-line convention fixed in Section 2, multiplication by s is a morphism*

$$\mathcal{O}(-\ell) \longrightarrow \mathcal{O}.$$

Hence, after tensoring by a residual character line $\mathcal{O}(a)$, the Koszul two-term factor is

$$\mathcal{O}(a - \ell) \xrightarrow{s} \mathcal{O}(a).$$

In particular, if x_i has residual weight $a_i > 0$, the corresponding term has weight $a - a_i$; if y_j has residual weight $-b_j < 0$, the corresponding term has weight $a + b_j$. For a subset $I \subseteq \{1, \dots, q\}$, the y -Koszul term resolving the locus $(y_1, \dots, y_q) = 0$, tensored by $\mathcal{O}(a)$, has residual weight

$$a + b_I, \quad b_I = \sum_{j \in I} b_j.$$

Proof. If s has residual character ℓ , it is a homogeneous section of $\mathcal{O}(\ell)$, equivalently a morphism $\mathcal{O} \rightarrow \mathcal{O}(\ell)$. After twisting by $\mathcal{O}(-\ell)$ this is the displayed multiplication map $\mathcal{O}(-\ell) \rightarrow \mathcal{O}$. Tensoring by $\mathcal{O}(a)$ gives $\mathcal{O}(a - \ell) \rightarrow \mathcal{O}(a)$. The final statements are the specializations $\ell = a_i$ and $\ell = -b_j$, and exterior products of the two-term Koszul factors add the displayed weights. □

Example 6.12 (the $p = q = 1$ residual sign check). Let the residual base be S , let x have residual weight $A > 0$, and let y have residual weight $-B < 0$. Then $\eta_+ = A$, $\eta_- = B$, and

$r = A - B$. The plus Hom cone is obtained by inverting x . Its local-cohomology monomials have the form

$$x^{-r_1} y^s, \quad r_1 \geq 1, s \geq 0,$$

and residual weight

$$-r_1 A - sB \leq -A = -\eta_+.$$

Thus two plus-window characters $n, m \in [w, w + A)$ have $m - n > -A$, so the Hom cone vanishes on that window.

Assume $A \geq B$. The supported plus locus is $Y_+ = \{y = 0, x \neq 0\}$. For a residual character line of weight a , the equation $y = 0$ is resolved by the two-term Koszul complex

$$\mathcal{O}(a + B) \xrightarrow{y} \mathcal{O}(a),$$

because y has weight $-B$. Hence a weight a supported block has ambient terms of weights a and $a + B$. If $w \leq a < w + r = w + A - B$, then

$$a < a + B \leq w + A - 1,$$

so the non-supported Koszul term lies in the next window $\mathcal{T}_{a+1}$. This is the elementary case of the peeling identity

$$\mathcal{T}_a = \langle \beta_a^+ \text{Perf}(S), \mathcal{T}_{a+1} \rangle.$$

The same computation with x and y interchanged gives the sign-reversed statement when $A < B$.

Lemma 6.13 (Koszul Hom calculation for supported residual blocks). *Let $s_+ : Y_+ \hookrightarrow \mathfrak{X}_+$ be the closed locus $y_1 = \dots = y_q = 0$, with $Y_+ = \mathfrak{X}_+$ if $q = 0$, and let $q_+ : Y_+ \rightarrow \mathfrak{Z}$ be the projection. For $F, G \in \text{Perf}(\mathfrak{Z})$ and $a, c \in \mathbb{Z}$, there are invertible residual-weight-zero tensor factors M_I on $\mathfrak{Z}$, with $M_\emptyset \simeq \mathcal{O}_\mathfrak{Z}$, such that*

$$\begin{aligned} & \mathbf{R} \text{Hom}_{\mathfrak{X}_+}(s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a)), s_{+*}(q_+^* G \otimes \mathcal{O}_{Y_+}(c))) \\ & \simeq \bigoplus_{I \subseteq \{1, \dots, q\}} \mathbf{R} \text{Hom}_{\mathfrak{Z}}(F, G \otimes M_I \otimes \mathbf{R}q_{+*} \mathcal{O}_{Y_+}(c - a - b_I)) [-|I|], \end{aligned}$$

where $b_I = \sum_{j \in I} b_j$. The differential between the displayed summands is zero. Moreover, for $H \in \text{Perf}(\mathfrak{Z})$ and $d, a \in \mathbb{Z}$,

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(\rho^* H \otimes \mathcal{O}_{\mathfrak{X}_+}(d), s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a))) \simeq \mathbf{R} \text{Hom}_{\mathfrak{Z}}(H, F \otimes \mathbf{R}q_{+*} \mathcal{O}_{Y_+}(a - d)).$$

Proof. The calculation is local on the fixed finite homogeneous Cox chart cover of $\mathfrak{Z}$. On such a chart the line bundles L_j^- are represented by graded free character shifts, and we use the double-graded polynomial model of Lemma 6.1. Thus $\mathfrak{X}_+$ is obtained by inverting the positive ideal and quotienting by a split diagonalizable group. The sequence $y_1, \dots, y_q$ is regular before quotienting, and the quotient calculation is degreewise in graded modules; hence it remains a regular sequence in the quotient-stack sense.

Resolve $s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a))$ by the Koszul complex on $y_1, \dots, y_q$. Its term indexed by $I \subseteq \{1, \dots, q\}$ is

$$\rho^*(F \otimes N_I) \otimes \mathcal{O}_{\mathfrak{X}_+}(a + b_I)[-|I|],$$

where N_I is an invertible residual-weight-zero tensor factor; the residual weight $a + b_I$ is exactly the sign convention of Lemma 6.11. Applying $\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(-, s_{+*}(q_+^* G \otimes \mathcal{O}_{Y_+}(c)))$, then adjunction for s_+ , and then the projection formula for q_+ , gives the displayed summands; the duals of the N_I 's are absorbed into the notation M_I . The Koszul differential in the Hom complex is precomposition by multiplication by the y_j 's. These functions act by zero on the target supported on Y_+ , so the induced differential is zero. This proves the first formula. When $q = 0$ the same formula has only the summand $I = \emptyset$.

For the second formula, use adjunction for the closed immersion:

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(\rho^* H \otimes \mathcal{O}(d), s_{+*} B) \simeq \mathbf{R} \text{Hom}_{Y_+}(s_+^*(\rho^* H \otimes \mathcal{O}(d)), B),$$

where $B = q_+^* F \otimes \mathcal{O}_{Y_+}(a)$. The source is a character-line tensor pullback from $\mathfrak{Z}$, so ordinary pullback agrees with derived pullback, and the projection formula for q_+ gives the stated expression. $\square$

Lemma 6.14 (residual supported peeling). *Assume $r = \eta_+ - \eta_- > 0$. For*

$$\mathcal{T}_a = \mathcal{C}_{[a, w+\eta_+]}(\mathfrak{X}_+), \quad w \leq a < w+r,$$

define

$$\beta_a^+(F) = s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a)), \quad F \in \text{Perf}(\mathfrak{Z}),$$

where $s_+ : Y_+ \hookrightarrow \mathfrak{X}_+$ is the locus $y_1 = \cdots = y_q = 0$, with $Y_+ = \mathfrak{X}_+$ if $q = 0$. Then

$$\mathcal{T}_a = \langle \beta_a^+ \text{Perf}(\mathfrak{Z}), \mathcal{T}_{a+1} \rangle.$$

More explicitly, $\beta_a^+ \text{Perf}(\mathfrak{Z}) \subset \mathcal{T}_a$, the functor β_a^+ is fully faithful, and $\mathcal{T}_{a+1} \perp \beta_a^+ \text{Perf}(\mathfrak{Z})$.

Proof. If $q = 0$, then $Y_+ = \mathfrak{X}_+$, s_+ is the identity, and $\beta_a^+(F) = q_+^* F \otimes \mathcal{O}(a)$. The inclusion $\beta_a^+ \text{Perf}(\mathfrak{Z}) \subset \mathcal{T}_a$ follows directly from the definition of $\mathcal{T}_a$: it is generated by objects $\rho^* F' \otimes \mathcal{O}(n)$ with $a \leq n < w + \eta_+$, and here $n = a$. The Hom and semiorthogonality computations below are the same weighted-projective pushforward calculation with no Koszul terms, so we write the proof uniformly, allowing the Koszul complex on the empty sequence.

For $q > 0$, s_+ is the regular closed immersion cut out by $y_1, \dots, y_q$. Hence s_{+*} sends perfect complexes on Y_+ to perfect complexes on $\mathfrak{X}_+$, because locally it is represented by the finite Koszul resolution of the regular sequence. The morphism q_+ is a weighted projective bundle stack, so q_+^* preserves perfect complexes. Thus β_a^+ is a well-defined exact functor $\text{Perf}(\mathfrak{Z}) \rightarrow \text{Perf}(\mathfrak{X}_+)$. The ambient terms of its Koszul resolution are

$$\rho^*(F \otimes N_I) \otimes \mathcal{O}_{\mathfrak{X}_+}(a + b_I), \quad b_I = \sum_{j \in I} b_j,$$

where $I \subseteq \{1, \dots, q\}$ and N_I is an invertible residual-weight-zero tensor factor on $\mathfrak{Z}$. Since $a \leq w + r - 1$ and $\eta_- = \eta_+ - r$, all these terms have weights at most

$$w + r - 1 + \eta_- = w + \eta_+ - 1.$$

Thus $\beta_a^+ \text{Perf}(\mathfrak{Z}) \subset \mathcal{T}_a$.

For full faithfulness, apply Lemma 6.13 with $c = a$. The summand indexed by $I = \emptyset$ is $\mathbf{R} \text{Hom}_{\mathfrak{Z}}(F, G)$, because Lemma 6.9 gives $\mathbf{R}q_{+*} \mathcal{O}_{Y_+} \simeq \mathcal{O}_{\mathfrak{Z}}$. If $I \neq \emptyset$, then

$$-\eta_+ < -b_I < 0,$$

because $0 < b_I \leq \eta_- < \eta_+$, so the same lemma gives $\mathbf{R}q_{+*} \mathcal{O}_{Y_+}(-b_I) = 0$. Therefore

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(\beta_a^+ F, \beta_a^+ G) \simeq \mathbf{R} \text{Hom}_{\mathfrak{Z}}(F, G),$$

and β_a^+ is fully faithful.

Now let $\rho^* G \otimes \mathcal{O}(d)$ be a generator of $\mathcal{T}_{a+1}$, so $a+1 \leq d < w + \eta_+$. The second formula of Lemma 6.13 reduces

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(\rho^* G \otimes \mathcal{O}(d), \beta_a^+ F)$$

to a complex involving $\mathbf{R}q_{+*} \mathcal{O}_{Y_+}(a-d)$. Since

$$-\eta_+ < a - d < 0,$$

Lemma 6.9 gives this Hom complex as zero. Hence $\mathcal{T}_{a+1} \perp \beta_a^+ \text{Perf}(\mathfrak{Z})$.

It remains to prove generation of $\mathcal{T}_a$. The category $\mathcal{T}_a$ is generated by $\mathcal{T}_{a+1}$ and the residual weight- a generators $\rho^* F \otimes \mathcal{O}(a)$. If $q = 0$, each such generator is already $\beta_a^+(F)$. If $q > 0$, the Koszul resolution of $s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a))$ has first ambient term $\rho^* F \otimes \mathcal{O}(a)$, while all other ambient terms have weights $a + b_I > a$ and at most $w + \eta_+ - 1$. Hence those other terms lie in $\mathcal{T}_{a+1}$, and the weight- a generator lies in $\langle \beta_a^+ \text{Perf}(\mathfrak{Z}), \mathcal{T}_{a+1} \rangle$. This proves the displayed semiorthogonal decomposition. $\square$

The residual theorem now follows by a fixed ledger of ingredients. Window generation is Lemma 6.6. Window restriction full-faithfulness is Lemma 6.8; its Hom cone is the local cohomology cone in the x -variables, respectively in the y -variables, and the base $\mathfrak{Z}$ -perfect complexes only contribute residual-weight-zero factors. The blocks are the functors β_a^+ , and their full faithfulness and semiorthogonality are exactly the two formulas in Lemma 6.13 plus the weighted-projective pushforward vanishing of Lemma 6.9: the summand $I = \emptyset$ gives the identity Hom on $\text{Perf}(\mathfrak{Z})$, while every nonempty I lies in the vanishing interval. Generation of each peeled step is the last paragraph of Lemma 6.14, where the first Koszul term is the weight- a character line and the remaining Koszul terms have strictly larger residual weight but still lie in the ambient window. The proof below only iterates this ledger.

Theorem 6.15 (residual wall theorem). *If $r \geq 0$, there is a fully faithful S -linear functor*

$$\Phi_{-\rightarrow+} : \text{Perf}(\mathfrak{X}_-) \rightarrow \text{Perf}(\mathfrak{X}_+)$$

with source zero if $q = 0$, and a semiorthogonal decomposition

$$\text{Perf}(\mathfrak{X}_+) = \langle \mathcal{B}_w^+, \dots, \mathcal{B}_{w+r-1}^+, \Phi_{-\rightarrow+} \text{Perf}(\mathfrak{X}_-) \rangle,$$

where the block list is empty for $r = 0$, and each nonempty block is equivalent to $\text{Perf}(\mathfrak{Z})$. If $r < 0$, the sign-reversed statement holds on $\mathfrak{X}_-$.

Proof. If $r = 0$, take $W = w + \eta_+$. Then $\eta_+ = \eta_-$, and the plus window $[w, w + \eta_+)$ equals the minus window

$$[W - \eta_-, W) = [w, w + \eta_+).$$

If $\eta_+ = \eta_- = 0$, then $p = q = 0$, both residual sides are empty by definition, and the assertion is the identity statement for the zero category. Otherwise $p, q > 0$, so Lemma 6.8 identifies this common window with both residual sides, giving the common-window equivalence.

If $r < 0$, set $\lambda' = -\lambda$. Then the old y -coordinates are positive for λ' , the old x -coordinates are negative for λ' , and the new integer is $\eta_- - \eta_+ > 0$. The $r > 0$ case for λ' , proved in the next paragraph, gives the asserted sign-reversed decomposition on $\mathfrak{X}_-$. It remains to prove the case $r > 0$. The positive window $\mathcal{C}_{[w, w + \eta_+)}(\mathfrak{X}_+)$ is all of $\text{Perf}(\mathfrak{X}_+)$ by Lemma 6.6. Let

$$\mathcal{T}_a = \mathcal{C}_{[a, w + \eta_+)}(\mathfrak{X}_+).$$

For $w \leq a < w + r$, Lemma 6.14 gives

$$\mathcal{T}_a = \langle \beta_a^+ \text{Perf}(\mathfrak{Z}), \mathcal{T}_{a+1} \rangle,$$

where the functor $\beta_a^+(F) = s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a))$ is fully faithful. Iterating for $a = w, \dots, w + r - 1$ leaves the common interval $[w + r, w + \eta_+) = [w + \eta_+ - \eta_-, w + \eta_+)$, of length η_- . If $q = 0$ this interval is empty. If $q > 0$, the negative window equivalence of Lemma 6.8, with $W = w + \eta_+$, identifies it with $\text{Perf}(\mathfrak{X}_-)$, producing $\Phi_{-\rightarrow+}$. $\square$

7. THE ACTUAL WALL THEOREM

We now pass from the residual vector-bundle model to an actual Cox wall. The notation covers both interior walls and boundary walls of the effective cone. Let $B = \mathbb{Z}[x_1, \dots, x_n]$ be graded by the free group A , and let $G = D(A)$. Let F be a nonzero codimension-one wall face, let $\theta_0 \in A$ be an integral character lying in its relative interior after positive scaling if necessary, and choose a primitive normal $\lambda : A \rightarrow \mathbb{Z}$ with $\text{span}_{\mathbb{R}} F = \ker(\lambda)_{\mathbb{R}}$. Put $A_0 = \ker \lambda$. In the interior case, θ_+ and θ_- are the two adjacent chamber characters on the positive and negative sides of F . In the one-sided boundary case, only one side is a chamber of the GIT fan and the other side is a nearby outside point with empty semistable locus. Here “boundary” means that F lies in a codimension-one supporting face of the effective cone Eff ; since the target chambers considered below are full-dimensional, Eff spans $A_{\mathbb{Q}}$. Thus, if the positive side is inside Eff , we orient λ so that $\text{Eff} \subset \{\lambda \geq 0\}$ and set $U_- = \emptyset$; if the negative side is inside Eff , we orient λ so that $\text{Eff} \subset \{\lambda \leq 0\}$ and set $U_+ = \emptyset$. This

supporting-face condition is exactly what implies that all Cox degrees have nonnegative, respectively nonpositive, λ -weight. Relabel the variables as

$$z_1, \dots, z_m, \quad u_1, \dots, u_p, \quad v_1, \dots, v_q$$

with

$$\lambda(\deg z_i) = 0, \quad \lambda(\deg u_i) = a_i > 0, \quad \lambda(\deg v_j) = -b_j < 0.$$

Put

$$\eta_+ = \sum_i a_i, \quad \eta_- = \sum_j b_j, \quad r = \eta_+ - \eta_-.$$

In the interior case $U_\pm = U_{\theta_\pm}$. In the one-sided boundary case the inside side $U_\pm$ is the semistable open of the unique adjacent full-dimensional chamber, and the missing outside side is, by definition, the empty semistable locus, as above. Thus the cases are:

wall type	normal variables	missing side	C
interior	$p, q \geq 1$, with both signs present	none	$U_+ \cap U_-$
positive boundary	$q = 0, p > 0$	$U_- = \emptyset$	$\emptyset$
negative boundary	$p = 0, q > 0$	$U_+ = \emptyset$	$\emptyset$.

Set

$$U_0 = U_{\theta_0}, \quad C = U_+ \cap U_-.$$

Let

$$B_Z = \mathbb{Z}[z_1, \dots, z_m], \quad I_Z = (z^K : \theta_0 \in \text{cone}(\deg z_k : k \in K)) \subset B_Z,$$

and set

$$V = D(I_Z B), \quad I_+ = (u_1, \dots, u_p), \quad I_- = (v_1, \dots, v_q).$$

Here $D(I_+)$ means the union of the principal opens $D(u_i)$, and is empty if $p = 0$; similarly for $D(I_-)$. These are the positive and negative normal-variable ideals. The corrected wall center is

$$\mathfrak{Z} = [D(I_Z)/D(A_0)].$$

In this section the two uses of the symbol D are always distinguished by their argument: $D(A)$ or $D(A_0)$ denotes the split diagonalizable group with character group A or A_0 , while $D(I)$ or $D(f)$ denotes the monomial open subset of the relevant affine Cox scheme. Since B_Z and B are reduced polynomial rings, $D(I_Z) = \emptyset$ is equivalent to $I_Z = 0$, and similarly for $I_Z B$. This equivalence is used only for monomial ideals in these reduced Cox rings.

There are three character-line conventions in this section. An ambient character $\chi \in A$ gives the line $\mathcal{O}(\chi)$ on the actual Cox quotients. Its restriction to the center remembers only its A_0 -component, giving a center character line L_α for $\alpha \in A_0$. After choosing a splitting $A = A_0 \oplus \mathbb{Z}\tau$, an ambient line with $\chi = \alpha + n\tau$ restricts on the residual model to

$$\rho^* L_\alpha \otimes \mathcal{O}(n),$$

where $\mathcal{O}(n)$ is the residual $\mathbb{G}_m^\lambda$ -character line of λ -weight n . Here

$$\mathbb{G}_m^\lambda = D(A/A_0) \simeq D(\mathbb{Z}) = \mathbb{G}_m$$

is the residual one-parameter diagonalizable quotient determined by λ ; the notation $\mathbb{G}_m^\lambda$ records this origin, while $\mathbb{G}_m$ is used after a splitting has identified A/A_0 with $\mathbb{Z}$. Lemmas 7.11, 7.12, and 7.13 prove that the quotient stacks, windows, and supported block functors do not depend on the chosen splitting; the splitting is only notation for the weight calculation. The notation is summarized as follows.

$\mathcal{O}(\chi)$	$[U_\bullet/D(A)]$	ambient character line for $\chi \in A$
L_α	$\mathfrak{Z} = [D(I_Z)/D(A_0)]$	center character line for $\alpha \in A_0$
$\mathcal{O}(n)$	residual $\mathbb{G}_m^\lambda$ -quotient	line of residual weight n
$\rho^* L_\alpha \otimes \mathcal{O}(n)$	$[V/G]$ in the residual model	restriction of an ambient line of weight n

For later reference we keep the empty cases separate from the ordinary interior case.

- (a) In an interior wall with $\mathfrak{Z} \neq \emptyset$, the residual neighborhood is $V = D(I_Z B)$, the common open is $C = U_+ \cap U_-$, and the supported loci used below are $Y_+ = \{v_1 = \dots = v_q = 0\} \subset V \cap U_+$ and $Y_- = \{u_1 = \dots = u_p = 0\} \subset V \cap U_-$. The wall theorem may call the outer induction on $\text{Perf}(\mathfrak{Z})$.
- (b) In an interior wall with $\mathfrak{Z} = \emptyset$, one has $V = \emptyset$. The open identities below then give $U_0 = U_+ = U_- = C$. There are no supported center blocks, and the relevant actual side is handled by the common-open Koszul argument of Lemma 7.22.
- (c) In a positive boundary wall, $q = 0$, $U_- = \emptyset$, and $C = \emptyset$. The only possible supported blocks are on the positive side and are attached to $\text{Perf}(\mathfrak{Z})$ if the center is nonempty.
- (d) In a negative boundary wall, $p = 0$, $U_+ = \emptyset$, and $C = \emptyset$. The sign-reversed statement holds on the negative side.

Equivalently, the boundary and zero cases used later are governed by the following conventions:

- (1) $p = 0$ means that there are no positive normal variables; in the negative-boundary case the positive side is empty.
- (2) $q = 0$ means that there are no negative normal variables; in the positive-boundary case the negative side is empty.
- (3) $\mathfrak{Z} = \emptyset$ means $D(I_Z) = \emptyset$; then $\text{Perf}(\mathfrak{Z}) = 0$ and all center blocks are omitted.
- (4) $C = \emptyset$ means that there is no common open; the two-open comparison reduces to the residual open.
- (5) $r = 0$ means $\eta_+ = \eta_-$; the displayed block interval is empty.

For the $r \geq 0$ orientation used in the directed induction, the operational case table is:

- (i) interior, $\mathfrak{Z} \neq \emptyset$: the source is $\text{Perf}(\mathfrak{X}_-)$; the remaining interval $[w + r, w + \eta_+)$ is the negative window with upper endpoint $w + \eta_+$; the displayed blocks are r copies of $\text{Perf}(\mathfrak{Z})$, one for each residual weight $w, \dots, w + r - 1$;
- (ii) interior, $\mathfrak{Z} = \emptyset$: $V = \emptyset$ and the same remaining-window statement holds on the common open C , but all supported center blocks are zero and are omitted;
- (iii) positive boundary: $q = 0$, $\eta_- = 0$, $\mathfrak{X}_- = \emptyset$, $C = \emptyset$, and the remaining interval after peeling is empty; if $\mathfrak{Z} = \emptyset$, the same statement has no supported blocks;
- (iv) negative boundary: this is the sign-reversed version. It is used only after replacing λ by $-\lambda$, so that the nonempty target side becomes the positive side for the applied theorem.

Remark 7.1 (proof map for the actual wall theorem). The rest of this section proves Proposition 7.26. The argument has five separate steps.

- (1) *Open identities.* Lemmas 7.2– 7.5 prove, stratum by stratum and then ideal-theoretically, the monomial identities

$$U_0 = V \cup C, \quad U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C.$$

This is the geometric input that separates the residual part V from the common open C .

- (2) *Hom comparison.* Lemmas 7.8 and 7.9 compare the two finite Čech totalizations attached to these open covers. The outcome is an explicit graded cone formula for character-line Homs on the actual side.
- (3) *Residual model.* If $\mathfrak{Z} \neq \emptyset$, Lemma 7.11 identifies the preceding Hom cone with the corresponding residual vector-bundle model over $\mathfrak{Z}$, and Lemmas 7.12– 7.13 remove the auxiliary splitting $A = A_0 \oplus \mathbb{Z}\tau$ from the notation. If $\mathfrak{Z} = \emptyset$, then $V = \emptyset$, the residual-model step is skipped, and the relevant cones are zero by the open identities.
- (4) *Window equivalence and supported peeling.* Lemmas 7.15– 7.23 prove generation on the actual side: the side windows generate by character lines, and the supported pieces are peeled off by Koszul triangles.
- (5) *Assembly.* Lemma 7.24 identifies the final window with the source side, and Proposition 7.26 assembles the blocks into the announced semiorthogonal decomposition.

The boundary corollary 7.28 is the one-sided specialization used when the directed path enters the effective cone from outside.

The point of separating these steps is that mapping-complex descent is used only in steps (2) and (3), while generation is proved in step (4) by the global character-line theorem and Koszul resolutions.

Lemma 7.2 (actual wall stratum criteria). *Let S_J be the coordinate stratum whose nonzero coordinates are exactly those indexed by J , and put*

$$J_0 = J \cap \{z_1, \dots, z_m\}, \quad J_+ = J \cap \{u_1, \dots, u_p\}, \quad J_- = J \cap \{v_1, \dots, v_q\}.$$

Then the following conditions hold.

- (i) $S_J \subset U_0$ if and only if $\theta_0 \in \omega_J$.
- (ii) $S_J \subset V$ if and only if there exists $K \subset J_0$ with $\theta_0 \in \text{cone}(\deg z_k : k \in K)$. Equivalently, the zero-weight support J_0 is θ_0 -semistable for the center Cox datum.
- (iii) $S_J \subset D(I_+)$ if and only if $J_+ \neq \emptyset$, and $S_J \subset D(I_-)$ if and only if $J_- \neq \emptyset$.
- (iv) In the interior-wall case, $S_J \subset U_+$ if and only if $\theta_+ \in \omega_J$, and $S_J \subset U_-$ if and only if $\theta_- \in \omega_J$. Hence $S_J \subset C$ if and only if both adjacent chamber characters lie in ω_J .
- (v) In a one-sided boundary case, the missing outside side is empty by definition, and the inside side satisfies the same criterion with the unique adjacent inside chamber character.

Proof. For any character χ , the semistable open U_χ is the union of principal opens $D(f)$ for homogeneous semi-invariants $f \in B_{d\chi}$, $d > 0$. This union is unchanged if one keeps only monomial semi-invariants: if $f = \sum_t c_t m_t$ and f is nonzero at a point, then at least one monomial term m_t is nonzero at that point. Thus U_χ is the union of the monomial opens $D(m)$ with $\deg m = d\chi$. A monomial open $D(m)$ contains S_J exactly when all variables of m lie in J . Therefore $S_J \subset U_\chi$ is equivalent to the existence of a monomial of degree $d\chi$, $d > 0$, using only variables in J , which is equivalent to $\chi \in \omega_J$. This proves (i) and, for chamber characters, (iv) and (v). The open $V = D(I_Z B)$ is the union of the $D(z^K)$'s appearing in the definition of I_Z ; such a principal open contains S_J exactly when $K \subset J_0$, proving (ii). Finally $D(I_+) = \bigcup_i D(u_i)$ and $D(I_-) = \bigcup_j D(v_j)$, which gives (iii). $\square$

Lemma 7.3 (strata near an actual wall). *Assume first that F is an interior wall, and let $S_J \subset U_0$ be a coordinate stratum. Let $\bar{\omega}_J$ be the image of ω_J in the quotient line*

$$A_{\mathbb{R}} / \text{span}_{\mathbb{R}} F.$$

Then the position of S_J is as follows.

$\bar{\omega}_J$	normal variables in J	consequence
$\{0\}$	none	$S_J \subset V$
$\mathbb{R}_{\geq 0}$, nonzero	$J_+ \neq \emptyset, J_- = \emptyset$	$S_J \subset V \cap D(I_+) \subset U_+, S_J \not\subset U_-$
$\mathbb{R}_{\leq 0}$, nonzero	$J_- \neq \emptyset, J_+ = \emptyset$	$S_J \subset V \cap D(I_-) \subset U_-, S_J \not\subset U_+$
$\mathbb{R}$	$J_+ \neq \emptyset, J_- \neq \emptyset$	$S_J \subset C = U_+ \cap U_-$

For a positive one-sided boundary wall, all Cox degrees have nonnegative λ -weight. If $S_J \subset U_0$, then $S_J \subset V$, and $S_J \subset U_+$ holds precisely when $J_+ \neq \emptyset$. The missing side and C are empty. The negative boundary case is obtained by reversing signs.

Proof. By Lemma 7.2, $S_J \subset U_0$ means $\theta_0 \in \omega_J$. Since θ_0 lies in the relative interior of the codimension-one cell F , Lemma 2.19 gives $F \subset \omega_J$. The image $\bar{\omega}_J$ is therefore a closed convex cone in a one-dimensional vector space, hence one of the four cones listed in the table.

If $\bar{\omega}_J$ is $\{0\}$, $\mathbb{R}_{\geq 0}$, or $\mathbb{R}_{\leq 0}$, all nonzero normal weights in J have one sign. Writing θ_0 as a nonnegative combination of degrees in J and applying λ forces all nonzero-weight coefficients to vanish. Thus θ_0 is generated by the zero-weight part J_0 , and Lemma 7.2 puts S_J in V . In the nonzero positive-ray case, some positive normal variable occurs in J and no negative normal variable does; hence $S_J \subset D(I_+)$ and $S_J \not\subset D(I_-)$. Since $\bar{\omega}_J$

contains the positive ray, choose a sufficiently small positive-normal perturbation $\theta' \in \omega_J$ of θ_0 . By the local one-dimensional fan of Lemma 2.20, the point θ' lies in the positive adjacent full-dimensional chamber cell. This is the same cell that contains the chosen chamber character θ_+ . Lemma 2.19 therefore gives $\theta_+ \in \omega_J$, hence $S_J \subset U_+$. It cannot lie in U_- , because the image of ω_J contains no negative ray. The negative-ray case is identical with signs reversed.

If $\bar{\omega}_J = \mathbb{R}$, then J contains normal variables of both signs. Choose points of ω_J with positive and negative λ -value. Small convex perturbations of θ_0 toward these points lie in ω_J and, by Lemma 2.20, in the two adjacent full-dimensional chamber cells. Applying Lemma 2.19 to those two cells shows that ω_J contains both adjacent chamber characters θ_+ and θ_- , and Lemma 7.2 gives $S_J \subset C$.

For a positive boundary wall, $\text{Eff} \subset \{\lambda \geq 0\}$. Since all Cox degrees lie in Eff , no negative normal variable exists. The same nonnegative-coefficient argument shows that every $S_J \subset U_0$ is already in V . It lies in the inside chamber exactly when a positive normal variable is present: if $J_+ = \emptyset$, the cone ω_J has zero transverse image and cannot meet the inside chamber; if $J_+ \neq \emptyset$, a small positive-normal perturbation of θ_0 can be chosen in ω_J and lands in the unique inside full-dimensional chamber cell of the local quotient fan. By Lemma 2.19, that entire inside chamber cell contains the chosen inside chamber character, so Lemma 7.2 gives $S_J \subset U_+$. The negative boundary case is the sign reversal. $\square$

Lemma 7.4 (monomial opens are determined by coordinate strata). *Let W_1, W_2 be finite monomial opens in a split affine space $\text{Spec } \mathbb{Z}[x_1, \dots, x_n]$. If, for every coordinate stratum S_J , one has*

$$S_J \subset W_1 \iff S_J \subset W_2,$$

then $W_1 = W_2$ as open subschemes, equivalently the defining monomial ideals have the same radical. This is a scheme-theoretic statement over $\mathbb{Z}$, and remains true after every base change to a residue field. More generally, if Z is a closed coordinate subscheme cut out by a subset of the variables and W is a finite monomial open, then $Z \times W = \emptyset$ scheme-theoretically if and only if no coordinate stratum contained in Z is contained in W . The same assertions hold after quotienting by a split diagonalizable group.

Proof. Write $W_\ell = D(I_\ell)$, where I_ℓ is a finitely generated monomial ideal. A point of the affine space has a well-defined support

$$J = \{i : x_i \text{ is nonzero in its residue field}\},$$

and membership in a principal monomial open $D(x^u)$ is exactly the condition $\text{supp}(x^u) \subset J$. Hence membership in $D(I_\ell)$ depends only on J , and $D(I_\ell)$ is the union of the coordinate strata satisfying the corresponding support condition. Equality on all coordinate strata therefore gives equality of the underlying open subsets. Equivalently, if $P_J = (x_i : i \notin J)$ is the squarefree monomial prime corresponding to the closure of S_J , then

$$S_J \subset D(I_\ell) \iff I_\ell \not\subset P_J.$$

Thus the two ideals are contained in exactly the same squarefree monomial primes. For monomial ideals this is equivalent to $\sqrt{I_1} = \sqrt{I_2}$, and therefore to equality of the open subschemes $D(I_1) = D(I_2)$, not merely equality of their sets of geometric points. Because radicals of monomial ideals are generated by squarefree monomials with coefficient 1, this radical equality is preserved by arbitrary base change from $\mathbb{Z}$. Thus the same criterion works uniformly over all residue characteristics.

For the closed coordinate subscheme $Z = V(x_i : i \in L)$, the fiber product $Z \times D(I)$ is the open $D(\bar{I})$ inside the coordinate polynomial ring obtained by setting the variables in L equal to zero. It is empty if and only if every monomial generator of I dies in this quotient, equivalently if and only if no support J compatible with the equations of Z satisfies the support condition for $D(I)$. This is the scheme-theoretic form of the stratum criterion used

below. Quotienting by $D(A)$ preserves equalities of invariant open subschemes and preserves the empty fiber product, so the quotient-stack statements follow. $\square$

Lemma 7.5 (radical form of the actual open identities). *Choose finitely generated monomial ideals*

$$\mathfrak{a}_0, \mathfrak{a}_+, \mathfrak{a}_-, \mathfrak{a}_C$$

with

$$U_0 = D(\mathfrak{a}_0), \quad U_+ = D(\mathfrak{a}_+), \quad U_- = D(\mathfrak{a}_-), \quad C = D(\mathfrak{a}_C),$$

where $\mathfrak{a}_C = 0$ if $C = \emptyset$. Then, in the full Cox polynomial ring,

$$\sqrt{\mathfrak{a}_0} = \sqrt{I_Z B + \mathfrak{a}_C},$$

and

$$\sqrt{\mathfrak{a}_+} = \sqrt{(I_Z B)I_+ + \mathfrak{a}_C}, \quad \sqrt{\mathfrak{a}_-} = \sqrt{(I_Z B)I_- + \mathfrak{a}_C}.$$

Here $I_+ = (u_1, \dots, u_p)$ and $I_- = (v_1, \dots, v_q)$, with an empty normal-variable list interpreted as the zero ideal.

Proof. For monomial ideals, equality of radicals is equivalent to equality of the corresponding monomial opens, by Lemma 7.4. We therefore check coordinate-stratum membership.

First note that V , C , U_+ , and U_- are all contained in U_0 . The inclusions for V and C are part of their definitions; for U_+ and U_- , if an adjacent chamber character lies in a closed support cone ω_J , then its boundary face F , and hence θ_0 , also lies in ω_J . Thus a stratum outside U_0 is outside every open appearing on the right below. On strata contained in U_0 , Lemma 7.3 gives

$$S_J \subset U_0 \iff S_J \subset V \text{ or } S_J \subset C.$$

Since $V = D(I_Z B)$ and $C = D(\mathfrak{a}_C)$, the right-hand side is equivalent to $S_J \subset D(I_Z B + \mathfrak{a}_C)$. Hence $D(\mathfrak{a}_0) = D(I_Z B + \mathfrak{a}_C)$, which is the first radical identity.

Similarly, for every coordinate stratum S_J , the same preliminary observation reduces to the case $S_J \subset U_0$, and the stratum table gives

$$S_J \subset U_+ \iff S_J \subset V \cap D(I_+) \text{ or } S_J \subset C.$$

Because

$$V \cap D(I_+) = D(I_Z B) \cap D(I_+) = D((I_Z B)I_+),$$

this is equivalent to

$$S_J \subset D((I_Z B)I_+ + \mathfrak{a}_C).$$

Thus $\sqrt{\mathfrak{a}_+} = \sqrt{(I_Z B)I_+ + \mathfrak{a}_C}$. The proof for U_- is identical, replacing I_+ by I_- . Boundary sides and empty centers are included because $D(0) = \emptyset$, $\mathfrak{a}_C = 0$ when $C = \emptyset$, and an empty normal-variable list gives the zero ideal. $\square$

The preceding uniform identities specialize as follows. This table is only an unpacking of Lemma 7.5, but it is useful for checking the empty and boundary cases where the conventions

$D(0) = \emptyset$ and $I_{\pm} = 0$ for an empty normal list are active.

case	$V = D(I_Z B)$	C	radical identities
interior, $\mathfrak{Z} \neq \emptyset$	nonempty	$U_+ \cap U_-$	$\sqrt{\mathfrak{a}_0} = \sqrt{I_Z B + \mathfrak{a}_C}$ $\sqrt{\mathfrak{a}_+} = \sqrt{(I_Z B)I_+ + \mathfrak{a}_C}$ $\sqrt{\mathfrak{a}_-} = \sqrt{(I_Z B)I_- + \mathfrak{a}_C}$
interior, $\mathfrak{Z} = \emptyset$	$\emptyset$ ($I_Z B = 0$)	U_0	$\sqrt{\mathfrak{a}_0} = \sqrt{\mathfrak{a}_C}$ $\sqrt{\mathfrak{a}_+} = \sqrt{\mathfrak{a}_C}$ $\sqrt{\mathfrak{a}_-} = \sqrt{\mathfrak{a}_C}$
positive boundary	$D(I_Z B)$	$\emptyset$ ($\mathfrak{a}_C = 0$)	$\sqrt{\mathfrak{a}_0} = \sqrt{I_Z B}$ $\sqrt{\mathfrak{a}_+} = \sqrt{(I_Z B)I_+}$ $\mathfrak{a}_- = 0$
negative boundary	$D(I_Z B)$	$\emptyset$ ($\mathfrak{a}_C = 0$)	$\sqrt{\mathfrak{a}_0} = \sqrt{I_Z B}$ $\mathfrak{a}_+ = 0$ $\sqrt{\mathfrak{a}_-} = \sqrt{(I_Z B)I_-}$

In the second row, $I_Z B = 0$ means precisely $D(I_Z B) = \emptyset$ in the reduced polynomial Cox ring. In the boundary rows the missing side is the empty semistable locus, so its defining monomial ideal is 0 under the convention $D(0) = \emptyset$.

The comparison used throughout this section is the following. The open $V \subset U_0$ is the residual neighborhood of the wall center, and the residual quotient is

$$[V/G] \simeq [\mathrm{Tot}_3(E_+ \oplus E_-)/\mathbb{G}_m].$$

The dictionary with the residual wall theorem is:

- $D(I_Z)$ in the zero-weight variables gives the corrected wall center $\mathfrak{Z}$.
- $V = D(I_Z B)$ is the actual neighborhood identified with $\mathrm{Tot}_3(E_+ \oplus E_-)$, where the residual theorem applies.
- $V \cap D(I_+)$ and $V \cap D(I_-)$ are the positive and negative residual sides $\mathfrak{X}_+$ and $\mathfrak{X}_-$.
- $C = U_+ \cap U_-$ is the common open away from the supported residual pieces; it is handled by common-window generation.

The complement of this residual neighborhood is controlled by the common open $C = U_+ \cap U_-$. The actual side covers are

$$U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C,$$

and the wall semistable locus itself is $U_0 = V \cup C$. Thus full-faithfulness and semiorthogonality are reduced to a comparison of finite Čech mapping complexes for these two-open covers; generation is proved separately by character-line generation and Koszul triangles.

For readability we separate the actual wall proof into five independent steps. First, Lemma 7.6 proves the open identities through the monomial radical identities of Lemma 7.5; the support-cone geometry and boundary wall cases are isolated in Lemma 7.3. Second, Lemma 7.7 records the elementary but crucial fact that a stratum in the common open contains normal variables of both signs. Third, Lemma 7.9 converts the actual-to-residual restriction of character-line Homs into a single homogeneous local-cohomology cone. Fourth, Lemma 7.11 applies the monomial weight bound to that cone, proving actual window full-faithfulness. Finally, Lemmas 7.20 and 7.23 identify the supported locus and peel off the residual blocks. Proposition 7.26 then only assembles these already-proved pieces.

Lemma 7.6 (support-cone open identities). *With the notation above,*

$$U_0 = V \cup C, \quad U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C.$$

Proof. Choose monomial ideals $\mathfrak{a}_0, \mathfrak{a}_+, \mathfrak{a}_-$, and $\mathfrak{a}_C$ defining U_0, U_+, U_- , and C . By Lemma 7.5,

$$\sqrt{\mathfrak{a}_0} = \sqrt{I_Z B + \mathfrak{a}_C}, \quad \sqrt{\mathfrak{a}_+} = \sqrt{(I_Z B)I_+ + \mathfrak{a}_C}, \quad \sqrt{\mathfrak{a}_-} = \sqrt{(I_Z B)I_- + \mathfrak{a}_C}.$$

Taking $D(-)$, and using $D(I + J) = D(I) \cup D(J)$ and $D(IJ) = D(I) \cap D(J)$, gives exactly

$$U_0 = V \cup C, \quad U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C.$$

□

Lemma 7.7 (common-open strata have both signs). *Assume F is an interior wall. If a coordinate stratum S_J is contained in $C = U_+ \cap U_-$, then J contains at least one positive-weight normal variable u_i and at least one negative-weight normal variable v_j . Equivalently, the ideals $(u_1, \dots, u_p)$ and $(v_1, \dots, v_q)$ generate the unit ideal on C . In the one-sided boundary cases $C = \emptyset$, so the assertion is vacuous.*

Proof. If $S_J \subset C$, then ω_J contains both adjacent chamber characters. In the quotient line $A_{\mathbb{R}}/\text{span}_{\mathbb{R}}(F)$, the image of ω_J therefore contains both positive and negative rays. A cone generated only by zero-weight and positive-weight variables cannot contain the negative ray, and similarly with the signs reversed. Thus J contains variables of both normal signs. The unit-ideal statement follows because it holds on every coordinate stratum of C : by Lemma 7.4, the stratumwise inclusions $C \subset D(I_+)$ and $C \subset D(I_-)$ are equivalent to the scheme-theoretic statements $I_+ \mathcal{O}_C = \mathcal{O}_C$ and $I_- \mathcal{O}_C = \mathcal{O}_C$. □

Lemma 7.8 (two-open totalization comparison). *Let*

$$\mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I, \quad \mathcal{H}_{V'} \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I$$

be two Čech diagrams of complexes which agree on the C -term and the intersection term, and differ only by a map $\mathcal{H}_V \rightarrow \mathcal{H}_{V'}$. Then the cofiber of the induced map of totalizations is canonically equivalent to

$$\text{cofib}(\mathcal{H}_V \rightarrow \mathcal{H}_{V'}).$$

Proof. The totalization of $\mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I$ is the fiber of the arrow $\mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I$. We therefore have a morphism of fiber sequences

$$\text{Tot}(\mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I) \rightarrow \mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I,$$

$$\text{Tot}(\mathcal{H}_{V'} \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I) \rightarrow \mathcal{H}_{V'} \oplus \mathcal{H}_C \rightarrow \mathcal{H}_I,$$

where the right vertical map is the identity. Taking cofibers and using the octahedral axiom, or equivalently the 3×3 lemma in a stable category, identifies the cofiber on the left with the cofiber of $\mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_{V'} \oplus \mathcal{H}_C$, which is $\text{cofib}(\mathcal{H}_V \rightarrow \mathcal{H}_{V'})$. □

For the Hom-cone calculation we use the following fixed diagram. On the positive side put $V_+ = V \cap D(I_+)$. Lemma 7.6 and Lemma 7.7 give a diagram of monomial opens

$$\begin{array}{ccccc} V \cap C & \subset & C & & V_+ \cap C = V \cap C & \subset & C \\ \cap & & \cap & & \cap & & \cap \\ V & \subset & U_0 = V \cup C & & V_+ & \subset & U_+ = V_+ \cup C. \end{array}$$

All arrows are restrictions of homogeneous monomial localizations, and the right vertical and top horizontal terms are literally the same in the two Čech covers. Therefore, for character lines $\mathcal{O}(\alpha), \mathcal{O}(\beta)$, the two mapping complexes are the $(\beta - \alpha)$ -homogeneous summands of finite graded Čech section complexes. Here $R\Gamma(W, \mathcal{O})$, for a finite monomial open W , means this finite graded Čech complex of sections in the quotient-stack graded model. With this convention the two complexes are

$$\text{Tot}(R\Gamma(V, \mathcal{O}) \oplus R\Gamma(C, \mathcal{O}) \rightarrow R\Gamma(V \cap C, \mathcal{O}))$$

and

$$\text{Tot}(R\Gamma(V_+, \mathcal{O}) \oplus R\Gamma(C, \mathcal{O}) \rightarrow R\Gamma(V \cap C, \mathcal{O})).$$

The only changed term is $R\Gamma(V, \mathcal{O}) \rightarrow R\Gamma(V_+, \mathcal{O})$. The negative-side diagram is the same with $V_- = V \cap D(I_-)$.

Lemma 7.9 (actual character-line Hom cone formula). *Let $\mathcal{O}(\alpha)$ and $\mathcal{O}(\beta)$ be two character lines on $[U_0/G]$, and put $\delta = \beta - \alpha$. Assume $p > 0$. The cone of the restriction map*

$$\mathbf{R} \operatorname{Hom}_{[U_0/G]}(\mathcal{O}(\alpha), \mathcal{O}(\beta)) \longrightarrow \mathbf{R} \operatorname{Hom}_{[U_+/G]}(\mathcal{O}(\alpha), \mathcal{O}(\beta))$$

is canonically equivalent to

$$[\operatorname{Cone}(R\Gamma(V, \mathcal{O}) \longrightarrow R\Gamma(V \cap D(I_+), \mathcal{O}))]_\delta.$$

If $q > 0$, the sign-reversed formula holds for the map to $[U_-/G]$, with I_- replacing I_+ .

Proof. We give the plus proof. The minus formula is obtained by replacing $V_+ = V \cap D(I_+)$ with $V_- = V \cap D(I_-)$, replacing the side $[U_+/G]$ with $[U_-/G]$, and using the same two-open Čech comparison for the covers $U_0 = V \cup C$ and $U_- = V_- \cup C$. In that case the common intersection condition is $V \cap C = V_- \cap C$, which follows from $V \cap C \subset D(I_-)$; for an interior wall this is another consequence of Lemma 7.7, and in a one-sided boundary case $C = \emptyset$. Put $V_+ = V \cap D(I_+)$. By Lemma 7.6,

$$U_0 = V \cup C, \quad U_+ = V_+ \cup C.$$

Moreover $V \cap C \subset D(I_+)$. Indeed, if $C = \emptyset$ this is because both sides of the desired equality below are empty, while for an interior wall it follows from Lemma 7.7; in the one-sided positive boundary case $C = \emptyset$. Hence the two covers have the same intersection

$$V \cap C = V_+ \cap C.$$

Mapping complexes of character lines on a split diagonalizable quotient are computed by homogeneous summands of the coordinate Čech complexes. Thus

$$\begin{aligned} \mathbf{R} \operatorname{Hom}_{[U_0/G]}(\mathcal{O}(\alpha), \mathcal{O}(\beta)) &\simeq \operatorname{Tot}(H_V \oplus H_C \rightarrow H_{VC}), \\ \mathbf{R} \operatorname{Hom}_{[U_+/G]}(\mathcal{O}(\alpha), \mathcal{O}(\beta)) &\simeq \operatorname{Tot}(H_{V_+} \oplus H_C \rightarrow H_{VC}), \end{aligned}$$

where $H_W = [R\Gamma(W, \mathcal{O})]_\delta$, and $VC = V \cap C$. The finite Čech differentials preserve the A -grading, so taking the δ -summand commutes with these totalizations by Lemma 2.11. The map between the two totalizations is induced by $H_V \rightarrow H_{V_+}$ and by the identity on the C - and VC -terms. Lemma 7.8 identifies its cone with

$$\operatorname{Cone}(H_V \rightarrow H_{V_+}) = [\operatorname{Cone}(R\Gamma(V, \mathcal{O}) \rightarrow R\Gamma(V \cap D(I_+), \mathcal{O}))]_\delta,$$

as claimed. $\square$

Lemma 7.10 (gluing the residual quotient presentation). *Assume $\mathfrak{Z} \neq \emptyset$, and choose a splitting $A = A_0 \oplus \mathbb{Z}\tau$ with $\lambda(\tau) = 1$. Write*

$$\deg u_i = \alpha_i + a_i\tau, \quad \deg v_j = \beta_j - b_j\tau, \quad \alpha_i, \beta_j \in A_0.$$

Let $E_+ = \bigoplus_i L_{\alpha_i}$ and $E_- = \bigoplus_j L_{\beta_j}$ on $\mathfrak{Z}$. Then, relative to this chosen splitting, there is a canonical isomorphism

$$[V/D(A)] \simeq [\operatorname{Tot}_3(E_+ \oplus E_-)/\mathbb{G}_m^\lambda],$$

where $\mathbb{G}_m^\lambda = D(\mathbb{Z})$ acts on the u_i -fiber coordinates with weights a_i and on the v_j -fiber coordinates with weights $-b_j$. Under this isomorphism $V \cap D(I_+)$ and $V \cap D(I_-)$ are the residual positive and negative semistable opens. The statement is independent of the splitting in the following sense: replacing τ by $\tau + \sigma$, $\sigma \in A_0$, changes the center characters by

$$\alpha_i \mapsto \alpha_i - a_i\sigma, \quad \beta_j \mapsto \beta_j + b_j\sigma,$$

equivalently tensors the corresponding summands by the indicated powers of the center character line L_σ . It gives an isomorphic quotient stack and the same residual open substacks and λ -window subcategories.

Proof. Choose a finite homogeneous monomial cover of $D(I_Z)$ by center charts $D(z^K)$. On such a chart put

$$R_K = \mathbb{Z}[z_1, \dots, z_m][(z^K)^{-1}],$$

with its A_0 -grading. The inverse image of this chart in V has coordinate ring

$$R_K[u_1, \dots, u_p, v_1, \dots, v_q],$$

graded by $A_0 \oplus \mathbb{Z}\tau$, with bidegrees

$$\deg u_i = (\alpha_i, a_i), \quad \deg v_j = (\beta_j, -b_j).$$

Quotienting first by $D(A_0)$ gives the base quotient chart of $\mathfrak{Z}$ and affine fiber coordinates transforming by the character lines L_{α_i} and L_{β_j} . The residual quotient by $D(\mathbb{Z}) = \mathbb{G}_m^\lambda$ records exactly the second grading coordinate, hence the displayed weights.

On an overlap $D(z^K) \cap D(z^L) = D(z^{K+L})$, both chart presentations are obtained by the same homogeneous localization of the A_0 -graded ring. The transition functions for the fiber coordinates are therefore precisely the transition functions of the center character lines. These transition functions satisfy the line-bundle cocycle condition on $\mathfrak{Z}$, so the chartwise quotient presentations glue to the asserted quotient-stack isomorphism. Since the ideals I_+ and I_- are generated by the same fiber coordinates before and after gluing, their principal-open unions identify with the residual semistable opens.

If $\tau' = \tau + \sigma$, then

$$\deg u_i = (\alpha_i - a_i\sigma) + a_i\tau', \quad \deg v_j = (\beta_j + b_j\sigma) - b_j\tau'.$$

This only changes the A_0 -character line factors in the above vector bundle description. The underlying A -graded coordinate ring, the quotient $[V/D(A)]$, the ideals $I_\pm$, and the intervals defined by λ -weights are unchanged. $\square$

Lemma 7.11 (residual model and actual Hom cones). *Assume $\mathfrak{Z} \neq \emptyset$. There are split toric vector bundles E_+ and E_- , each a direct sum of toric character line bundles on $\mathfrak{Z}$, such that*

$$[V/G] \simeq [\mathrm{Tot}_{\mathfrak{Z}}(E_+ \oplus E_-)/\mathbb{G}_m],$$

with residual weights a_i on the u_i -directions and $-b_j$ on the v_j -directions. Under this identification $[V \cap U_+/G]$ and $[V \cap U_-/G]$ are the residual sides of Section 6; the closed substacks cut out by $(v_1, \dots, v_q)$ and by $(u_1, \dots, u_p)$ are the residual supported loci Y_+ and Y_- , their projections to $\mathfrak{Z}$ are the weighted-projective bundle maps q_+ and q_- , and their regular closed-immersion pushforwards are the s_{+} and s_{-*} used in the supported block functors. Moreover restriction $[U_0/G] \rightarrow [U_+/G]$ is fully faithful on every plus window $w \leq \lambda(\chi) < w + \eta_+$, and restriction $[U_0/G] \rightarrow [U_-/G]$ is fully faithful on every minus window $W - \eta_- \leq \lambda(\chi) < W$.*

Proof. Choose $\tau \in A$ with $\lambda(\tau) = 1$, so $A = A_0 \oplus \mathbb{Z}\tau$. Write

$$\deg u_i = \alpha_i + a_i\tau, \quad \deg v_j = \beta_j - b_j\tau, \quad \alpha_i, \beta_j \in \ker \lambda.$$

Let $E_+ = \bigoplus_i L_{\alpha_i}$ and $E_- = \bigoplus_j L_{\beta_j}$. Lemma 7.10 gives the quotient-stack isomorphism

$$[V/G] \simeq [\mathrm{Tot}_{\mathfrak{Z}}(E_+ \oplus E_-)/\mathbb{G}_m]$$

and identifies $V \cap U_\pm$ with the residual semistable opens. Under this same equivalence, the equation $v_1 = \dots = v_q = 0$ in the actual Cox chart is the zero section in the negative summand E_- , hence gives the residual supported locus Y_+ ; similarly $u_1 = \dots = u_p = 0$ gives Y_- . Forgetting the remaining positive, respectively negative, fiber coordinates gives the morphisms $q_+ : Y_+ \rightarrow \mathfrak{Z}$ and $q_- : Y_- \rightarrow \mathfrak{Z}$. Because these descriptions are given by the same homogeneous coordinate ideals before and after the quotient equivalence, the regular closed immersions and their pushforward functors agree with the residual $s_{\pm*}$'s after restriction to V .

If $p = 0$, then $\eta_+ = 0$ and the plus window is empty, so plus-window full faithfulness is vacuous. Assume $p > 0$ for the plus calculation below. For character lines $\mathcal{O}(\alpha), \mathcal{O}(\beta)$, the

cone of the plus restriction map is the $(\beta - \alpha)$ -graded summand computed in Lemma 7.9. On a monomial chart of $D(I_Z)$, the relative Čech cohomology for inverting the u_i 's is represented by monomials

$$u_1^{-r_1} \cdots u_p^{-r_p} v_1^{s_1} \cdots v_q^{s_q} z^\mu, \quad r_i \geq 1, \quad s_j \geq 0,$$

whose λ -weights are at most $-\eta_+$, as in Lemma 6.2. Thus the plus Hom cone has no cohomology in degrees δ with $\lambda(\delta) > -\eta_+$. This proves plus-window full faithfulness on line bundles. The calculation is local on the finite monomial cover of $D(I_Z)$; finite Čech totalization over the zero-coordinate charts preserves the displayed λ -weight bounds, so the local computations glue. Lemma 2.14 gives full faithfulness for the window subcategory. If $q = 0$, the minus window is empty and the minus assertion is vacuous. For $q > 0$, the minus calculation inverts the v_j 's and gives cohomology weights at least η_- . If α, β lie in a minus window $[W - \eta_-, W)$, then $\lambda(\beta - \alpha) < \eta_-$; hence none of these cone weights can contribute, and the same thick-closure argument gives full faithfulness on minus windows. $\square$

Lemma 7.12 (ambient lifts of residual weights). *Assume $\mathfrak{Z} \neq \emptyset$. Choose a splitting $A = A_0 \oplus \mathbb{Z}\tau$ with $\lambda(\tau) = 1$. If $\chi = \alpha + n\tau$, $\alpha \in \ker \lambda$, then under the residual identification*

$$[V/G] \simeq [\mathrm{Tot}_{\mathfrak{Z}}(E_+ \oplus E_-)/\mathbb{G}_m]$$

the ambient character line $\mathcal{O}(\chi)$ restricts to

$$\rho^* L_\alpha \otimes \mathcal{O}(n),$$

where L_α is the character line on $\mathfrak{Z}$ and $\mathcal{O}(n)$ is the residual $\mathbb{G}_m$ -character line of weight $n = \lambda(\chi)$. If χ' is another lift with $\lambda(\chi') = n$, then $\chi' - \chi \in \ker \lambda$, and replacing L_α by $L_{\alpha+\chi'-\chi}$ gives the same residual-weight block. Replacing the splitting τ changes only this $\ker \lambda$ -character bookkeeping; the quotient stack, residual opens, and thick subcategories defined by a fixed λ -window are unchanged.

Proof. In the chart notation of Lemma 7.11, the A -grading is the bidegree

$$A = A_0 \oplus \mathbb{Z}\tau.$$

The character line associated to $\chi = \alpha + n\tau$ is therefore the graded free rank-one module shifted by bidegree (α, n) . After quotienting first by $D(A_0)$, the A_0 -component is the line L_α on the center, while the second component is the residual $\mathbb{G}_m$ -weight n . This gives $\rho^* L_\alpha \otimes \mathcal{O}(n)$ on every monomial center chart, and the identifications agree on overlaps because they are induced by the same graded localization maps. The statements about changing the lift and the splitting are immediate from the decomposition of the character χ : only the $\ker \lambda$ -component changes, and the center category is generated by all such character twists. $\square$

Lemma 7.13 (intrinsic supported block functors). *Assume $\mathfrak{Z} \neq \emptyset$. The functors*

$$\beta_a^+(F) = s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a)), \quad \beta_a^-(F) = s_{-*}(q_-^* F \otimes \mathcal{O}_{Y_-}(a))$$

are intrinsic to the wall normal λ and do not depend on a choice of splitting $A = A_0 \oplus \mathbb{Z}\tau$ or on a choice of ambient lift of a residual weight. The splitting is used only to express ambient character lines as $\rho^ L_\alpha \otimes \mathcal{O}(n)$ when checking containment in windows and generation by character-line generators.*

Proof. The residual character lattice is the quotient

$$A/A_0 \simeq \mathbb{Z}$$

defined by λ , so the residual character line $\mathcal{O}(a)$ on $[V/G]$, and hence on $Y_\pm$, is canonical once the orientation of λ is fixed. The substacks $Y_\pm \subset [V/G]$ are cut out by the homogeneous ideals generated by the opposite normal variables; these ideals are defined before any splitting of A . Therefore the expression

$$s_{\pm*}(q_\pm^* F \otimes \mathcal{O}_{Y_\pm}(a))$$

defines an exact functor $\mathrm{Perf}(\mathfrak{Z}) \rightarrow \mathrm{Perf}(\mathfrak{X}_\pm)$ intrinsically in the quotient stack.

If $\tau' = \tau + \sigma$ is another splitting, then an ambient lift $\chi = \alpha + a\tau$ of residual weight a is rewritten as

$$\chi = (\alpha - a\sigma) + a\tau'.$$

Thus only the center character factor changes. Since F ranges over all of $\text{Perf}(\mathfrak{Z})$, and the actual functor above uses the canonical residual line $\mathcal{O}(a)$, the splitting affects only the bookkeeping in the proof that character-line generators are peeled off; it does not alter the functor $\beta_a^\pm$. $\square$

In the following proposition the open-cover identities are not only set-theoretic statements on geometric points. By Lemma 7.4 and Lemma 7.6, they are equalities of homogeneous monomial open subschemes, equivalently radical monomial ideal identities, before passing to quotient stacks.

Proposition 7.14 (actual-residual comparison package). *For an actual wall with notation as above, the following statements hold. If $\mathfrak{Z} = \emptyset$, then $V = \emptyset$, all assertions involving the residual vector-bundle model are to be read as vacuous, and the Hom-cone formulas below have zero right-hand side; this empty-center interpretation is proved in the first paragraph of the proof. If $\mathfrak{Z} \neq \emptyset$, the residual model is the one in Lemma 7.11.*

(a) *The two-open covers*

$$U_0 = V \cup C, \quad U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C$$

have the same intersections in the relevant comparisons:

$$V \cap C = (V \cap D(I_+)) \cap C = (V \cap D(I_-)) \cap C$$

whenever the corresponding side is nonempty.

(b) *If $\mathfrak{Z} \neq \emptyset$, then on V , after choosing a splitting $A = A_0 \oplus \mathbb{Z}\tau$, the quotient is the residual vector-bundle model*

$$[V/G] \simeq [\text{Tot}_{\mathfrak{Z}}(E_+ \oplus E_-)/\mathbb{G}_m],$$

*and an ambient character $\chi = \alpha + n\tau$ restricts to $\rho^*L_\alpha \otimes \mathcal{O}(n)$. Changing the lift or the splitting only changes the $\ker \lambda$ -character factor on $\mathfrak{Z}$. If $\mathfrak{Z} = \emptyset$, then $V = \emptyset$, this residual-model statement is vacuous, and every subsequent cone formula involving V is the zero cone.*

(c) *If $p > 0$, then for character lines $\mathcal{O}(\alpha), \mathcal{O}(\beta)$, the cone of the plus restriction*

$$\mathbf{R} \text{Hom}_{[U_0/G]}(\mathcal{O}(\alpha), \mathcal{O}(\beta)) \rightarrow \mathbf{R} \text{Hom}_{[U_+/G]}(\mathcal{O}(\alpha), \mathcal{O}(\beta))$$

is the $(\beta - \alpha)$ -graded summand of

$$\text{Cone}(R\Gamma(V, \mathcal{O}) \rightarrow R\Gamma(V \cap D(I_+), \mathcal{O})),$$

and, if $q > 0$, the sign-reversed formula holds on the minus side. If one of the normal lists is empty, the corresponding side/window assertion below is vacuous and no Hom-cone formula is used for that side.

(d) *When $p > 0$, the plus cone has no cohomology in λ -weights $> -\eta_+$, and when $q > 0$, the minus cone has no cohomology in λ -weights $< \eta_-$. Consequently restriction from U_0 to U_+ is fully faithful on every nonempty plus window $[w, w + \eta_+)$, and restriction from U_0 to U_- is fully faithful on every nonempty minus window $[W - \eta_-, W)$.*

(e) *The preceding full-faithfulness statements extend from character lines to the whole window subcategories by Lemma 2.14, because the windows are defined as thick subcategories generated by the character lines in the indicated intervals.*

Proof. Part (a) is the open-identity statement of Lemma 7.6, together with Lemma 7.7: in the interior case every stratum of C contains both positive and negative normal variables, so it lies in both $D(I_+)$ and $D(I_-)$; in a boundary case $C = \emptyset$. Since all opens and intersections here are monomial opens, the stratum statement implies the scheme-theoretic equalities of

intersections by Lemma 7.4. If $\mathfrak{J} = \emptyset$, then $D(I_Z) = \emptyset$, hence $V = \emptyset$. The open identities give

$$U_0 = C, \quad U_+ = C \quad \text{whenever } U_+ \neq \emptyset, \quad U_- = C \quad \text{whenever } U_- \neq \emptyset.$$

In a boundary case $C = \emptyset$, this simply says that the corresponding nonexistent side has empty semistable locus. Thus all residual-model statements are vacuous, the cones in part (c) are zero cones, and the window full-faithfulness statements in part (d) are identities on the nonempty common-open side or vacuous on an empty side. This proves the proposition in the empty-center case.

Assume from now on that $\mathfrak{J} \neq \emptyset$. Part (b) is Lemma 7.11 for the residual model and Lemma 7.12 for characters. For part (c), apply Lemma 7.9 on each side whose normal list is nonempty. If $p = 0$, then $\eta_+ = 0$ and there is no nonempty plus window interval; in the boundary convention the plus side is the missing side. Thus the plus assertion is vacuous. The same argument with signs reversed handles $q = 0$. On the nonempty-list sides, the proof of Lemma 7.9 is precisely the two-open Čech totalization comparison of Lemma 7.8. The weight bounds in part (d) are the monomial local-cohomology bounds of Lemma 6.2, applied to the cones in part (c); the displayed window inequalities are exactly $\lambda(\beta - \alpha) > -\eta_+$ on a plus window and $\lambda(\beta - \alpha) < \eta_-$ on a minus window. Part (e) is Lemma 2.14. $\square$

Lemma 7.15 (common-open window generation). *On the common open $C = U_+ \cap U_-$, the plus and minus windows generate:*

$$\text{Perf}([C/G]) = \text{thick}\{\mathcal{O}_C(\chi) : w \leq \lambda(\chi) < w + \eta_+\}$$

for every $w \in \mathbb{Z}$, and

$$\text{Perf}([C/G]) = \text{thick}\{\mathcal{O}_C(\chi) : W - \eta_- \leq \lambda(\chi) < W\}$$

for every $W \in \mathbb{Z}$. In particular, when $r \geq 0$ and $W = w + \eta_+$, the interval $[w + r, w + \eta_+)$ is the common remaining window used in the $r \geq 0$ wall crossing and it generates $\text{Perf}([C/G])$.

Proof. If $C = \emptyset$, the assertion is vacuous. Otherwise, Lemma 7.7 shows that the ideals $I_+ = (u_1, \dots, u_p)$ and $I_- = (v_1, \dots, v_q)$ generate the unit ideal on C .

By Theorem 5.4, the character lines $\mathcal{O}_C(\chi)$ generate $\text{Perf}([C/G])$. Since I_+ is the unit ideal on C , Lemma 6.3 applied to $u_1, \dots, u_p$ gives the plus-window statement. Since I_- is also the unit ideal on C , the same lemma applied to $v_1, \dots, v_q$ with $-\lambda$ gives generation by every interval $W - \eta_- \leq \lambda(\chi) < W$. $\square$

Lemma 7.16 (normal ideals are units on the actual sides). *If $\mathfrak{X}_+ = [U_+/G]$ is nonempty, then*

$$I_+ \mathcal{O}_{U_+} = (u_1, \dots, u_p) \mathcal{O}_{U_+} = \mathcal{O}_{U_+}.$$

If $\mathfrak{X}_- = [U_-/G]$ is nonempty, then

$$I_- \mathcal{O}_{U_-} = (v_1, \dots, v_q) \mathcal{O}_{U_-} = \mathcal{O}_{U_-}.$$

Consequently the Koszul complexes on $u_1, \dots, u_p$ over U_+ , and on $v_1, \dots, v_q$ over U_- , are exact complexes of equivariant vector bundles.

Proof. We prove the plus statement. If $U_+ \neq \emptyset$, then $p > 0$. Lemma 7.6 gives

$$U_+ = (V \cap D(I_+)) \cup C.$$

On $V \cap D(I_+)$, the ideal I_+ is the unit ideal by definition of $D(I_+)$. On C , Lemma 7.7 says that every coordinate stratum contains a positive normal variable; by the scheme-theoretic monomial-open criterion of Lemma 7.4, this is exactly the assertion $C \subset D(I_+)$, or $I_+ \mathcal{O}_C = \mathcal{O}_C$. In a boundary wall case $C = \emptyset$, so there is nothing to check on the second open. Hence I_+ is the unit ideal on both members of the two-open cover of U_+ , and therefore on U_+ . The minus statement is identical, using

$$U_- = (V \cap D(I_-)) \cup C$$

and the negative-variable part of Lemma 7.7. The exactness of the Koszul complexes follows from the standard algebraic fact that the Koszul complex of a finite sequence generating the unit ideal is exact; here the terms are equivariant vector bundles because the coordinates are homogeneous. $\square$

Lemma 7.17 (mapping descent for actual wall covers). *The covers in Lemma 7.6 give finite Čech descent for perfect complexes and mapping complexes on the quotient stacks. Consequently, isomorphisms and vanishing of mapping complexes may be proved by proving the corresponding statements on $V \cap U_{\pm}$, on C , and on their intersection $V \cap C$.*

Proof. Choose finite monomial generating sets for I_Z , I_+ , I_- , and the irrelevant ideals defining C . The covers

$$U_+ = (V \cap D(I_+)) \cup C, \quad U_- = (V \cap D(I_-)) \cup C$$

refine to finite covers by homogeneous principal monomial opens, and all finite intersections are again homogeneous principal monomial opens. Proposition 2.4 gives Čech descent for perfect complexes and computes mapping complexes by the same finite totalization. For $P, Q \in \text{Perf}([U_+/G])$, this gives

$$\mathbf{R} \text{Hom}_{[U_+/G]}(P, Q) \simeq \text{Tot}(\mathcal{H}_V \oplus \mathcal{H}_C \rightarrow \mathcal{H}_{VC}),$$

where

$$\mathcal{H}_V = \mathbf{R} \text{Hom}_{[V \cap U_+/G]}(P_V, Q_V),$$

$$\mathcal{H}_C = \mathbf{R} \text{Hom}_{[C/G]}(P_C, Q_C),$$

$$\mathcal{H}_{VC} = \mathbf{R} \text{Hom}_{[V \cap C/G]}(P_{VC}, Q_{VC}).$$

Thus full faithfulness and semiorthogonality can be checked on the two members of the cover and on their intersection: the totalization of a diagram of zero complexes is zero, and an isomorphism of such diagrams induces an isomorphism on totalizations. For the negative cover, replace $V \cap U_+$ by $V \cap U_-$ and $D(I_+)$ by $D(I_-)$. The finite monomial refinements and the Čech totalization formula are unchanged. This lemma is a mapping-complex descent statement. Generation below is proved directly from global character-line generators and Koszul triangles. $\square$

Lemma 7.18 (supported objects reduce to the residual open). *Let $P, Q \in \text{Perf}(\mathfrak{X}_+)$, and suppose that at least one of P, Q restricts to zero on the common open $[C/G]$. Then its restriction to $[V \cap C/G]$ is also zero, and the Čech formula for the cover*

$$U_+ = (V \cap U_+) \cup C$$

reduces to

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(P, Q) \simeq \mathbf{R} \text{Hom}_{[V \cap U_+/G]}(P|_{V \cap U_+}, Q|_{V \cap U_+}).$$

The same statement holds on the negative side with U_- in place of U_+ .

Proof. The restriction to $V \cap C$ is obtained by further restricting the zero object on C . In the notation of Lemma 7.17, both the C -term and the VC -term of the two-open mapping totalization are therefore zero. The totalization is

$$\text{Tot}(\mathcal{H}_{V \cap U_+} \oplus 0 \rightarrow 0) \simeq \mathcal{H}_{V \cap U_+},$$

which is the displayed residual-open mapping complex. The negative side is identical. $\square$

Corollary 7.19 (actual side window equivalences). *If $\mathfrak{X}_+ \neq \emptyset$, then for every $w \in \mathbb{Z}$ the restriction of the positive ambient window*

$$\mathcal{W}_+^w = \text{thick}\{\mathcal{O}_{[U_0/G]}(\chi) : w \leq \lambda(\chi) < w + \eta_+\}$$

to $\mathfrak{X}_+$ is an equivalence onto $\text{Perf}(\mathfrak{X}_+)$. If $\mathfrak{X}_- \neq \emptyset$, then for every $W \in \mathbb{Z}$ the restriction of the negative ambient window

$$\mathcal{W}_-^W = \text{thick}\{\mathcal{O}_{[U_0/G]}(\chi) : W - \eta_- \leq \lambda(\chi) < W\}$$

to $\mathfrak{X}_-$ is an equivalence onto $\text{Perf}(\mathfrak{X}_-)$.

Proof. Full faithfulness is Proposition 7.14. We prove essential surjectivity on the positive side. If $\mathfrak{X}_+ \neq \emptyset$, then $p > 0$, hence $\eta_+ > 0$; otherwise there is no positive normal coordinate and the positive side of the wall is empty by the support-cone description. By Theorem 5.4, the character lines generate $\text{Perf}([U_+/G])$. By Lemma 7.6,

$$U_+ = (V \cap D(I_+)) \cup C, \quad I_+ = (u_1, \dots, u_p).$$

By Lemma 7.16, the sections $u_1, \dots, u_p$ generate the unit ideal on all of U_+ , and their Koszul complex is exact on the quotient stack. Lemma 6.3 moves every character-line generator into the positive window. This generation is generation by restrictions of ambient window objects: every character line $\mathcal{O}_{[U_+/G]}(\chi)$ is the restriction of the character line $\mathcal{O}_{[U_0/G]}(\chi)$, because $U_+ \subset U_0$, and the Koszul complexes used to move weights are obtained by restricting the ambient Koszul complexes on the homogeneous functions u_i . Hence every character line on $\mathfrak{X}_+$ lies in the essential image of the restricted positive ambient window. Thus the positive window generates.

For the negative side, assume $\mathfrak{X}_- \neq \emptyset$. Then $q > 0$, and Lemma 7.6 gives

$$U_- = (V \cap D(I_-)) \cup C, \quad I_- = (v_1, \dots, v_q).$$

By Lemma 7.16, the sections $v_1, \dots, v_q$ generate the unit ideal on all of U_- , and their Koszul complex is exact on the quotient stack. Applying Lemma 6.3 to the homomorphism $-\lambda$ therefore moves all character-line generators into the negative window $[W - \eta_-, W)$. As on the positive side, each term is the restriction of the same ambient character line on U_0 , and the Koszul complexes are ambient homogeneous Koszul complexes restricted to U_- . This proves essential surjectivity of the negative window restriction. $\square$

Lemma 7.20 (actual supported locus). *Assume $\mathfrak{Z} \neq \emptyset$. Let $Y_+ \subset \mathfrak{X}_+$ be the closed locus $v_1 = \dots = v_q = 0$, with $Y_+ = \mathfrak{X}_+$ if $q = 0$. Then*

$$Y_+ \subset [V \cap U_+/G], \quad Y_+ \cap [C/G] = \emptyset.$$

Under the residual identification

$$[V/G] \simeq [\text{Tot}_3(E_+ \oplus E_-)/\mathbb{G}_m],$$

the substack Y_+ identifies with the residual supported locus

$$[(\text{Tot}_3(E_+) \setminus 0)/\mathbb{G}_m] \subset [V \cap U_+/G].$$

If $q > 0$, its ideal in $\mathfrak{X}_+$ is generated by the regular sequence $v_1, \dots, v_q$. The sign-reversed statement holds for Y_- .

Proof. If $q = 0$, then no support cone has negative image in the transverse quotient line. Thus the negative side is empty in this notation, so $C = U_+ \cap U_- = \emptyset$, and this is the positive boundary/empty negative-side case in the case table above; an interior two-sided wall has $p, q \geq 1$. Hence Lemma 7.6 gives $U_+ = V \cap D(I_+)$. Thus $Y_+ = \mathfrak{X}_+$ lies in $[V \cap U_+/G]$ and is disjoint from C .

Assume $q > 0$. If a stratum of U_+ satisfies $v_1 = \dots = v_q = 0$, it cannot lie in C , because Lemma 7.7 says that every stratum of C contains at least one negative-weight normal variable. The open identity

$$U_+ = (V \cap D(I_+)) \cup C$$

therefore places this stratum in $V \cap D(I_+) = V \cap U_+$. This proves the containment in $[V \cap U_+/G]$ and the disjointness from $[C/G]$. The disjointness is scheme-theoretic, not only set-theoretic: both C and the closed locus $v_1 = \dots = v_q = 0$ are built from coordinate monomial opens and coordinate closed subschemes, and Lemma 7.4 applies. Thus the absence of a coordinate stratum lying simultaneously in C and in the closed locus $v_1 = \dots = v_q = 0$ is equivalent to the pullback of Y_+ to $[C/G]$ being the empty quotient stack.

On V , Lemma 7.11 identifies the quotient with the residual vector-bundle model. The equations $v_j = 0$ set the negative-weight vector-bundle coordinates to zero, while the condition $D(I_+)$ is precisely that the positive coordinates are not all zero. Hence Y_+ is the displayed

residual supported locus. Finally, on every homogeneous monomial chart of $D(I_Z)$ and $D(I_+)$, the functions $v_1, \dots, v_q$ are part of a polynomial coordinate system. They are therefore a regular sequence degreewise in the graded coordinate ring, and regularity descends to the quotient stack. $\square$

Lemma 7.21 (Koszul model for the actual supported locus). *Assume $\mathfrak{Z} \neq \emptyset$. In the notation of Lemma 7.20, the closed immersion*

$$s_+ : Y_+ \hookrightarrow \mathfrak{X}_+$$

is the identity if $q = 0$, and is a regular closed immersion cut out by the homogeneous regular sequence $v_1, \dots, v_q$ if $q > 0$. For $F \in \text{Perf}(\mathfrak{Z})$ and $a \in \mathbb{Z}$, the object

$$s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a))$$

is perfect on $\mathfrak{X}_+$. If $F = L_\alpha$ is a center character line and $\chi \in A$ is an ambient lift with residual weight $\lambda(\chi) = a$, then this pushforward is represented by the Koszul complex on $v_1, \dots, v_q$, tensored by $\mathcal{O}_{\mathfrak{X}_+}(\chi)$. Its terms are ambient character lines whose residual weights are

$$a + b_I, \quad I \subseteq \{1, \dots, q\}, \quad b_I = \sum_{j \in I} b_j.$$

The sign-reversed statement holds for Y_- .

Proof. The case $q = 0$ is immediate, since then $Y_+ = \mathfrak{X}_+$ and s_+ is the identity. Assume $q > 0$. Work on a homogeneous monomial chart

$$D(z^K) \cap D(u_i) \subset V \cap U_+.$$

The coordinate algebra is a homogeneous localization of the polynomial algebra in the zero-weight variables z , positive normal variables u , and negative normal variables v . On this chart, Y_+ is defined by the ideal $(v_1, \dots, v_q)$. These variables are part of a polynomial coordinate system, hence form a regular sequence in the localized graded ring. The associated Koszul complex is a finite complex of graded free modules resolving the quotient by $(v_1, \dots, v_q)$. The construction is homogeneous and compatible with further monomial localization, so the local Koszul complexes glue through the finite graded Čech descent of Proposition 2.8. Passing to the quotient by the split diagonalizable group preserves this description because the quotient category is the graded derived category.

Thus s_+ has finite Tor-amplitude and $s_{+*}\mathcal{O}_{Y_+}$ is represented locally by a finite complex of vector bundles. Since pullback of perfect objects is perfect and tensoring with a character line is invertible, $s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a))$ is perfect for every $F \in \text{Perf}(\mathfrak{Z})$. For $F = L_\alpha$, choose an ambient lift χ . Tensoring the Koszul resolution by $\mathcal{O}_{\mathfrak{X}_+}(\chi)$ gives the stated complex. Each coordinate v_j has residual weight $-b_j$; with the shift convention of Lemma 6.11, the source term involving the subset I has residual weight $a + b_I$. The negative statement is obtained by interchanging the u - and v -variables and replacing λ by $-\lambda$. $\square$

Lemma 7.22 (empty-center actual peeling). *Assume $r = \eta_+ - \eta_- \geq 0$ and $\mathfrak{Z} = \emptyset$. Put*

$$\mathcal{T}_a^+ = \text{thick}\{\mathcal{O}_{\mathfrak{X}_+}(\chi) : a \leq \lambda(\chi) < w + \eta_+\}.$$

Then, for $w \leq a < w + r$,

$$\mathcal{T}_a^+ = \mathcal{T}_{a+1}^+.$$

Equivalently, the supported peeling identity holds with the center block omitted. For $r < 0$, the sign-reversed statement holds on $\mathfrak{X}_-$.

Proof. Since $\mathfrak{Z} = \emptyset$, the zero-weight semistable locus $D(I_Z)$ is empty, hence $V = \emptyset$. Lemma 7.6 therefore gives $U_+ = C$. If $U_+ = \emptyset$, then $\mathcal{T}_a^+ = \mathcal{T}_{a+1}^+ = 0$, so assume $U_+ \neq \emptyset$.

If $q = 0$, there is no negative normal coordinate. No support cone has negative image in the transverse quotient line, so the negative side is empty and $C = U_+ \cap U_- = \emptyset$. This contradicts $U_+ = C$ unless $U_+ = \emptyset$, already handled.

Thus $q > 0$. Since $U_+ = C$, every coordinate stratum of U_+ lies in the common open. In the plus-target orientation, the one-sided boundary case has $q = 0$; hence $q > 0$ is the interior-wall case, and Lemma 7.7 applies. Every coordinate stratum of U_+ contains at least one negative-weight normal coordinate, so the ideal $I_- = (v_1, \dots, v_q)$ is the unit ideal on U_+ . The Koszul complex on $v_1, \dots, v_q$ is therefore exact on $\mathfrak{X}_+$.

Tensor this exact Koszul complex by a character line $\mathcal{O}_{\mathfrak{X}_+}(\chi)$ of λ -weight a . Its non-augmentation terms have weights

$$a + b_I, \quad \emptyset \neq I \subseteq \{1, \dots, q\}.$$

This is the specialization of Lemma 6.11 to coordinates of weights $-b_j$. For $w \leq a < w + r$,

$$a < a + b_I \leq a + \eta_- \leq w + r - 1 + \eta_- = w + \eta_+ - 1.$$

Thus every weight a character line lies in $\mathcal{T}_{a+1}^+$. By definition, $\mathcal{T}_a^+$ is generated by $\mathcal{T}_{a+1}^+$ and the weight a character lines, so $\mathcal{T}_a^+ = \mathcal{T}_{a+1}^+$. $\square$

Lemma 7.23 (actual supported peeling). *Assume $r = \eta_+ - \eta_- \geq 0$. Put*

$$\mathcal{T}_a^+ = \text{thick}\{\mathcal{O}_{\mathfrak{X}_+}(\chi) : a \leq \lambda(\chi) < w + \eta_+\}.$$

For $w \leq a < w + r$, the following hold.

(a) *If $\mathfrak{Z} = \emptyset$, then*

$$\mathcal{T}_a^+ = \mathcal{T}_{a+1}^+.$$

(b) *If $\mathfrak{Z} \neq \emptyset$, let*

$$\beta_a^+(F) = s_{+*}(q_+^* F \otimes \mathcal{O}_{Y_+}(a)), \quad F \in \text{Perf}(\mathfrak{Z}),$$

where $Y_+ \hookrightarrow \mathfrak{X}_+$ is the closed locus $v_1 = \dots = v_q = 0$ and $q_+ : Y_+ \rightarrow \mathfrak{Z}$ is the natural weighted projective morphism, as in Lemma 7.20. If $q = 0$, this means $Y_+ = \mathfrak{X}_+$. Here $\mathcal{O}_{Y_+}(a)$ denotes the residual character line of $\mathbb{G}_m^\lambda$ -weight a under the residual model of Lemma 7.11; for a full A -character χ , the corresponding residual weight is $\lambda(\chi)$.

$$\mathcal{T}_a^+ = \langle \beta_a^+ \text{Perf}(\mathfrak{Z}), \mathcal{T}_{a+1}^+ \rangle.$$

For $r < 0$, the sign-reversed statement holds on $\mathfrak{X}_-$.

Proof. If $\mathfrak{Z} = \emptyset$, this is Lemma 7.22. Assume from now on that $\mathfrak{Z} \neq \emptyset$. The functor β_a^+ is intrinsic. The residual character line $\mathcal{O}_{Y_+}(a)$ is defined by the quotient character

$$A/A_0 \simeq \mathbb{Z}$$

given by λ . A choice of a splitting $A = A_0 \oplus \mathbb{Z}\tau$ is used only to write an ambient character of weight a as $\alpha + a\tau$; changing τ twists the center character α by another character of A_0 , and Lemma 7.13 identifies the resulting functor. Consequently the generation argument below may be checked on center character lines, because Theorem 5.4 thickly generates $\text{Perf}(\mathfrak{Z})$ by such lines. By Lemma 7.20, the supported locus Y_+ is contained in $V \cap U_+$, is disjoint from C , and agrees there with the residual supported locus. Lemma 7.21 gives the regular closed immersion and finite homogeneous Koszul resolution; in particular s_{+*} sends the displayed supported objects to perfect objects. Pullback q_+^* preserves perfect complexes, and Lemma 7.13 shows that this definition is independent of the auxiliary splitting used later for weight bookkeeping. Thus β_a^+ is a well-defined exact functor on $\text{Perf}(\mathfrak{Z})$.

We now spell out the actual-to-residual reduction used for full faithfulness and semiorthogonality.

- (i) A generator $\mathcal{O}_{\mathfrak{X}_+}(\chi)$ of $\mathcal{T}_{a+1}^+$ has residual weight $\lambda(\chi)$ satisfying $a+1 \leq \lambda(\chi) < w + \eta_+$. On the residual open $[V \cap U_+/G]$ it restricts to the corresponding residual character line, hence lies in the residual window subcategory with the same interval used in Lemma 6.14.
- (ii) The supported block $\beta_a^+(F)$ restricts to the residual supported block on $[V \cap U_+/G]$. On the common open $[C/G]$ it restricts to zero, scheme-theoretically, by Lemma 7.20.

- (iii) Therefore every mapping complex whose source or target is a supported block is computed by the residual open: the C -term and the $V \cap C$ -term in the two-open Čech totalization vanish by Lemma 7.18.

After these three checks, the relevant actual Hom complexes are exactly the residual Hom complexes in the same direction as in Lemma 6.14. Concretely, for $F, G \in \text{Perf}(\mathfrak{Z})$,

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(\beta_a^+ F, \beta_a^+ G) \simeq \mathbf{R} \text{Hom}_{\mathfrak{X}_+^{\text{res}}}(\beta_a^{+, \text{res}} F, \beta_a^{+, \text{res}} G),$$

where $\mathfrak{X}_+^{\text{res}} = [V \cap U_+/G]$ and $\beta_a^{+, \text{res}}$ is the supported block functor in the residual model. This is the block–block Hom used for full faithfulness. If $\mathcal{O}_{\mathfrak{X}_+}(\chi)$ is a character-line generator of $\mathcal{T}_{a+1}^+$, then

$$\mathbf{R} \text{Hom}_{\mathfrak{X}_+}(\mathcal{O}_{\mathfrak{X}_+}(\chi), \beta_a^+ F) \simeq \mathbf{R} \text{Hom}_{\mathfrak{X}_+^{\text{res}}}(\mathcal{O}_{\mathfrak{X}_+^{\text{res}}}(\chi), \beta_a^{+, \text{res}} F).$$

This is the later-window-to-block direction $\mathcal{T}_{a+1} \perp \beta_a$, not the opposite Hom direction. The weighted-projective pushforward vanishing in Lemma 6.14 is applied to this residual Hom, so it gives semiorthogonality from $\mathcal{T}_{a+1}^+$ to $\beta_a^+ \text{Perf}(\mathfrak{Z})$. The remaining assertion, $\beta_a^+ \text{Perf}(\mathfrak{Z}) \subset \mathcal{T}_a^+$, is a generation/containment statement and is proved below from the actual Koszul resolution, not from a Hom vanishing statement.

We next prove that the supported block is actually contained in $\mathcal{T}_a^+$. By Theorem 5.4, it is enough to check $F = L_\alpha$ for center character lines. Indeed, the category $\text{Perf}(\mathfrak{Z})$ is thickly generated by those lines, β_a^+ is exact, and Lemma 2.15 applies to the thick subcategory $\mathcal{T}_a^+ \subset \text{Perf}(\mathfrak{X}_+)$. Choose $\tau \in A$ with $\lambda(\tau) = 1$, and write the corresponding ambient character as $\chi = \alpha + a\tau$. By Lemma 7.12, the residual line $L_\alpha \otimes \mathcal{O}(a)$ is represented by this ambient lift, and a different lift would only twist L_α by another center character. By Lemma 7.21, the Koszul resolution of $\beta_a^+(L_\alpha)$ has terms that are character lines with λ -weights

$$a + b_I, \quad I \subseteq \{1, \dots, q\}.$$

The sign is the one fixed in Lemma 6.11: each coordinate v_j has residual weight $-b_j$, so the corresponding Koszul source term has weight shifted by $+b_j$. If $q = 0$ this is only the weight a term. If $q > 0$, then for all I

$$a \leq a + b_I \leq a + \eta_- \leq w + r - 1 + \eta_- = w + \eta_+ - 1.$$

Hence every term of the Koszul resolution lies in $\mathcal{T}_a^+$, and $\beta_a^+(L_\alpha) \in \mathcal{T}_a^+$. Thus $\beta_a^+ \text{Perf}(\mathfrak{Z}) \subset \mathcal{T}_a^+$.

It remains to prove generation of $\mathcal{T}_a^+$. The category $\mathcal{T}_a^+$ is generated by $\mathcal{T}_{a+1}^+$ and the character lines $\mathcal{O}_{\mathfrak{X}_+}(\chi)$ with $\lambda(\chi) = a$. Choose $\tau \in A$ with $\lambda(\tau) = 1$, and write $\chi = \alpha + a\tau$, $\alpha \in A_0$. Let L_α be the corresponding character line on $\mathfrak{Z}$. Lemma 7.12 identifies this ambient lift with the residual weight- a line attached to L_α . By Lemma 7.21, the Koszul complex resolving $\beta_a^+(L_\alpha)$ has first ambient term $\mathcal{O}_{\mathfrak{X}_+}(\chi)$; all other ambient terms have weights

$$a + b_I, \quad \emptyset \neq I \subseteq \{1, \dots, q\}.$$

Again this follows from Lemma 6.11. If $q = 0$, there are no other terms and $\beta_a^+(L_\alpha) = \mathcal{O}_{\mathfrak{X}_+}(\chi)$. If $q > 0$, then

$$a < a + b_I \leq a + \eta_- \leq w + r - 1 + \eta_- = w + \eta_+ - 1,$$

so every other term lies in $\mathcal{T}_{a+1}^+$. Hence each weight a character line lies in $\langle \beta_a^+ \text{Perf}(\mathfrak{Z}), \mathcal{T}_{a+1}^+ \rangle$, proving the claimed decomposition of $\mathcal{T}_a^+$. The negative case is the same argument with u and v interchanged and λ replaced by $-\lambda$. $\square$

Lemma 7.24 (remaining window as source image). *Assume $r = \eta_+ - \eta_- \geq 0$ and that both actual sides $\mathfrak{X}_+$ and $\mathfrak{X}_-$ are nonempty. Put $W = w + \eta_+$ and let*

$$\mathcal{W}_{\text{rem}} = \text{thick}\{\mathcal{O}_{[U_0/G]}(\chi) : w + r \leq \lambda(\chi) < w + \eta_+\}.$$

Then restriction to $\mathfrak{X}_-$ identifies $\mathcal{W}_{\text{rem}}$ with $\text{Perf}(\mathfrak{X}_-)$, and restriction to $\mathfrak{X}_+$ is fully faithful. The composite

$$\text{Perf}(\mathfrak{X}_-) \xleftarrow[\sim]{\text{res}_-} \mathcal{W}_{\text{rem}} \xrightarrow{\text{res}_+} \text{Perf}(\mathfrak{X}_+)$$

has image exactly the remaining positive window

$$\mathcal{T}_{w+r}^+ = \text{thick}\{\mathcal{O}_{\mathfrak{X}_+}(\chi) : w+r \leq \lambda(\chi) < w+\eta_+\}.$$

On the common open C , and on the residual open V , the two side restrictions are the restrictions of the same ambient character lines in $\mathcal{W}_{\text{rem}}$.

Proof. The interval

$$[w+r, w+\eta_+) = [w+\eta_+ - \eta_-, w+\eta_+) = [W - \eta_-, W)$$

is the negative window with upper endpoint W . By Corollary 7.19, restriction of this ambient window to $\mathfrak{X}_-$ is an equivalence. The same interval is contained in the positive window $[w, w+\eta_+)$, and Proposition 7.14 gives full faithfulness of restriction to $\mathfrak{X}_+$ on the positive window, hence on $\mathcal{W}_{\text{rem}}$. Its image is exactly the thick subcategory generated by the restrictions of the displayed ambient character lines, which is the definition of $\mathcal{T}_{w+r}^+$.

The compatibility on C and V is not an additional descent assertion: an object of $\mathcal{W}_{\text{rem}}$ is built from ambient character lines on $[U_0/G]$, and all restrictions in the square

$$\begin{array}{ccc} V, & C, & V \cap C, \\ & & U_{\pm} \end{array}$$

are ordinary restrictions of those same character lines. On V , the identification with the residual model is the one in Lemma 7.11 and Lemma 7.12; on C , supported blocks restrict to zero and only the common ambient window remains. $\square$

Lemma 7.25 (*S*-linearity of wall functors). *All functors used in the residual and actual wall decompositions are $\text{Perf}(S)$ -module functors. More precisely:*

- (i) *character-line tensor functors, open restrictions, pullbacks along the projections to wall centers, and regular closed-immersion pushforwards are $\text{Perf}(S)$ -linear;*
- (ii) *the residual and actual window subcategories are stable under the $\text{Perf}(S)$ -action;*
- (iii) *a window restriction equivalence has a $\text{Perf}(S)$ -linear inverse;*
- (iv) *the transported wall functors $\Phi_{\rightarrow+}$ and the supported block functors $\beta_a^{\pm}$ are $\text{Perf}(S)$ -linear.*

Proof. All quotient stacks in the wall arguments are stacks over $S = \text{Spec } \mathbb{Z}$, and their $\text{Perf}(S)$ -module structures are given by tensoring with pullbacks from S . Tensoring by a character line commutes with such pullbacks. Open restriction and ordinary pullback commute with pullback from S by functoriality. For the regular closed immersions appearing in the supported blocks, pushforward is $\text{Perf}(S)$ -linear by the projection formula; locally this is the projection formula for the finite Koszul resolution of the regular sequence. The projections to the center are morphisms over S , so their pullbacks are also $\text{Perf}(S)$ -linear.

The window categories are thick subcategories generated by character lines and by pullbacks of perfect objects from the center. Tensoring with an object of $\text{Perf}(S)$ preserves these thick subcategories because the base object has residual and wall weight zero. Hence the window restrictions are functors between $\text{Perf}(S)$ -module subcategories.

Let $T : \mathcal{W} \rightarrow \mathcal{C}$ be any of the window restriction equivalences. The functor T is $\text{Perf}(S)$ -linear by the preceding paragraph. Its inverse is $\text{Perf}(S)$ -linear as well: for $M \in \text{Perf}(S)$ and $C \in \mathcal{C}$, apply T to the canonical comparison

$$T^{-1}(M \otimes C) \longrightarrow M \otimes T^{-1}(C).$$

After applying T , this comparison becomes the identity comparison

$$M \otimes C \longrightarrow M \otimes C,$$

and T is conservative. Thus the comparison is an equivalence. The wall functors are composites of these window equivalences, their inverses, restrictions, pullbacks, character-line twists, and supported closed-immersion pushforwards; hence they are $\text{Perf}(S)$ -linear. $\square$

The actual wall proposition below uses the preceding lemmas in the following strict order.

- (1) *Open comparison.* Lemma 7.6 gives the two-open covers $U_0 = V \cup C$ and $U_{\pm} = (V \cap D(I_{\pm})) \cup C$, and Lemma 7.7 identifies the common intersections scheme-theoretically by monomial radical equality.
- (2) *Hom computation.* Lemma 7.9 compares the two finite Čech mapping diagrams. The only changed term is $V \rightarrow V \cap D(I_{\pm})$, so the cone is the homogeneous local-cohomology cone controlled by Lemma 6.2.
- (3) *Window full faithfulness.* Proposition 7.14 converts that Hom computation into full faithfulness on the relevant windows; this is a mapping-complex statement, not a generation argument.
- (4) *Window generation.* Corollary 7.19 uses global character-line generation and Koszul triangles for the unit ideals $I_{\pm}$ on the actual sides to prove essential surjectivity of the window restrictions.
- (5) *Supported blocks.* Lemmas 7.20 and 7.23 show that the supported block functors are fully faithful, semiorthogonal to later weights, and peel off one residual weight at a time.

Thus Proposition 7.26 only assembles already-proved full faithfulness, semiorthogonality, and generation statements; it does not use a descent argument for generation by right orthogonals.

For reference, in the $r \geq 0$ orientation the categories and functors assembled in the proposition are the following. The ambient wall category is

$$\mathcal{A}_0 = \text{Perf}([U_0/G]).$$

Inside it we use the positive window

$$\mathcal{W}_+^w = \text{thick}\{\mathcal{O}_{[U_0/G]}(\chi) : w \leq \lambda(\chi) < w + \eta_+\},$$

and, if $\mathfrak{X}_- \neq \emptyset$, the remaining window

$$\mathcal{W}_{\text{rem}} = \text{thick}\{\mathcal{O}_{[U_0/G]}(\chi) : w + r \leq \lambda(\chi) < w + \eta_+\}.$$

The restriction functors are

$$\text{Perf}(\mathfrak{X}_-) \xleftarrow[\sim]{\text{res}_-} \mathcal{W}_{\text{rem}} \xrightarrow{\text{res}_+} \text{Perf}(\mathfrak{X}_+),$$

where res_- is an equivalence by the negative window theorem and res_+ is fully faithful by actual window full faithfulness. The transported source functor is

$$\Phi_{-\rightarrow+} = \text{res}_+ \circ (\text{res}_-)^{-1} : \text{Perf}(\mathfrak{X}_-) \longrightarrow \text{Perf}(\mathfrak{X}_+),$$

with the convention that it is the zero functor if $\mathfrak{X}_- = \emptyset$. For each block weight $a = w, \dots, w + r - 1$, the supported block functor is

$$\beta_a^+ : \text{Perf}(\mathfrak{Z}) \longrightarrow \text{Perf}(\mathfrak{X}_+),$$

defined by the regular closed-immersion pushforward from the actual supported locus $Y_+ \subset \mathfrak{X}_+$, after identifying Y_+ with the residual weighted-projective bundle over $\mathfrak{Z}$. If $\mathfrak{Z} = \emptyset$, this functor has zero domain and the block is omitted. Finally,

$$\mathcal{T}_a^+ = \text{thick}\{\mathcal{O}_{\mathfrak{X}_+}(\chi) : a \leq \lambda(\chi) < w + \eta_+\}$$

is a subcategory of $\text{Perf}(\mathfrak{X}_+)$, not of the ambient wall category. Lemma 7.23 is precisely the assertion

$$\mathcal{T}_a^+ = \langle \beta_a^+ \text{Perf}(\mathfrak{Z}), \mathcal{T}_{a+1}^+ \rangle.$$

The boundary and empty cases used in the proposition can be checked from the following case checklist. It is a summary of the radical identities and conventions proved above, not an additional input. The entries are written in the $r \geq 0$ orientation; the $r < 0$ case is obtained by reversing the signs.

- (a) *Interior wall, nonempty center.* Here $V = D(I_Z B)$, $C = U_+ \cap U_-$, and $U_+ = (V \cap D(I_+)) \cup C$. The remaining window is $\mathcal{T}_{w+r}^+ = \Phi_{-\rightarrow+} \text{Perf}(\mathfrak{X}_-)$, and the supported blocks are $\beta_a^+ \text{Perf}(\mathfrak{Z})$ for $w \leq a < w + r$.

- (b) *Interior wall, empty center.* Here $V = \emptyset$, $C = U_0$, and $U_+ = C$ when the positive side is nonempty. The remaining window is killed by Čech conservativity, and all center blocks are omitted.
- (c) *Positive boundary wall.* Here $C = \emptyset$, $U_0 = V$, $U_+ = V \cap D(I_+)$, and $\mathfrak{X}_- = \emptyset$. In the boundary case used by the global induction one has $q = 0$, so the remaining interval is empty. The one-sided center blocks are included when $\mathfrak{Z} \neq \emptyset$ and omitted when $\mathfrak{Z} = \emptyset$.
- (d) *The case $r = 0$.* The block interval $w \leq a < w + r$ is empty, and $\mathcal{T}_w^+$ is the source image. The wall step is an equivalence onto the target side.

Proposition 7.26 (actual wall crossing). *Assume every nonempty adjacent side is a split smooth proper quotient stack over $\mathbb{Z}$ with finite split diagonalizable stabilizers, and assume that the corrected center $\mathfrak{Z}$, if nonempty, has the same properties. Empty sides and empty centers are interpreted as zero categories and carry no geometric hypothesis; supported blocks attached to an empty center are zero and are omitted. Let $r = \eta_+ - \eta_-$. If $r \geq 0$, and $\mathfrak{X}_+ \neq \emptyset$, then*

$$\mathrm{Perf}(\mathfrak{X}_+) = \langle \mathcal{B}_w^+, \dots, \mathcal{B}_{w+r-1}^+, \Phi_{-\rightarrow+} \mathrm{Perf}(\mathfrak{X}_-) \rangle,$$

with empty block list when $r = 0$. Each displayed block is the image of $\mathrm{Perf}(\mathfrak{Z})$; if $\mathfrak{Z} = \emptyset$ this image is zero, and otherwise it is equivalent to $\mathrm{Perf}(\mathfrak{Z})$. If $r \geq 0$ and $\mathfrak{X}_+ = \emptyset$, the proposition asserts only $\mathrm{Perf}(\mathfrak{X}_+) = 0$. If $r < 0$, the sign-reversed assertion holds on $\mathfrak{X}_-$, again only as a displayed decomposition when the target side is nonempty.

Remark 7.27 (case-by-case reading of Proposition 7.26). For checking boundary and empty-center cases, the proposition should be read as the following three statements.

- (i) In an interior wall with $\mathfrak{Z} \neq \emptyset$, the supported blocks are the images of $\mathrm{Perf}(\mathfrak{Z})$ under the functors $\beta_a^\pm$, and the remaining window is identified with the source side by Lemma 7.24.
- (ii) In an interior wall with $\mathfrak{Z} = \emptyset$, $V = \emptyset$ and all supported center blocks are omitted. The peeling step is Lemma 7.22; no recursive center category is invoked.
- (iii) In a one-sided boundary wall, the outside side is the zero category and $C = \emptyset$. The nonempty side is handled by the same actual comparison, and Corollary 7.28 records the boundary crossing used in the global induction.

When $r = 0$, the block interval is empty and Lemma 7.24 identifies the remaining window with the source image. Thus an $r = 0$ wall step is an equivalence and introduces no center blocks.

Proof. Write $\mathfrak{X}_\pm = [U_\pm/G]$, and assume first that $r \geq 0$. If $\mathfrak{X}_+ = \emptyset$, then $\mathrm{Perf}(\mathfrak{X}_+) = 0$, which is the asserted zero-category case. Thus suppose the target side is nonempty. We follow the proof map in Remark 7.1: the open identities and residual comparison have already been proved in the preceding lemmas, and the present argument only assembles the resulting window and supported-block statements. The proof uses two different mechanisms. Full faithfulness and semiorthogonality are mapping-complex statements, checked by the residual Hom calculations and finite Čech descent. Generation comes from the actual side window equivalence and the supported peeling lemma, both ultimately using character-line generation and Koszul triangles. Corollary 7.19 gives

$$\mathrm{Perf}(\mathfrak{X}_+) = \mathcal{T}_w^+ = \mathrm{thick}\{\mathcal{O}_{\mathfrak{X}_+}(\chi) : w \leq \lambda(\chi) < w + \eta_+\}.$$

Iterating Lemma 7.23 for $a = w, \dots, w + r - 1$ gives

$$\mathrm{Perf}(\mathfrak{X}_+) = \langle \mathcal{B}_w^+, \dots, \mathcal{B}_{w+r-1}^+, \mathcal{T}_{w+r}^+ \rangle,$$

with no supported blocks when $r = 0$. The remaining interval is

$$[w + r, w + \eta_+) = [w + \eta_+ - \eta_-, w + \eta_+),$$

the negative window with upper endpoint $w + \eta_+$.

If $\mathfrak{X}_- \neq \emptyset$, Corollary 7.19 identifies this negative window with $\text{Perf}(\mathfrak{X}_-)$. Equivalently, Lemma 7.24 defines $\Phi_{-\rightarrow+}$ as the inverse of the negative window equivalence followed by restriction to $\mathfrak{X}_+$, and identifies

$$\mathcal{T}_{w+r}^+ = \Phi_{-\rightarrow+} \text{Perf}(\mathfrak{X}_-).$$

If $\mathfrak{X}_- = \emptyset$, then $C \subset U_-$ is empty and $V \cap U_- = \emptyset$. The restriction of $\mathcal{T}_{w+r}^+$ to $[V \cap U_+/G]$ is exactly the remaining residual window

$$[w+r, w+\eta_+) = [w+\eta_+ - \eta_-, w+\eta_+).$$

In the directed boundary crossing used in the global induction, this is the positive boundary case: $q = 0$, $\eta_- = 0$, and the displayed interval is empty. The following argument is the uniform convention which also covers interior empty-source notation. If $V = \emptyset$, this restriction is already zero. If $V \neq \emptyset$, then the residual center is nonempty and, since $V \cap U_- = \emptyset$, the residual wall theorem identifies this remaining window with $\text{Perf}([V \cap U_-/G]) = 0$. Its restriction to C is zero because $C = \emptyset$. The finite monomial cover in Lemma 7.6 is conservative by Proposition 2.4, so $\mathcal{T}_{w+r}^+ = 0$. This is object-level Čech conservativity for perfect complexes, not a descent argument for generation by a right orthogonal. It is the same as the image of the unique functor from the zero source category.

Thus the displayed sequence is a semiorthogonal decomposition for $r \geq 0$. If $r < 0$, set $\lambda' = -\lambda$. Then the old negative variables are the λ' -positive variables, the old positive variables are the λ' -negative variables, and the new integer $r' = \eta_- - \eta_+$ is nonnegative. Applying the $r' \geq 0$ case just proved to λ' gives the same construction with $+$ and $-$ interchanged, which is precisely the asserted wall crossing in the original orientation. All functors in the displayed decomposition are $\text{Perf}(\mathbb{Z})$ -linear by Lemma 7.25. $\square$

Corollary 7.28 (one-sided boundary actual crossing). *In the situation and under the geometric hypotheses of Proposition 7.26, let F be a codimension-one boundary wall face of the GIT fan contained in ∂Eff , let θ_{in} be the adjacent full-dimensional chamber inside Eff , and let the other side be a nearby outside point, so its semistable locus is empty. Orient the primitive normal λ so that $\lambda > 0$ on θ_{in} . Then the $r \geq 0$ wall crossing applies with $\mathfrak{X}_+ = \mathfrak{X}_{\theta_{\text{in}}}$ and $\mathfrak{X}_- = \emptyset$:*

$$\text{Perf}(\mathfrak{X}_{\theta_{\text{in}}}) = \langle \mathcal{B}_w^+, \dots, \mathcal{B}_{w+r-1}^+ \rangle,$$

with empty blocks omitted.

Proof. Since F is a boundary wall face of the GIT fan contained in ∂Eff , and λ is positive on the inside side, all Cox degrees have nonnegative λ -weight and at least one has positive weight. Hence $q = 0$, $\eta_- = 0$, and the outside side has empty semistable locus by Lemma 2.17. The boundary case of Lemma 7.6 gives

$$U_0 = V, \quad U_+ = V \cap D(I_+), \quad U_- = \emptyset.$$

Thus the actual wall theorem is precisely the one-sided residual crossing with zero source, glued through the same actual wall comparison. Applying Proposition 7.26 gives the displayed decomposition. In the global induction, the hypotheses on the nonempty inside chamber are supplied by Lemma 2.29, and the hypotheses on the corrected center are supplied by Proposition 3.2. $\square$

8. DIRECTED GLOBAL INDUCTION

Let

$$\mathfrak{X}_\theta = [U_\theta/D(A)]$$

be a Cox chamber quotient with A free and $B_0 = \mathbb{Z}$. Write

$$\text{Eff} = \sum_i \mathbb{Q}_{\geq 0} \deg(x_i) \subset A_{\mathbb{Q}}.$$

The induction protocol is as follows. Choose a generic segment from a rational outside point to the target chamber. On an open interval outside Eff the category is zero. When the

segment crosses a boundary wall from the outside into Eff , use the one-sided boundary actual crossing, whose nonempty chamber hypothesis is supplied by Lemma 2.29 and whose center hypothesis is supplied by Proposition 3.2. When the segment crosses an interior wall, use Proposition 7.26; its adjacent nonempty chamber quotients are proper by Lemma 2.29, and its centers are in the smaller inductive class by Propositions 3.2 and 4.2. For two-sided centers the relative dimension drops; for one-sided centers with one normal variable the dimension may stay the same but the rational character rank drops. The wall orientation lemma below ensures that the already-constructed side is always the source side of the $r \geq 0$ wall theorem.

Geometrically, the vector

$$\kappa = \sum_i \deg(x_i)$$

records the total normal weight of the Cox coordinates at every wall: $\lambda(\kappa) = \eta_+ - \eta_-$. The directed segment is chosen with direction d in the interior of Eff , and close enough to κ with respect to the finite wall arrangement that every wall seeing a nonzero value of $\lambda(\kappa)$ is crossed with the same sign. Removing the finitely many wall hyperplanes, codimension-at-least-two spans, and the line through the target leaves a nonempty rational open subset of $\text{int}(\text{Eff})$; this is why a rational d with all required genericity conditions exists. Boundary crossings are exactly the steps from the outside empty interval into the effective cone, while interior crossings occur between adjacent chambers already inside Eff . Thus the sign convention is not chosen independently at each wall. Once the global direction d is fixed, a wall normal λ is oriented so that $\lambda(d) > 0$ on the next chamber met by the segment. Because d was chosen to have the same wall signs as κ , one has

$$r = \eta_+ - \eta_- = \lambda(\kappa) \geq 0$$

for every nontrivial crossing in the directed induction. In a two-dimensional secondary fan picture, this says that the segment always crosses a wall in the direction in which the total Cox-coordinate weight lies on the target side; the $r \geq 0$ wall theorem is therefore always applied with the already constructed chamber as the source.

Lemma 8.1 (generic outside segment). *Assume $\text{rk } A_{\mathbb{Q}} \geq 2$, $B_0 = \mathbb{Z}$, and χ lies in a full-dimensional support chamber. Put*

$$\kappa = \sum_i \deg x_i.$$

There are a rational vector $d \in A_{\mathbb{Q}}$ and a rational number $T > 0$ such that

$$\theta_{\text{out}} = \chi - Td \notin \text{Eff},$$

the segment $[\theta_{\text{out}}, \chi]$ crosses only codimension-one walls and crosses them transversely, neither endpoint lies on a wall, the segment avoids the origin, and after multiplying θ_{out} by a positive integer it is an integral character defining the same empty semistable locus. For every crossed wall normal λ ,

$$\lambda(\kappa) \neq 0 \implies \text{sign } \lambda(d) = \text{sign } \lambda(\kappa).$$

More strongly, d is chosen before the segment is drawn: for every codimension-one wall cell W of the support-cone decomposition and every rational equation $\alpha_W = 0$ for its span, the signs of $\alpha_W(d)$ and $\alpha_W(\kappa)$ agree whenever $\alpha_W(\kappa) \neq 0$. The displayed assertion for crossed primitive normals is the corresponding specialization.

Proof. The cone Eff is a closed rational polyhedral cone. It is strongly convex because $B_0 = \mathbb{Z}$: if $v, -v \in \text{Eff}$, then after clearing denominators a nontrivial nonnegative relation among the degrees would give a nonconstant degree-zero monomial. The target chamber is full-dimensional, so Eff spans $A_{\mathbb{Q}}$. For every facet functional of Eff , at least one Cox degree has positive value and all have nonnegative value; hence κ lies in the interior of Eff .

Let $\mathcal{W}$ be the finite set of codimension-one wall cells, and for each W choose a rational linear form α_W whose kernel is the linear span of W . The set

$$\text{int}(\text{Eff}) \cap \bigcap_{\alpha_W(\kappa) > 0} \{\alpha_W > 0\} \cap \bigcap_{\alpha_W(\kappa) < 0} \{\alpha_W < 0\}$$

is a nonempty open neighborhood of κ . Remove from it the finitely many rational proper subspaces consisting of the wall hyperplanes, the spans of χ together with each codimension at least two cell, and the line $\mathbb{Q}\chi$. Rational points are dense and a finite union of proper subspaces has empty interior, so we may choose a rational

$$d \in \text{int}(\text{Eff})$$

outside all these subspaces and preserving the displayed signs.

The affine line $\chi - \mathbb{R}d$ is therefore not contained in any wall hyperplane and meets wall hyperplanes transversely. Let E be a codimension-at-least-two cell and let $L_E = \text{span}_{\mathbb{Q}}(E)$. Since E has codimension at least two in $A_{\mathbb{Q}}$, the rational subspace L_E has dimension at most $\rho - 2$, where $\rho = \text{rk } A_{\mathbb{Q}}$. The target χ lies in a full-dimensional chamber, so $\chi \notin L_E$; therefore $\text{span}_{\mathbb{Q}}(\chi, L_E)$ has dimension at most $\rho - 1$ and is one of the proper rational subspaces removed above. If the line $\chi - \mathbb{R}d$ met L_E , then $t \neq 0$ and for some $t \in \mathbb{R}$ one would have $\chi - td \in L_E$. Hence

$$d = t^{-1}\chi - t^{-1}(\chi - td) \in \text{span}_{\mathbb{R}}(\chi, L_E),$$

contradicting the choice of d . Thus the line misses L_E , and therefore misses the cell E itself. The same argument with $L_E = 0$ shows that the line avoids the origin because $d \notin \mathbb{Q}\chi$. Since $d \in \text{int}(\text{Eff})$ and Eff is strongly convex, $-d \notin \text{Eff}$; by separation, $\chi - Td \notin \text{Eff}$ for all sufficiently large T . Avoid the finitely many rational values of T for which $\chi - Td$ lies on a wall, and put $\theta_{\text{out}} = \chi - Td$. Then $[\theta_{\text{out}}, \chi]$ has the required genericity. The character θ_{out} is rational. Choose $N > 0$ with $N\theta_{\text{out}} \in A$. The support-cone semistable locus is unchanged by positive scaling, so $U_{N\theta_{\text{out}}} = U_{\theta_{\text{out}}} = \emptyset$ because $\theta_{\text{out}} \notin \text{Eff}$. The sign statement is exactly the way d was chosen, after replacing α_W by the primitive normal λ of the crossed wall. If that replacement requires multiplying the chosen linear equation of the wall by -1 , then both $\alpha_W(d)$ and $\alpha_W(\kappa)$ change sign, so the displayed sign equality is unchanged for the oriented normal λ . $\square$

Lemma 8.2 (path admissibility consequences). *For the segment supplied by Lemma 8.1, the following assertions hold.*

- (i) *The segment meets only finitely many walls, and every crossing is transverse through the relative interior of a nonzero codimension-one wall face.*
- (ii) *At every crossing, the local support-cone decomposition modulo the linear span of the crossed wall is a one-dimensional rational polyhedral fan. Hence the forward nonempty chamber, when it exists, is the unique chamber in the forward transverse direction.*
- (iii) *Boundary crossings are precisely the crossings where the interval just before the wall is outside Eff . In such a crossing the backward semistable locus is empty and the forward side is the unique inside chamber.*
- (iv) *The segment avoids the origin and all codimension-at-least-two cells. Thus no higher-rank wall crossing is taken through the origin; the only rank-one origin case used in the proof is the separate base case Lemma 8.4.*
- (v) *At a boundary entry with primitive normal positive on the inside side, all normal Cox weights are nonnegative and at least one is positive. Hence $q = 0$, $p > 0$, and $r = \eta_+ > 0$. In particular, an $r = 0$ wall step is never an outside-to-inside boundary entry.*

Proof. The support-cone decomposition is finite. By construction of the line in Lemma 8.1, it is not contained in a wall hyperplane, meets no codimension-at-least-two cell, and avoids the origin. Therefore its intersections with the walls are finite and transverse, and the point of

intersection lies in the relative interior of a unique codimension-one wall face. This proves (i) and (iv).

Assertion (ii) is Lemma 2.20 applied at each crossing: near a point in the relative interior of the wall face, the decomposition is the inverse image of the star fan modulo the wall span, and that quotient fan is one-dimensional. The statement about the unique forward chamber follows by applying this local description to the transverse tangent vector of the segment.

For (iii), Lemma 2.17 identifies the complement of Eff with empty semistable locus for the purposes of the segment. Since Eff is a closed convex cone, the intersection $[\theta_{\text{out}}, \chi] \cap \text{Eff}$ is a closed interval ending at χ . Hence the directed segment can enter Eff at most once and cannot leave it again. A transverse crossing from its complement into Eff occurs through a boundary wall face, and the local one-dimensional fan has only the inside ray on the effective side. Conversely an interior wall crossing has nonempty adjacent chambers on both sides and is contained in Eff .

For (v), at a boundary entry choose the primitive normal λ positive on the inside side. The effective cone lies in the half-space $\lambda \geq 0$ near the crossed face, so every Cox degree has nonnegative λ -weight; otherwise the ray generated by that Cox degree would give semistable support on the outside side. Since the forward side is nonempty and the wall is not the whole effective cone, at least one normal variable has positive λ -weight. Thus the negative normal list is empty, the positive normal list is nonempty, and

$$r = \eta_+ - \eta_- = \eta_+ > 0.$$

□

Lemma 8.3 (directed wall orientation). *Let a directed segment cross a nonzero wall from the already reached chamber to the next chamber. Choose the primitive wall normal λ positive on the next chamber. Let*

$$\kappa = \sum_i \deg x_i, \quad r = \eta_+ - \eta_-.$$

Then $r = \lambda(\kappa)$. For the outside-to-target directed segment used in the Cox secondary fan, $r \geq 0$. If $r = 0$, the wall step is an equivalence.

Proof. The identity $r = \lambda(\kappa)$ is obtained by summing the positive, negative, and zero normal weights of all Cox variables. Let d be the direction chosen in Lemma 8.1. Parametrize the directed segment as

$$\gamma(t) = \theta_{\text{out}} + td, \quad 0 \leq t \leq T,$$

so that increasing t moves from the outside endpoint toward the target. If t_0 is a crossing time and the primitive normal λ is chosen positive on the next chamber, then for $0 < \varepsilon \ll 1$

$$\lambda(\gamma(t_0 - \varepsilon)) < 0, \quad \lambda(\gamma(t_0 + \varepsilon)) > 0.$$

Since $\lambda(\gamma(t_0 + \varepsilon)) - \lambda(\gamma(t_0 - \varepsilon)) = 2\varepsilon\lambda(d)$, we have $\lambda(d) > 0$. If $\lambda(\kappa) \neq 0$, the sign preservation in Lemma 8.1 gives $\lambda(\kappa) > 0$. If $\lambda(\kappa) = 0$, then $r = 0$. Hence the directed crossing is always in the $r \geq 0$ orientation, and Proposition 7.26 has an empty block list when $r = 0$, so that step is an equivalence. The same parameterization also covers the boundary entry from the outside empty interval: the missing side is the $\lambda < 0$ side by this convention, and the nonempty inside side is the $\lambda > 0$ target side. □

Lemma 8.4 (rank-one base case). *Let $\mathfrak{X} = [U_\theta/D(A)]$ be a free-lattice Cox chamber quotient over $\mathbb{Z}$, with $B_0 = \mathbb{Z}$, $\text{rk } A_{\mathbb{Q}} = 1$, and $\mathfrak{X}$ split smooth proper with finite split diagonalizable stabilizers. Then $\text{Perf}(\mathfrak{X})$ has a finite S -linear semiorthogonal decomposition by copies of $\text{Perf}(S)$.*

Proof. Since A is free of rank one, choose a generator and view all Cox degrees as integers. The condition $B_0 = \mathbb{Z}$ forces every variable to have nonzero degree and all degrees to have the same sign: a zero-degree variable is a nonconstant element of B_0 , while variables of both signs give a nonconstant degree-zero monomial after clearing the two positive integers.

After replacing the generator by its negative if necessary, all variable degrees are positive. The opposite side of the unique wall is outside Eff, hence has empty semistable locus by Lemma 2.17.

The wall is the $q = 0$ residual weighted-projective wall with center $\mathfrak{z} = S$. Indeed, the primitive wall normal has $\ker \lambda = 0$, and $B_0 = \mathbb{Z}$ excludes zero-weight variables. Thus $\eta_- = 0$, $r = \eta_+ = \sum_i \deg x_i$, and Theorem 6.15 gives

$$\mathrm{Perf}(\mathfrak{X}) = \langle \mathcal{B}_w, \dots, \mathcal{B}_{w+\eta_+-1} \rangle, \quad \mathcal{B}_t \simeq \mathrm{Perf}(S).$$

The positive residual weights allow stacky stabilizers; the number of blocks is the weighted sum η_+ . $\square$

Example 8.5 (projective space). For $X = \mathbb{P}_{\mathbb{Z}}^n$, the Cox ring is

$$B = \mathbb{Z}[x_0, \dots, x_n], \quad A = \mathbb{Z}, \quad \deg x_i = 1.$$

The effective cone is $\mathbb{Q}_{\geq 0}$, and the directed induction is the rank-one boundary crossing from the outside empty side into the unique positive chamber. Here $q = 0$, $\eta_+ = n + 1$, and the center is $\mathrm{Spec} \mathbb{Z}$. Lemma 8.4 gives

$$\mathrm{Perf}(\mathbb{P}_{\mathbb{Z}}^n) = \langle \mathcal{O}(w), \mathcal{O}(w+1), \dots, \mathcal{O}(w+n) \rangle$$

up to the shift convention for character lines. Thus $N = n + 1$, matching the number of maximal cones in the fan and the usual Beilinson block count. For a weighted rank-one Cox datum with positive weights a_i , the same calculation gives $\eta_+ = \sum_i a_i$ residual grading blocks; this is the stacky boundary case recorded in Remark 6.10.

Example 8.6 (a non-étale rank-one stabilizer check). Consider the rank-one Cox datum

$$B = \mathbb{Z}[x_0, x_1], \quad A = \mathbb{Z}, \quad \deg x_0 = \deg x_1 = 2.$$

The positive chamber quotient is the weighted projective stack $[(\mathbb{A}_{\mathbb{Z}}^2 \setminus 0)/\mathbb{G}_m]$ with weights $(2, 2)$. Its generic stabilizer is μ_2 , which is not étale over the fiber of characteristic 2. The proof above nevertheless applies without change, because the quotient is treated as a split diagonalizable quotient and all calculations are performed in the $\mathbb{Z}$ -graded derived category. Here $\eta_+ = 4$, so the residual boundary theorem gives four residual grading blocks

$$\mathcal{O}(w), \mathcal{O}(w+1), \mathcal{O}(w+2), \mathcal{O}(w+3).$$

This example is not the final smooth toric scheme case; it is included to show where non-étale finite split diagonalizable stabilizers enter the intermediate wall quotients and why the graded-module formalism is used instead of an étale representation-theoretic argument.

Example 8.7 ($\mathbb{P}^1 \times \mathbb{P}^1$). Let

$$B = \mathbb{Z}[x_0, x_1, y_0, y_1], \quad A = \mathbb{Z}^2,$$

with $\deg x_i = (1, 0)$ and $\deg y_j = (0, 1)$. The effective cone is the first quadrant, and the ample chamber is its interior. Choose an outside starting point with first coordinate negative and second coordinate positive, and direct a generic segment to $(1, 1)$. The segment enters Eff through the boundary wall $a_1 = 0$, away from the origin. For this wall the zero-weight variables are y_0, y_1 , so the corrected center is $\mathbb{P}_{\mathbb{Z}}^1$; the positive normal variables are x_0, x_1 , each of weight 1, so $\eta_+ = 2$. The boundary wall crossing gives two supported blocks equivalent to $\mathrm{Perf}(\mathbb{P}_{\mathbb{Z}}^1)$. Refining those two center blocks by the rank-one case for $\mathbb{P}_{\mathbb{Z}}^1$ gives four $\mathrm{Perf}(\mathbb{Z})$ -blocks, as expected from the four maximal cones of the fan of $\mathbb{P}^1 \times \mathbb{P}^1$.

Example 8.8 (a two-wall path: the Hirzebruch surface $\mathbb{F}_1$). This example illustrates how the directed induction can use both a boundary crossing and an interior crossing. Consider the Cox datum

$$B = \mathbb{Z}[x_0, x_1, y_0, y_1], \quad A = \mathbb{Z}^2,$$

with

$$\deg x_0 = \deg x_1 = (1, 0), \quad \deg y_0 = (0, 1), \quad \deg y_1 = (1, 1).$$

The GIT fan in $A_{\mathbb{Q}}$ has effective cone the first quadrant, subdivided by the ray $\mathbb{Q}_{\geq 0}(1, 1)$. The chamber cone $((1, 0), (1, 1))^{\circ}$ gives the smooth toric surface $\mathbb{F}_1$, equivalently the blow-up of $\mathbb{P}_{\mathbb{Z}}^2$ at a torus fixed point. Take the target character $\theta = (2, 1)$. Starting from an outside point $\theta_{\text{out}} = (-\varepsilon, 1)$, with $0 < \varepsilon \ll 1$ rational, choose a rational generic segment arbitrarily close to the horizontal segment from θ_{out} to θ . It first crosses the boundary wall $s = 0$ and then the interior wall $s = t$. The perturbation is chosen small enough to preserve the signs of the two wall normals and to avoid the higher-codimension cells.

At the boundary wall $s = 0$, choose $\lambda(s, t) = s$, positive on the inside side. The zero-weight variable is y_0 , while x_0, x_1, y_1 have positive normal weight 1. Hence the corrected center is

$$[D(y_0)/D(\ker \lambda)] \simeq \text{Spec } \mathbb{Z},$$

and $\eta_+ = 3, \eta_- = 0$. The one-sided boundary wall theorem adds three $\text{Perf}(\mathbb{Z})$ -blocks and reaches the intermediate chamber cone $((0, 1), (1, 1))^{\circ}$.

At the interior wall $s = t$, orient the primitive normal as $\lambda(s, t) = s - t$, positive on the target chamber. The normal weights are

$$\lambda(\deg x_0) = \lambda(\deg x_1) = 1, \quad \lambda(\deg y_0) = -1, \quad \lambda(\deg y_1) = 0.$$

Thus $\eta_+ = 2, \eta_- = 1$, and $r = 1$. The zero-weight center is again $[D(y_1)/D(\ker \lambda)] \simeq \text{Spec } \mathbb{Z}$, so the interior wall crossing adds one further $\text{Perf}(\mathbb{Z})$ -block. The directed path therefore produces $3 + 1 = 4$ blocks, matching the four maximal cones of the fan of $\mathbb{F}_1$. The example is meant only as bookkeeping for the global induction: different generic directed paths can present the same final integer through different intermediate chamber quotients, but every step is governed by the same boundary/interior wall statements used in Theorem 8.10.

Before proving the global induction theorem, we record the dependency structure to make the induction non-circular. The residual wall theorem, Theorem 6.15, is proved independently of global induction. The actual wall theorem, Proposition 7.26, uses the residual theorem and the geometric hypotheses on the adjacent chambers and corrected center, but it does not assume any block decomposition for those categories. In the directed induction below, the source-side decomposition is supplied only by the finite induction along the chosen path, while a center decomposition is supplied only by the outer lexicographic induction after Propositions 3.2 and 4.2 have shown that the center has smaller induction invariant.

Proposition 8.9 (wall centers are inductive). *Let a nonzero wall crossed by the directed segment occur in a free-lattice Cox datum with $B_0 = \mathbb{Z}$, and let*

$$d = \#\{\text{Cox variables}\} - \text{rk } A_{\mathbb{Q}}$$

be the relative dimension of the adjacent chamber quotients. If the corrected center is nonempty, then it is again a free-lattice Cox quotient with degree-zero Cox part $\mathbb{Z}$, split smooth proper over $\mathbb{Z}$ with finite split diagonalizable stabilizers, and its induction invariant is strictly smaller than that of the adjacent chamber:

- (a) *for a two-sided wall, its relative dimension is $d - (p + q - 1) < d$;*
- (b) *for a one-sided wall with s normal variables, its relative dimension is $d - (s - 1)$, so either the dimension drops when $s \geq 2$, or $s = 1$ and the dimension is unchanged but $\text{rk}(\ker \lambda)_{\mathbb{Q}} = \text{rk } A_{\mathbb{Q}} - 1$.*

Proof. The geometric hypotheses and the assertion on degree-zero Cox parts are Proposition 3.2 in the one-sided case and Proposition 4.2 in the two-sided case. Their dimension formulae are respectively

$$\dim \mathfrak{X} - \dim \mathfrak{Z} = s - 1 \quad \text{and} \quad \dim \mathfrak{X} - \dim \mathfrak{Z} = p + q - 1.$$

For a two-sided wall $p, q \geq 1$, hence $p + q - 1 \geq 1$. In the one-sided case, if $s = 1$ then the dimension formula gives no drop, but $\lambda : A \rightarrow \mathbb{Z}$ is primitive and A is free, so $\ker \lambda$ is a free direct summand of rational rank one less than A . This is exactly the lexicographic decrease used in the global induction. $\square$

The recursive bookkeeping used in the next theorem is therefore as follows.

- (i) An outside-to-outside step uses no wall theorem and makes no recursive call.
- (ii) A boundary crossing with empty center uses the zero or empty-block case of Corollary 7.28; no center category is invoked. This case is kept in the statement to match the empty-center convention of the actual wall theorem. In the ordinary nonempty outside-to-inside step, if the corrected center is nonempty then Proposition 3.2 supplies its induction hypotheses; if it is empty, there is simply no recursive branch.
- (iii) A boundary crossing with nonempty center uses Corollary 7.28. The recursive input is $\text{Perf}(\mathfrak{Z})$, supplied by the outer induction; the dimension drops if the one-sided normal set has size $s \geq 2$, while for $s = 1$ the dimension is unchanged and the rational rank drops.
- (iv) An interior crossing with empty center uses Proposition 7.26 but omits the supported blocks and makes no recursive center call.
- (v) An interior crossing with nonempty center uses Proposition 7.26. The recursive input is $\text{Perf}(\mathfrak{Z})$, supplied by the outer induction, and the dimension drops by $p + q - 1$.

Thus only the nonempty-center cases call the outer induction, and those are precisely the cases covered by Proposition 8.9.

Theorem 8.10 (directed induction theorem). *Let $\mathfrak{X} = [U_\theta/D(A)]$ be a free-lattice Cox chamber quotient over $\mathbb{Z}$, with $B_0 = \mathbb{Z}$, θ in a full-dimensional chamber, and $\mathfrak{X}$ split smooth proper with finite split diagonalizable stabilizers. Then there are fully faithful S -linear functors $\text{Perf}(S) \rightarrow \text{Perf}(\mathfrak{X})$ whose images form a finite S -linear semiorthogonal decomposition of $\text{Perf}(\mathfrak{X})$.*

Proof. We prove the statement by lexicographic induction on pairs (d, ρ) , where $d = \dim(\mathfrak{X}/S)$ is the relative dimension over $S = \text{Spec } \mathbb{Z}$, computed from a quotient presentation as “number of Cox variables minus $\text{rk } A$ ”, and $\rho = \text{rk } A_{\mathbb{Q}}$. The induction claim $P(d, \rho)$ is that every chamber quotient satisfying the hypotheses with dimension at most d , and with equality in dimension only when the rank is at most ρ , has the asserted decomposition. For a fixed target with invariants (d, ρ) , after choosing the directed outside segment below, we use an ordinary finite induction on the number of wall crossings from an empty initial interval on the segment to the target chamber. Rank zero gives $\mathfrak{X} \simeq S$. Rank one is Lemma 8.4.

Assume $\text{rk } A_{\mathbb{Q}} \geq 2$ and assume $P(d', \rho')$ is known for all $(d', \rho') < (d, \rho)$. Apply Lemma 8.1 with $\chi = \theta$, and orient the generic segment from the outside endpoint $\theta_{\text{out}} \notin \text{Eff}$ to θ . Subdivide this segment by its intersections with the finitely many wall faces of the GIT fan. On every open subinterval lying outside Eff , the semistable locus is empty by Lemma 2.17. We start the finite induction along the directed path with this zero category. By the convention in Section 2, it carries the empty semiorthogonal decomposition. Since Eff is convex, Lemma 8.2 shows that the intersection $[\theta_{\text{out}}, \theta] \cap \text{Eff}$ is a single terminal interval: the path enters the effective cone at most once and never leaves it again. Thus the only outside-to-inside step is a boundary crossing; after that, all nonempty-to-nonempty crossings are interior wall crossings. The segment was chosen to avoid the origin and all cells of codimension at least two. Hence every wall actually crossed in the finite induction is a nonzero codimension-one wall face, so the hypotheses of the actual and boundary wall statements apply after choosing the corresponding primitive normal. The possible origin wall in rational rank one is not treated by this high-rank path argument; it is exactly the separate base case Lemma 8.4. All support-cone cells are rational polyhedral, so for each chamber and wall face met by the segment we choose, after positive scaling, integral characters in the relevant relative interiors; these choices define the same semistable opens. Since the Cox ring is fixed and has $B_0 = \mathbb{Z}$, Lemma 2.29 shows that every nonempty full-dimensional chamber quotient met by the segment is split smooth proper over $\mathbb{Z}$, with finite split diagonalizable stabilizers. At each crossed wall, Propositions 3.2 and 4.2 show that the corrected center, when nonempty, has the same geometric properties and has degree-zero Cox part $\mathbb{Z}$. More precisely, Proposition 8.9 shows that every nonempty center lies in the smaller lexicographic induction range. Thus

the geometric hypotheses of Proposition 7.26 are satisfied at each crossed wall, with empty sides interpreted as zero categories. For reference, the induction ledger at a crossed wall is the following.

- (a) If the target interval is still outside Eff, no wall theorem is used and the category remains zero.
- (b) If the wall is a one-sided boundary wall from the outside into Eff, Corollary 7.28 applies. Its nonempty inside chamber is covered by Lemma 2.29, and every nonempty center is covered by Proposition 8.9.
- (c) If the wall is an interior two-sided wall, the nonempty adjacent chambers are covered by Lemma 2.29. A nonempty corrected center is covered by Proposition 8.9.
- (d) If the corrected center is empty, the supported blocks at that wall are omitted by Proposition 7.26; no recursive call on a center category is made.

We now induct along the finitely many crossed walls. Suppose the already reached chamber has the desired decomposition. At the next wall, Lemma 8.3 says that the crossing is the $r \geq 0$ orientation of Proposition 7.26. There are three cases. First, if the target side of this crossing is still empty, its perfect category is zero and the empty decomposition remains in force; this empty-to-empty step does not use the nonempty-target part of Proposition 7.26. Second, if the wall is an interior wall between two adjacent GIT chambers and the target is nonempty, then Proposition 7.26 applies in its nonempty-target case. Third, if this is the boundary crossing from the empty outside interval into Eff, then Corollary 7.28 applies. In the second and third cases the next chamber is obtained from the previous one by adding finitely many blocks equivalent to the corrected center; when $r = 0$ the block list is empty and the step is an equivalence.

By Proposition 8.9, every nonempty center has invariant $(d', \rho') < (d, \rho)$, so its perfect category has a finite $\text{Perf}(S)$ -block decomposition by the outer induction hypothesis. The fully faithful wall functor transports the already constructed source SOD to its image by Lemma 2.13; likewise each fully faithful center block functor β_a transports the outer-induction SOD of $\text{Perf}(3)$ to the corresponding supported block. Applying Lemma 2.12 to the wall decomposition and then to these transported decompositions gives the decomposition for the next chamber. The finite inner induction over all crossings reaches the target chamber. $\square$

9. FROM COX QUOTIENTS TO TORIC SCHEMES

Proposition 9.1 (Cox presentation over $\mathbb{Z}$). *Let $X = X_\Sigma$ be a split smooth projective toric scheme over $\mathbb{Z}$. Then X is obtained from its Cox polynomial ring*

$$B = \mathbb{Z}[x_\rho : \rho \in \Sigma(1)]$$

graded by $\text{Pic}(X)$, with irrelevant open U_Σ , as

$$X \simeq [U_\Sigma / D(\text{Pic}(X))].$$

The group $\text{Pic}(X)$ is free, and $B_0 = \mathbb{Z}$.

Proof. For a smooth split toric scheme the invariant divisor sequence is

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{\Sigma(1)} \longrightarrow \text{Pic}(X) \longrightarrow 0.$$

For a smooth fan, the sublattice $M \subset \mathbb{Z}^{\Sigma(1)}$ is saturated: equivalently, the invariant prime divisors meeting each affine chart contain a $\mathbb{Z}$ -basis dual to the corresponding smooth cone. Hence the quotient $\text{Pic}(X)$ is free; this is the usual smooth toric divisor sequence. The Cox construction and the quotient presentation commute with base change from $\text{Spec } \mathbb{Z}$, because the affine toric charts are $\text{Spec } \mathbb{Z}[\sigma^\vee \cap M]$ and the irrelevant open is defined by monomials indexed by maximal cones. Smoothness of the fan makes the Cox action on U_Σ scheme-theoretically free. Indeed, on the affine Cox chart attached to a maximal cone σ , all variables x_ρ with $\rho \notin \sigma(1)$ are invertible. The divisor classes of these $\rho \notin \sigma(1)$ generate

$\mathrm{Pic}(X)$, by the smooth divisor sequence and the fact that the rays of σ form part of a lattice basis. More explicitly, in

$$0 \rightarrow M \rightarrow \mathbb{Z}^{\Sigma(1)} \rightarrow \mathrm{Pic}(X) \rightarrow 0,$$

the coordinates corresponding to $\sigma(1)$ split off integrally because σ is smooth; quotienting by them identifies $\mathrm{Pic}(X)$ with the free abelian group generated by the classes of the divisors $\rho \notin \sigma(1)$. Hence the stabilizer

$$D(\mathrm{Pic}(X)/\langle [D_\rho] : x_\rho \neq 0 \rangle)$$

is the trivial group scheme on every Cox chart. Thus the quotient stack is the toric scheme itself rather than a nontrivial stacky toric quotient. This gives the displayed quotient presentation; see [CLS, Theorem 5.1.11 and Chapter 14]. Finally $B_0 = \mathbb{Z}$. Since $X \rightarrow \mathrm{Spec} \mathbb{Z}$ is a split projective toric scheme with connected geometric fibers, the Stein factorization gives

$$\pi_* \mathcal{O}_X = \mathcal{O}_{\mathrm{Spec} \mathbb{Z}}.$$

Equivalently, after base change to every field, completeness of the fan implies that the only torus characters extending to global regular functions on the complete toric variety are constants. A degree-zero Cox monomial is a $D(\mathrm{Pic} X)$ -invariant regular function on U_Σ , hence descends through the Cox quotient to a global regular function on X . The previous paragraph therefore excludes every nonconstant degree-zero monomial. Since the Cox monomials form a $\mathbb{Z}$ -basis of B , the entire invariant degree-zero part is exactly $\mathbb{Z}$. $\square$

Proposition 9.2 (object extraction). *Let $F_i : \mathrm{Perf}(S) \rightarrow \mathrm{Perf}(X)$, $1 \leq i \leq N$, be fully faithful S -linear functors whose images form a semiorthogonal generating sequence. Put $E_i = F_i(\mathcal{O}_S)$. Then*

$$F_i(M) \simeq \pi^* M \otimes E_i$$

for every $M \in \mathrm{Perf}(S)$, and

$$\mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_i, E_i) \simeq \mathcal{O}_S, \quad \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_j, E_i) \simeq 0 \quad (j > i).$$

The objects E_i generate $\mathrm{Perf}(X)$.

Proof. The object $\mathcal{O}_S$ thickly generates $\mathrm{Perf}(S)$, because every perfect $\mathbb{Z}$ -complex is a bounded complex of finite free modules up to retracts. The functor F_i is an exact $\mathrm{Perf}(S)$ -module functor. Hence S -linearity gives a natural transformation

$$M \star F_i(\mathcal{O}_S) \longrightarrow F_i(M), \quad M \star F_i(\mathcal{O}_S) = \pi^* M \otimes E_i.$$

It is an equivalence for $M = \mathcal{O}_S$, and the class of M 's for which it is an equivalence is a thick subcategory of $\mathrm{Perf}(S)$. Thus it is an equivalence for every $M \in \mathrm{Perf}(S)$.

For every $P \in \mathrm{Perf}(S)$,

$$\begin{aligned} \mathrm{Map}_S(P, \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_i, E_i)) &\simeq \mathrm{Map}_X(\pi^* P \otimes E_i, E_i) \\ &\simeq \mathrm{Map}_X(F_i(P), F_i(\mathcal{O}_S)) \simeq \mathrm{Map}_S(P, \mathcal{O}_S), \end{aligned}$$

where the last equivalence is full faithfulness. The perfect complexes on $S = \mathrm{Spec} \mathbb{Z}$ compactly generate $D_{\mathrm{qc}}(S)$, so testing against all perfect P identifies objects of $D_{\mathrm{qc}}(S)$. Hence $\mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_i, E_i) \simeq \mathcal{O}_S$. For $j > i$, the same calculation gives

$$\mathrm{Map}_S(P, \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_X(E_j, E_i)) \simeq \mathrm{Map}_X(F_j(P), F_i(\mathcal{O}_S)) = 0$$

by semiorthogonality, hence the cross-Ext complex is zero. Finally, the thick subcategory generated by the E_i 's contains every $F_i(M)$, since it contains $F_i(\mathcal{O}_S)$ and is closed under the $\mathrm{Perf}(S)$ -action generated by $\mathcal{O}_S$. The images of the F_i 's generate $\mathrm{Perf}(X)$. $\square$

Lemma 9.3 (admissibility of relative blocks). *Let $T = \mathrm{Spec} R$, let $\pi : Y \rightarrow T$ be smooth and proper, and let $E \in \mathrm{Perf}(Y)$. The T -linear functor*

$$\Phi_E : \mathrm{Perf}(R) \longrightarrow \mathrm{Perf}(Y), \quad M \longmapsto \pi^* M \otimes E$$

has right adjoint

$$\Psi_E^R(P) = \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_Y(E, P).$$

It also has a left adjoint, given by relative duality. In particular, if Φ_E is fully faithful, then its image is an admissible subcategory of $\text{Perf}(Y)$.

Consequently, a finite semiorthogonal decomposition of $\text{Perf}(Y)$ whose components are images of fully faithful functors of this form gives a finite iterated split exact filtration in $\text{Cat}_\infty^{\text{perf}}$.

Proof. The displayed formula comes from tensor-Hom adjunction on Y , followed by the definition of the derived pushforward mapping object over R :

$$\text{Map}_Y(\pi^*M \otimes E, P) \simeq \text{Map}_R(M, \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_Y(E, P)).$$

Because π is smooth and proper, it is perfect and proper, so the right-hand object is perfect when P is perfect. The existence of the left adjoint follows from relative Serre duality for the smooth proper morphism π . Since E is perfect, $\mathbf{R}\mathcal{H}om_Y(E, P) \simeq P \otimes E^\vee$. The left adjoint is obtained by first applying $P \mapsto P \otimes E^\vee$, and then applying the left adjoint to π^* , namely

$$Q \longmapsto \mathbf{R}\pi_*(Q \otimes \pi^! \mathcal{O}_T).$$

Here $\pi^! \mathcal{O}_T$ is a perfect invertible object up to shift because π is smooth. When π has constant relative dimension d , it is $\omega_\pi[d]$, and the left adjoint may be written as

$$\Psi_E^L(P) \simeq \mathbf{R}\pi_*(P \otimes E^\vee \otimes \omega_\pi[d]).$$

Thus a fully faithful Φ_E has an admissible image.

The functor and its adjoints are R -linear. For the right adjoint, if $N \in \text{Perf}(R)$, then the tensor-Hom adjunction and the projection formula give

$$\Psi_E^R(P \otimes \pi^*N) \simeq \mathbf{R}\pi_*(\mathbf{R}\mathcal{H}om_Y(E, P) \otimes \pi^*N) \simeq \Psi_E^R(P) \otimes N.$$

The same calculation, with the additional fixed perfect factor $\pi^! \mathcal{O}_T$, gives

$$\Psi_E^L(P \otimes \pi^*N) \simeq \Psi_E^L(P) \otimes N.$$

Thus the inclusions and projection functors attached to such blocks live in the $\text{Perf}(R)$ -linear Morita category, not merely in the underlying Morita category.

For the last assertion, write the decomposition as $\text{Perf}(Y) = \langle \mathcal{A}_1, \dots, \mathcal{A}_N \rangle$, with $\mathcal{C}_m = \text{thick}\langle \mathcal{A}_1, \dots, \mathcal{A}_m \rangle$. The admissible projection from $\text{Perf}(Y)$ to $\mathcal{A}_m$ kills $\mathcal{C}_{m-1}$ by semiorthogonality and is the identity on $\mathcal{A}_m$; hence it preserves $\mathcal{C}_m$. Therefore $\mathcal{A}_m$ is admissible in $\mathcal{C}_m$, and the projection $\mathcal{C}_m \rightarrow \mathcal{A}_m$ has kernel $\mathcal{C}_{m-1}$ and is split by the inclusion $\mathcal{A}_m \hookrightarrow \mathcal{C}_m$. Hence

$$\mathcal{C}_{m-1} \longrightarrow \mathcal{C}_m \longrightarrow \mathcal{A}_m$$

is split exact in the $\text{Perf}(R)$ -linear Morita category. Iterating over m gives the claimed split exact filtration. $\square$

Proof of Theorem 1.1. By Proposition 9.1, the toric scheme X is a free-lattice Cox chamber quotient with $B_0 = \mathbb{Z}$. Choose the class $\theta \in \text{Pic}(X)$ of an ample toric line bundle. In the Cox construction for a projective fan, θ lies in the full-dimensional ample chamber of the secondary fan and its semistable locus is the irrelevant open U_Σ ; hence

$$X \simeq [U_\theta / D(\text{Pic } X)].$$

Lemma 2.29 supplies the finite split diagonalizable Artin hypotheses for all nonempty chambers encountered in the directed induction. The directed induction theorem, Theorem 8.10, gives a finite S -linear semiorthogonal decomposition of $\text{Perf}(X)$ by fully faithful copies of $\text{Perf}(S)$. Applying Proposition 9.2 to those copies gives the objects E_i , the tensor formula for the functors, the relative Ext identities, and generation. $\square$

Lemma 9.4 (perfect Morita and Ext base change). *Let $R_0 \rightarrow R$ be a morphism of commutative rings, let $\pi : Y \rightarrow \text{Spec } R_0$ be smooth and proper, and set*

$$Y_R = Y \times_{\text{Spec } R_0} \text{Spec } R.$$

Then the ordinary fiber product agrees with the derived fiber product, and there is a Morita base-change equivalence

$$\mathrm{Perf}(Y_R) \simeq \mathrm{Perf}(Y) \otimes_{\mathrm{Perf}(R_0)} \mathrm{Perf}(R).$$

Moreover, for $E, F \in \mathrm{Perf}(Y)$, with pullbacks $E_R, F_R \in \mathrm{Perf}(Y_R)$, one has

$$\mathbf{R}\pi_{R*} \mathbf{R}\mathcal{H}om_{Y_R}(E_R, F_R) \simeq R \otimes_{R_0}^{\mathbf{L}} \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_Y(E, F).$$

Proof. Smoothness implies flatness, so the classical fiber product is the derived fiber product. The Morita equivalence is an instance of the standard perfect-stack tensor product theorem: smooth proper schemes are perfect stacks, and for perfect stacks the tensor product of perfect categories computes perfect complexes on derived fiber products. In the present affine-base situation this gives

$$\mathrm{Perf}(Y) \otimes_{\mathrm{Perf}(R_0)} \mathrm{Perf}(R) \simeq \mathrm{Perf}(Y \times_{\mathrm{Spec} R_0}^{\mathbf{L}} \mathrm{Spec} R) = \mathrm{Perf}(Y_R),$$

by the tensor-product/fiber-product theorem of Ben-Zvi–Francis–Nadler [BZFN, Theorem 1.2]; the same affine tensor identity is the form of derived Morita theory appearing in Toën [Toe, Theorem 1.4], with the tensor product of small idempotent-complete stable categories as in Blumberg–Gepner–Tabuada [BGT, §3]. We do not use an independent claim that tensoring small stable categories commutes with an arbitrary Čech limit. The finite affine Čech calculation is only a compatibility check: on an affine open $\mathrm{Spec} C \subset Y$, flatness gives

$$C \otimes_{R_0} R = C \otimes_{R_0}^{\mathbf{L}} R,$$

and the theorem restricts to the elementary Morita identity

$$\mathrm{Perf}(C) \otimes_{\mathrm{Perf}(R_0)} \mathrm{Perf}(R) \simeq \mathrm{Perf}(C \otimes_{R_0}^{\mathbf{L}} R).$$

The hypotheses needed for this application are exactly the following: $\mathrm{Spec} R_0$, $\mathrm{Spec} R$, and Y are perfect stacks; the morphism $Y \rightarrow \mathrm{Spec} R_0$ is represented by a smooth proper scheme, hence is a perfect morphism; and the tensor product is the relative tensor product of small idempotent-complete stable $\mathrm{Perf}(R_0)$ -module categories. The derived fiber product appearing in the theorem is the one displayed above, and smoothness identifies it with the ordinary fiber product Y_R . If the tensor-product theorem is read in a presentable QCoh -linear form, we apply it to these perfect stacks and then pass to compact objects: for perfect stacks the compact objects of QCoh are precisely Perf , and in the smooth proper situation the functors involved preserve compact objects. Thus the resulting equivalence is the displayed equivalence of small idempotent-complete $\mathrm{Perf}(R_0)$ -module categories. Equivalently, the external theorem input is precisely the relative formula

$$\mathrm{Perf}(X) \otimes_{\mathrm{Perf}(S)} \mathrm{Perf}(T) \simeq \mathrm{Perf}(X \times_S^{\mathbf{L}} T)$$

for the perfect stacks $X = Y$, $S = \mathrm{Spec} R_0$, and $T = \mathrm{Spec} R$. No stronger base-change statement for arbitrary stacks or arbitrary limits of small categories is used. This is the only Morita base-change input used in the sequel, and it is applied only in this smooth proper scheme setting.

For the Ext formula, the complex $\mathbf{R}\mathcal{H}om_Y(E, F)$ is perfect on Y . Since $Y \rightarrow \mathrm{Spec} R_0$ is smooth and proper, it is a perfect proper morphism; proper perfect base change for this perfect complex gives the displayed formula for the relative Ext complexes. No noetherian or flatness hypothesis on R is used. $\square$

Proposition 9.5 (base change of relative block decompositions). *Let $R_0 \rightarrow R$ be a morphism of commutative rings, let $\pi : Y \rightarrow \mathrm{Spec} R_0$ be smooth and proper, and put*

$$Y_R = Y \times_{\mathrm{Spec} R_0} \mathrm{Spec} R.$$

Since π is smooth, it is flat; hence this ordinary fiber product agrees with the derived fiber product. Equivalently, the derived base-change formula below may be read with $Y_R = Y \times_{\mathrm{Spec} R_0}^{\mathbf{L}} \mathrm{Spec} R$, and flatness identifies it with the displayed classical scheme. Assume that $\mathrm{Perf}(Y)$

has a finite R_0 -linear semiorthogonal decomposition whose components are the images of functors

$$\Phi_i : \text{Perf}(R_0) \rightarrow \text{Perf}(Y), \quad M \mapsto \pi^* M \otimes E_i, \quad E_i \in \text{Perf}(Y),$$

and assume these blocks are admissible. Then the base-changed objects $E_{i,R} = Lg^* E_i \in \text{Perf}(Y_R)$ define fully faithful R -linear functors

$$\Phi_{i,R} : \text{Perf}(R) \rightarrow \text{Perf}(Y_R), \quad M \mapsto \pi_R^* M \otimes E_{i,R},$$

whose images form a finite R -linear semiorthogonal decomposition of $\text{Perf}(Y_R)$. The relative Ext identities base change:

$$\mathbf{R}\pi_{R*} \mathbf{R}\mathcal{H}om_{Y_R}(E_{j,R}, E_{i,R}) \simeq R \otimes_{R_0}^{\mathbf{L}} \mathbf{R}\pi_* \mathbf{R}\mathcal{H}om_Y(E_j, E_i).$$

Proof. Apply Lemma 9.4. This Morita base-change equivalence is the only external input in the proposition: after it is applied, generation follows from split exactness of the admissible filtration, while full faithfulness and semiorthogonality are checked by the relative Ext base-change formula from the same lemma.

By admissibility, the given SOD determines a finite iterated split exact filtration in the Morita category of small idempotent-complete stable categories and exact functors. Each split exact step is represented by an exact inclusion together with exact left and right projection functors. Tensoring this diagram over $\text{Perf}(R_0)$ with $\text{Perf}(R)$ preserves the splitting data, hence preserves the split exact sequences, and identifies the graded pieces with $\text{Perf}(R)$. Thus the base-changed blocks generate $\text{Perf}(Y_R)$. The ordered semiorthogonal structure is then checked by the relative Ext complexes below: their diagonal terms give full faithfulness of each block and their upper-triangular vanishing gives semiorthogonality in the same order as the original decomposition.

The displayed formula for relative Ext complexes is the one in Lemma 9.4. Since the original block functors are fully faithful and semiorthogonal, the right-hand side is R for $i = j$ and zero for $j > i$. The projection formula then gives, for $M, N \in \text{Perf}(R)$,

$$\text{Map}_{Y_R}(\pi_R^* M \otimes E_{j,R}, \pi_R^* N \otimes E_{i,R}) \simeq \text{Map}_R(M, N \otimes_R^{\mathbf{L}} \mathbf{R}\pi_{R*} \mathbf{R}\mathcal{H}om_{Y_R}(E_{j,R}, E_{i,R})).$$

This identifies the $i = j$ mapping complexes with $\text{Map}_R(M, N)$ and the $j > i$ mapping complexes with zero, proving full faithfulness and semiorthogonality. $\square$

Corollary 9.6 (arbitrary split base). *Let R be a commutative ring, put $S_R = \text{Spec } R$, and let*

$$X_R = X_{\Sigma, \mathbb{Z}} \times_{\text{Spec } \mathbb{Z}} S_R$$

be the split smooth projective toric R -scheme associated with a smooth projective fan Σ . Then there are perfect complexes

$$E_{1,R}, \dots, E_{N,R} \in \text{Perf}(X_R)$$

such that the functors

$$\Phi_{i,R} : \text{Perf}(R) \longrightarrow \text{Perf}(X_R), \quad M \longmapsto \pi_R^* M \otimes E_{i,R}$$

are fully faithful and their images form a finite S_R -linear semiorthogonal decomposition

$$\text{Perf}(X_R) = \langle \Phi_{1,R} \text{Perf}(R), \dots, \Phi_{N,R} \text{Perf}(R) \rangle.$$

Moreover

$$\mathbf{R}\pi_{R*} \mathbf{R}\mathcal{H}om_{X_R}(E_{i,R}, E_{i,R}) \simeq \mathcal{O}_{S_R}, \quad \mathbf{R}\pi_{R*} \mathbf{R}\mathcal{H}om_{X_R}(E_{j,R}, E_{i,R}) \simeq 0 \quad (j > i),$$

and the $E_{i,R}$'s generate $\text{Perf}(X_R)$.

Proof. Apply Theorem 1.1 to the integral split toric model $X_{\Sigma, \mathbb{Z}}$, obtaining relative block objects $E_i \in \text{Perf}(X_{\Sigma, \mathbb{Z}})$. The blocks are admissible by Lemma 9.3. Since $X_{\Sigma, \mathbb{Z}} \rightarrow \text{Spec } \mathbb{Z}$ is smooth and proper, Proposition 9.5 applies to the morphism $\mathbb{Z} \rightarrow R$, giving the base-changed block decomposition and the displayed relative Ext identities. $\square$

Remark 9.7. The preceding corollary is a statement about split toric schemes, or equivalently toric schemes obtained from a fan by base change from $\mathbb{Z}$. For a general base ring the absolute group $\text{Pic}(X_R)$ may also contain classes pulled back from $\text{Pic}(R)$; the relevant Cox grading is the relative toric class lattice attached to the fan. Non-split toric forms are not covered by this corollary; in that setting one should expect possible twisted forms of the blocks.

Corollary 9.8 (additive invariants). *In the situation of Corollary 9.6, let N be the number of blocks in the semiorthogonal decomposition. If*

$$E : \text{Cat}_\infty^{\text{perf}} \longrightarrow \text{Sp}$$

is any Morita-invariant additive invariant of small idempotent-complete stable categories, then

$$E(\text{Perf}(X_R)) \simeq \bigoplus_{i=1}^N E(\text{Perf}(R)).$$

This splitting is the one determined by the chosen semiorthogonal decomposition and its admissible projection functors, hence it is tied to the directed VGIT path and window choices used above. We assert the existence of this finite additive splitting, not a canonical identification of its summands independent of those choices.

Proof. By Lemma 9.3, the decomposition of Corollary 9.6 is a finite iterated split exact filtration in $\text{Cat}_\infty^{\text{perf}}$, with graded pieces equivalent to $\text{Perf}(R)$. Additive invariants send split exact sequences to direct sums, so the displayed formula follows for every additive invariant. $\square$

Remark 9.9 (on the TC model). For TC below we fix the spectral-category model used in the localization theorem of Blumberg–Mandell [BM, Theorems 5.9, 7.1, and 7.2]. Thus TC is regarded as a Morita-invariant functor on spectral presentations of small idempotent-complete stable categories, and we use only its behavior on the finite split exact filtrations produced by the admissible semiorthogonal decomposition. The statement is not a comparison theorem among all possible models of topological cyclic homology; it is the additive splitting for this fixed functorial model.

Corollary 9.10 (K-theory, THH, and a fixed model of TC). *In the situation of Corollary 9.6, the scheme notation below means the value of the corresponding invariant on $\text{Perf}(-)$. For nonconnective algebraic K-theory, topological Hochschild homology, and the chosen spectral-category model of topological cyclic homology satisfying the Blumberg–Mandell localization and split-additivity theorem, viewed as a Morita-invariant functor on small idempotent-complete stable categories via spectral-category presentations, one has*

$$K(X_R) \simeq \bigoplus_{i=1}^N K(R), \quad \text{THH}(X_R) \simeq \bigoplus_{i=1}^N \text{THH}(R), \quad \text{TC}(X_R) \simeq \bigoplus_{i=1}^N \text{TC}(R).$$

Since the sums are finite, $\bigoplus$ denotes the finite biproduct in spectra, equivalently the finite product. Consequently

$$K_m(X_R) \cong K_m(R)^{\oplus N}$$

for all $m \in \mathbb{Z}$ for nonconnective K-theory, and similarly for the homotopy groups of THH and TC. If, in addition, one fixes a natural cyclotomic trace transformation from the chosen nonconnective K-theory functor to the chosen TC-functor on this Morita category, then under these decompositions the trace

$$K(X_R) \longrightarrow \text{TC}(X_R)$$

is the direct sum of N copies of the cyclotomic trace $K(R) \rightarrow \text{TC}(R)$. Hence its homotopy fiber also splits:

$$\text{fib}(K(X_R) \rightarrow \text{TC}(X_R)) \simeq \bigoplus_{i=1}^N \text{fib}(K(R) \rightarrow \text{TC}(R)).$$

All splittings in this corollary are splittings of functorial spectra attached to $\mathrm{Perf}(-)$ through the chosen admissible semiorthogonal decomposition.

Proof. Nonconnective K -theory is a localizing, hence additive, invariant in the sense of Blumberg–Gepner–Tabuada [BGT, Theorems 8.7 and 9.8]. The analogous localization and split-additivity statements for THH and for the spectral-category model of TC are the localization theorems of Blumberg–Mandell [BM, Theorems 7.1 and 7.2]. Applying Corollary 9.8 to these invariants gives the displayed decompositions. These decompositions are not chosen independently for the different invariants: they are induced by the same exact inclusion functors of the admissible blocks and by the same exact projection functors supplied by Lemma 9.3. The assertion about the cyclotomic trace is conditional on having fixed a natural transformation between the chosen nonconnective K -theory functor and the chosen TC-functor on this same Morita category. Natural transformations commute with the exact inclusion and projection functors defining the splittings. Therefore, for each block $\mathcal{A}_i \simeq \mathrm{Perf}(R)$, the square

$$\begin{array}{ccc} K(\mathcal{A}_i) & \longrightarrow & K(\mathrm{Perf}(X_R)) \\ \downarrow & & \downarrow \\ \mathrm{TC}(\mathcal{A}_i) & \longrightarrow & \mathrm{TC}(\mathrm{Perf}(X_R)) \end{array}$$

commutes, and the analogous square for the projection back to $\mathcal{A}_i$ also commutes. Under the finite direct-sum decompositions defined by these inclusion/projection pairs, the trace on $\mathrm{Perf}(X_R)$ is therefore the direct sum of the traces on the N graded pieces $\mathcal{A}_i \simeq \mathrm{Perf}(R)$. This proves the asserted splitting of the trace and hence of its homotopy fiber. $\square$

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